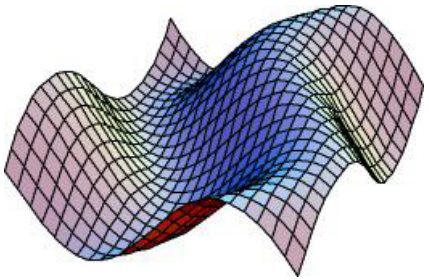
THE UNIVERSITY OF MELBOURNE

SCHOOL OF MATHEMATICS AND STATISTICS

MAST10006 Calculus 2

Lecture Notes



STUDENT NAME:	**EMAIL:**

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Section 0 - Notation used in MAST10006 Calculus 2

Standard Abbreviations

1. such that or given that: $|$
2. for all: $\forall$
3. there exists: $\exists$
4. equivalent to: $\equiv$
5. that is: $i.e$
6. approximate: $\approx$
7. much smaller than: $\ll$

Standard Notation for Sets of Numbers

1. natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$
2. integers: $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$
3. rational numbers: $\mathbb{Q} = \{\frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0\}$
4. real numbers: $\mathbb{R}$ (rational numbers plus irrational numbers)
5. complex numbers: $\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R},\ i^2 = -1\}$
6. $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$ (xy plane)
7. $\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ (3 dimensional space)

Standard Notation for Intervals

1. element of: $\in$
   so $a \in X$ means "a is an element of the set X"

2. open interval: (a, b)
   so $x \in (0, 1)$ means "$0 < x < 1$"

3. closed interval: $[a, b]$
   so $x \in [0, 1]$ means "$0 \leq x \leq 1$"

4. partial open and closed interval: $(a, b]$ or $[a, b)$
   so $x \in [0, 1)$ means "$0 \leq x < 1$"

5. not including: $\setminus$
   so $x \in \mathbb{R} \setminus \{0\}$ means "x is any real number excluding 0".
   Alternatively, we could write $(-\infty, 0) \cup (0, \infty)$ where $\cup$ means the "union of the two intervals".

More Standard Notation

1. natural logarithm (base e): $\log x$
   base 10 logarithm: $\log_{10} x$

   Alternative notations for natural logarithms used in textbooks: $\log_e x, \ln x$

2. inverse trigonometric functions: $\arcsin x$, $\arctan x$ etc

   Alternative notations used in textbooks: $\sin^{-1} x$, $\tan^{-1} x$ etc

3. implies: $\Rightarrow$
   so $p \Rightarrow q$ means "p implies q"

4. if and only if (iff): $\Leftrightarrow$ (means both $\Leftarrow$ and $\Rightarrow$)
   so $p \Leftrightarrow q$ means "p implies q" AND "q implies p"

5. approaches: $\rightarrow$
   so $f(x) \rightarrow 1$ as $x \rightarrow 0$ means "$f(x)$ approaches 1 as x approaches 0"

Greek Alphabet

α	alpha	ν	nu
β	beta	ξ	xi
γ	gamma	o	omicron
δ	delta	π	pi
ϵ or ε	epsilon	ρ	rho
ζ	zeta	σ	sigma
η	eta	τ	tau
θ	theta	υ	upsilon
ι	iota	ϕ	phi
κ	kappa	χ	chi
λ	lambda	ψ	psi
μ	mu	ω	omega

Section 1: Limits, Continuity, Sequences, Series

Why do we study limits and continuity in calculus?

Definition of limit of a function

Let $f : D \to \mathbb{R}$ be a real-valued function, whose domain D is a subset of $\mathbb{R}$.

The limit of $f(x)$ as x approaches a is L, written

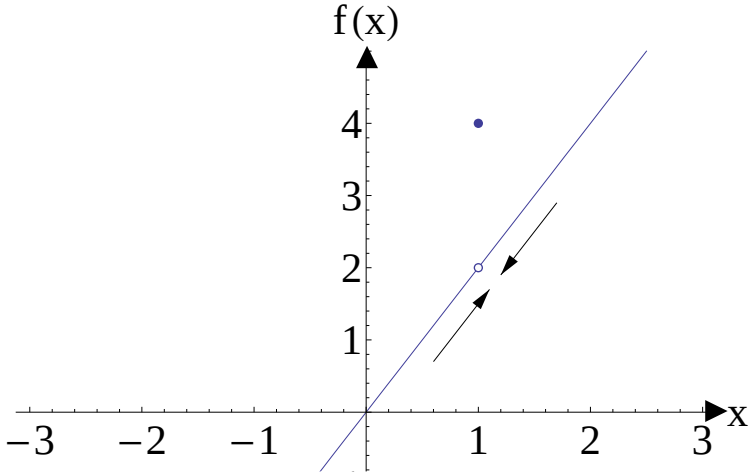
$$\lim_{x \to a} f(x) = L$$

if $f(x)$ gets arbitrarily close to L whenever x is close enough to a but $x \neq a$.

Note:

If it exists, the limit L must be a unique finite real number.

Example 1.1: Let $f(x) = \begin{cases} 2x & x \neq 1 \\ 4 & x = 1 \end{cases}$. Evaluate $\lim_{x \to 1} f(x)$.

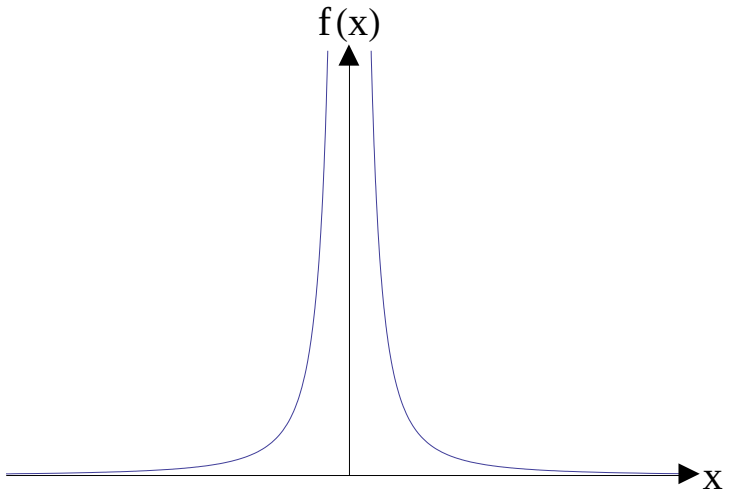


Solution:

Note:

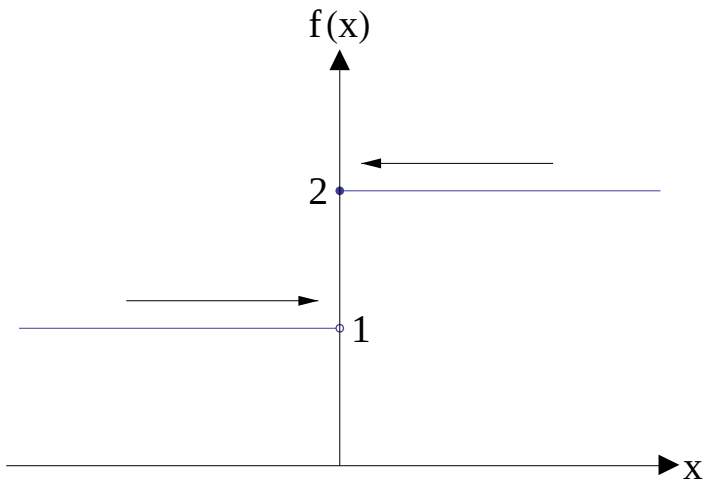
The limit of f as x approaches a does not depend on $f(a)$. The limit can exist even if f is undefined at $x = a$.

Example 1.2: Let $f(x) = \dfrac{1}{x^2}$. Evaluate $\lim_{x \to 0} f(x)$.



Solution:

Example 1.3: Let $f(x) = \begin{cases} 1 & x < 0 \\ 2 & x \geq 0 \end{cases}$. Evaluate $\lim_{x \to 0} f(x)$.



Solution:

We can describe this behaviour in terms of one-sided limits. We write

Theorem:

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$.

Thus the limit exists if and only if the left and right hand limits exist and are equal.

Limit Laws

Let f and g be real-valued functions and let $c \in \mathbb{R}$ be a constant. If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ exist, then

1. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$.
2. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x)$.
3. $\lim_{x \to a} [f(x)g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$.
4. $\lim_{x \to a} \left[\dfrac{f(x)}{g(x)}\right] = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ provided $\lim_{x \to a} g(x) \neq 0$.
5. $\lim_{x \to a} c = c$.
6. $\lim_{x \to a} x = a$.

Example 1.4: Use the limit laws to evaluate $\lim_{x \to 2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

Solution:

Limits as x Approaches Infinity

The limit of $f(x)$ as x approaches positive infinity is L,

$$\lim_{x\to\infty} f(x) = L$$

if $f(x)$ gets arbitrarily close to L whenever x is sufficiently large and positive.

The limit of $f(x)$ as x approaches negative infinity is M,

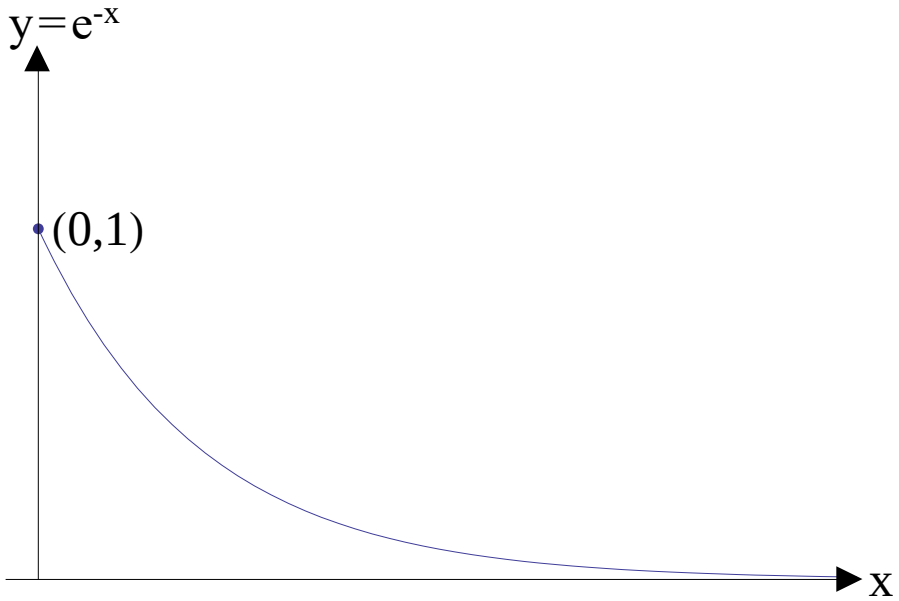
$$\lim_{x\to-\infty} f(x) = M$$

if $f(x)$ gets arbitrarily close to M whenever x is sufficiently large and negative.

Note:

1. L and M must be finite.
2. Limit laws (1)-(5) apply to limits as $x \to \pm\infty$.
3. $\lim_{x\to-\infty} f(x) = \lim_{x\to\infty} f(-x)$.

Example 1.5: Evaluate $\lim_{x\to\infty} e^{-x}$.



Solution:

Some Standard Limits

We can use the following standard limits without further proof:

(1) $\lim_{x\to\infty} \frac{1}{x^p} = 0 \quad (p > 0)$

(2) $\lim_{x\to\infty} r^x = 0 \quad (0 \leq r < 1)$

We will see more standard limits later.

Terminology

The following are all equivalent ways of expressing $\lim_{x \to a} f(x) = L$:

- The limit of $f(x)$ as x approaches a is L
- $f(x)$ converges to L as x approaches a
- $f(x) \to L$ as $x \to a$

These all mean the same thing.

If the limit $\lim_{x \to a} f(x)$ exists, we say that
$f(x)$ converges as x approaches a.

If the limit $\lim_{x \to a} f(x)$ does not exist, we say that
$f(x)$ diverges as x approaches a.

Similarly for limits as $x \to \pm\infty$.

Note:

Diverges simply means does not converge.

More about divergence

How do we show that a limit $\lim_{x \to a} f(x)$ diverges?

Example 1.6: Explain why the following limits diverge.

1. $\lim_{x \to \infty} \sin x$
2. $\lim_{x \to 0} \frac{1}{x^2}$

Solution:

Notation and ∞

∞ is not a number, and should not appear as a number in your writing.

In other textbooks, you will often see the notation $\lim_{x \to 0} \frac{1}{x^2} = \infty$ to mean that $\frac{1}{x^2}$ diverges to infinity.

You should not use this notation in Calculus 2 and will lose notation marks for this!

Why?

Example 1.7: Evaluate $\lim_{x\to\infty} \frac{3x^2 - 2x + 3}{x^2 + 4x + 4}$.

Solution:

This technique of *dividing numerator and denominator by the fastest growing term* sometimes works for limits as $x \to \infty$ when the limit has indeterminate form $\dfrac{\infty}{\infty}$.

Example 1.8: Evaluate $\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right)$.

Solution:

We will finish this example later.

This technique of *multiplying and dividing by the conjugate* sometimes works for limits as $x \to \infty$ when the limit has indeterminate form $\infty - \infty$.

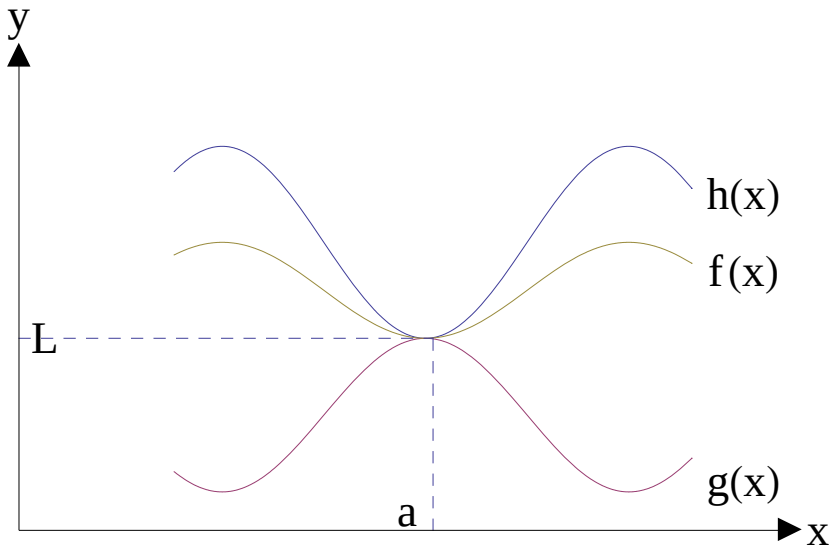
Sandwich Theorem (version for $x \to a$)

If $g(x) \leq f(x) \leq h(x)$ when x is near a but $x \neq a$, and

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} f(x) = L.$$



Note: "x is near a but $x \neq a$" means that $x \in (b, a) \cup (a, c)$ for some $b < a < c$.

Sandwich Theorem (version for $x \to \infty$)

A version of the Sandwich Theorem also works for limits as $x \to \infty$.

If $g(x) \leq f(x) \leq h(x)$ when x is sufficiently large, and

$$\lim_{x\to\infty} g(x) = \lim_{x\to\infty} h(x) = L$$

then

$$\lim_{x\to\infty} f(x) = L.$$

Similarly, the theorem holds for $x \to -\infty$.

Note: "x is sufficiently large" means $x \in (c, \infty)$ for some real number c.

Example 1.9: Evaluate $\lim_{x \to 0} \left[x^2 \sin\left(\frac{1}{x}\right) \right]$.

Solution:

Example 1.10: Evaluate $\lim_{x \to 0} \left[x \sin\left(\frac{1}{x}\right) \right]$.

Solution:

Example 1.8 (continued)

$$\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right) = \lim_{x\to\infty}\frac{1}{\sqrt{x^2+1}+x}$$

Continuity

Definition of continuity

Let f be a real-valued function.

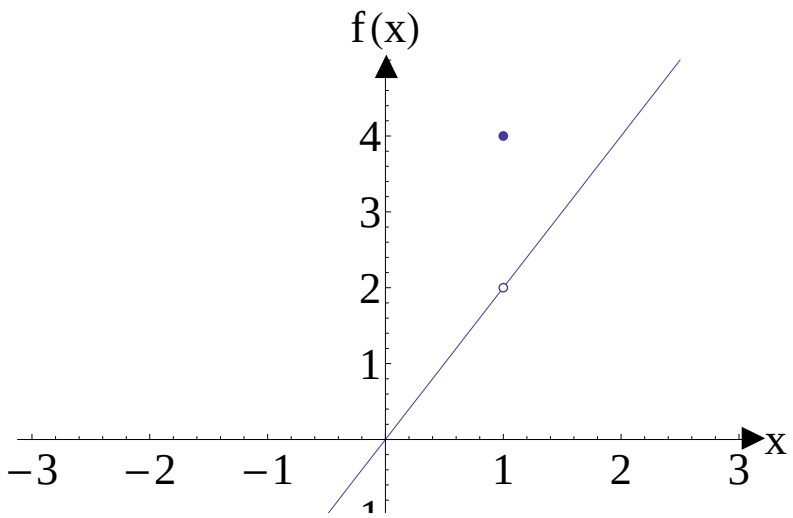
The function f is continuous at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Example 1.11: Let

$$f(x) = \begin{cases} 2x & x \neq 1 \\ 4 & x = 1. \end{cases}$$

Is f continuous at $x = 1$?



Solution:

Example 1.12: Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2} & x \neq 2 \\ 4 & x = 2. \end{cases}$

Is f continuous at $x = 2$?

Solution:

Continuity Theorem 1:

The following function types are continuous at every point in their domains:

- polynomials
- trigonometric functions: $\sin x$, $\cos x$, $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$, $\arcsin x$, $\arccos x$, $\arctan x$
- exponential functions: a^x for $a > 0$
- logarithm functions: $\log_a x$ for $a > 0$
- nth root functions: $\sqrt[n]{x}$ for $n \in \{2, 3, 4, ...\}$
- absolute value function: $|x|$
- hyperbolic functions: $\sinh x$, $\cosh x$, $\tanh x$, $\operatorname{sech} x$, $\operatorname{cosech} x$, $\coth x$, $\operatorname{arcsinh} x$, $\operatorname{arccosh} x$, $\operatorname{arctanh} x$

Let f and g be real-valued functions and $c \in \mathbb{R}$ be a constant.

Continuity Theorem 2:

If the functions f and g are continuous at $x = a$, then the following functions are continuous at $x = a$:

1. $f + g$,
2. cf,
3. fg,
4. $\dfrac{f}{g}$ if $g(a) \neq 0$.

Recall that $(g \circ f)(x) = g(f(x))$.

Continuity Theorem 3:

If f is continuous at $x = a$ and g is continuous at $x = f(a)$, then $g \circ f$ is continuous at $x = a$.

Example 1.13: Let $h(x) = \sin(2\log x)$.

Prove that h is continuous at all $x > 0$.

Solution:

Example 1.14: Let $f(x) = \begin{cases} x+2 & x \leq 1 \\ (x-3)^2 + c & x > 1 \end{cases}$

For which values of c is f continuous at all $x \in \mathbb{R}$? Justify your answer.

See here for an interactive graph of $f(x)$:
`https://desmos.com/c/urqgrxiui0`

Solution:

Continuity rule for limits

Theorem:

Let f and g be real-valued functions and $b \in \mathbb{R}$.

If $\lim_{x \to a} g(x) = b$ and f is continuous at b then

$$\lim_{x \to a} f\big(g(x)\big) = f\left(\lim_{x \to a} g(x)\right) = f(b).$$

Note:

This theorem also holds for limits as $x \to \infty$.

Example 1.15: Evaluate $\lim_{x \to \infty} \sin\left(e^{-x}\right)$.

Solution:

Differentiability

Definition of derivative

Let f be a real-valued function whose domain contains an open interval around $x = a$. The derivative of f at $x = a$ is defined by

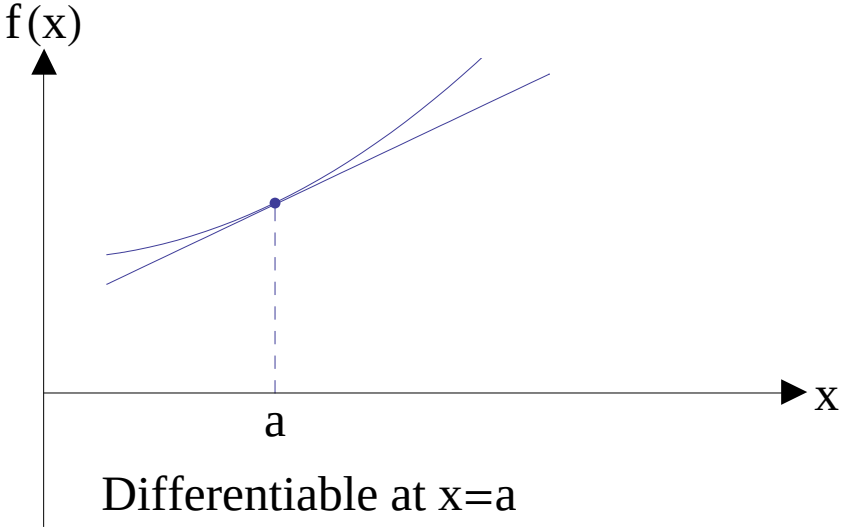
$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

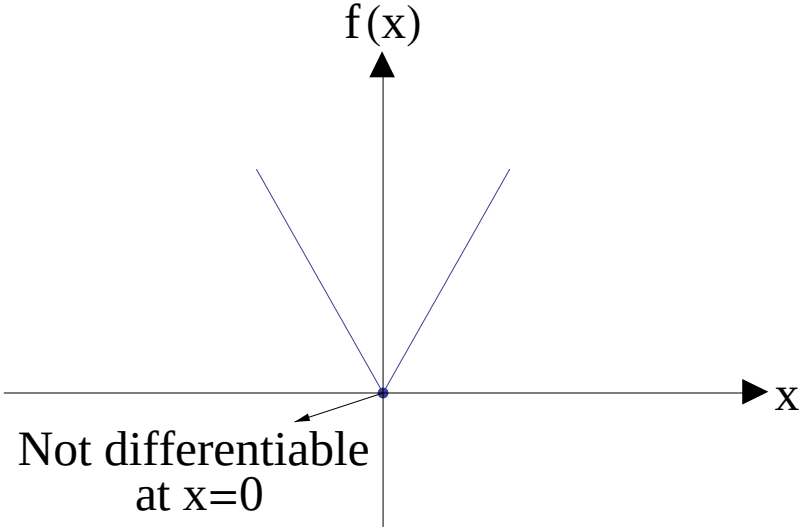
The function f is differentiable at $x = a$ if this limit exists.

Where does this definition come from?

Before seeing what is filled in from the lectures, try to come up with a picture that explains where the definition comes from.

Geometrically, f is differentiable at $x = a$ if the graph $y = f(x)$ has a *tangent line* at $x = a$.





If f is differentiable at $x = a$, then

$$y = f(a) + f'(a)(x - a)$$

is the tangent line to the curve $y = f(x)$ at the point $x = a$, and

$$f(x) \approx f(a) + f'(a)(x - a)$$

when x is close to a.

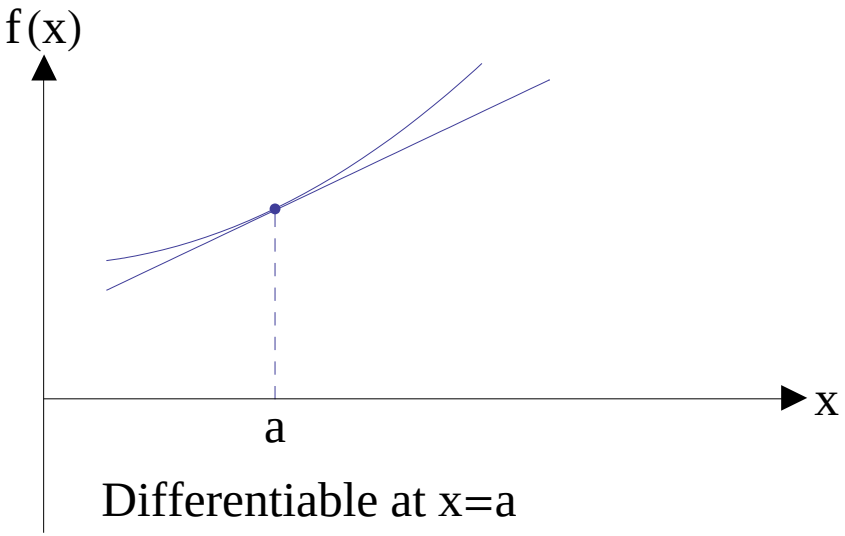
That is, the tangent line is a linear approximation for the function close to the point $x = a$.

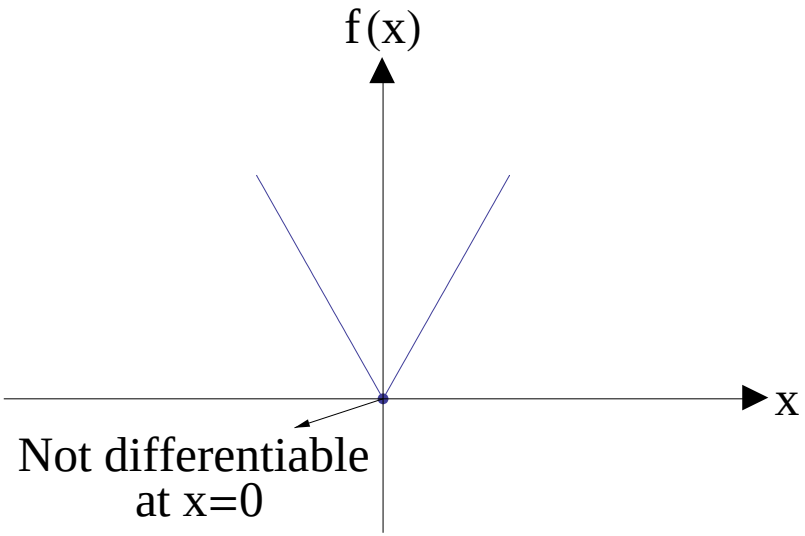
Differentiability and Continuity

Theorem:

If f is differentiable at $x = a$, then f is continuous at $x = a$.

What about the converse?





L'Hôpital's Rule

L'Hôpital's rule is a technique for evaluating limits of the form $\lim_{x \to a} \frac{f(x)}{g(x)}$ when f and g are differentiable.

Theorem:

Let f, g be real-valued functions. If

- $\lim_{x \to a} \frac{f(x)}{g(x)}$ has the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and
- f and g are differentiable near $x = a$, and
- $g'(x) \neq 0$ at all points x near a with $x \neq a$, and
- $\lim_{x \to a} \frac{f'(x)}{g'(x)}$ exists,

then

$$\boxed{\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.}$$

L'Hôpital's Rule

Note:

- L'Hôpital's Rule can only be used to show that a limit exists. It cannot be used to show that a limit does not exist.
- Remember to check that the limit is of the form 0/0 or ∞/∞ before using L'Hôpital's Rule.
- L'Hôpital's Rule also holds for limits as $x \to \infty$, and for one-sided limits $x \to a^+$ and $x \to a^-$ for $a \in \mathbb{R}$.

Example 1.16: Evaluate $\lim_{x \to 0} \frac{\sin x}{x}$.

Solution:

Example 1.17: Evaluate $\lim_{x\to\infty} \frac{3x^2 - 2x + 3}{x^2 + 4x + 4}$.

Solution:

Example 1.18: Evaluate $\lim_{x\to\infty}\left(x^{-\frac{1}{3}}\log x\right)$. $\qquad (0\cdot\infty)$

Solution:

Aside - What is a limit really?*

Recall our definition of limit:

$\lim_{x \to a} f(x) = L$ if $f(x)$ gets arbitrarily close to L whenever x is close enough to a but $x \neq a$.

What do '*arbitrarily close*' and '*close enough*' mean?

More formally,

for any arbitrary positive real number ε, there is a positive real number δ such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad 0 < |x - a| < \delta.$$

This formal definition of limit is covered in MAST20026 Real Analysis, but as a taster we do two examples.

** Slides 52-54 are not examinable in MAST10006.*

Aside - What is a limit really?*

Example 1.19: Using the definition, prove that

$$\lim_{x \to 1} 2x = 2$$

Solution:

For an arbitrary positive real number ε,

$$|f(x) - 2| = 2|x - 1| < \varepsilon \text{ if and only if } |x - 1| < \frac{1}{2}\varepsilon = \delta.$$

This shows that $|f(x) - 2|$ can be arbitrarily small whenever $|x - 1|$ is small enough but not equal to 0.

In other words, $f(x)$ can be arbitrarily close to 2 whenever x is close enough to 1 but $x \neq 1$.

Therefore,

$$\lim_{x \to 1} f(x) = 2.$$

Aside - What is a limit really?*

Example 1.20: Sketch a proof of the limit law

$$\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$$

Solution:

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$.

For an arbitrary positive real number ε, to make

$$|f(x)+g(x)-(L+M)| < \varepsilon$$

we only need to make $|f(x)-L| < \frac{\varepsilon}{2}$ and $|g(x)-M| < \frac{\varepsilon}{2}$.

These will be satisfied whenever x is close enough to a but $x \neq a$ since $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$.

Hence $f(x)+g(x)$ can be arbitrarily close to $L+M$ whenever x is close enough to a but $x \neq a$, which means that

$$\lim_{x\to a}[f(x)+g(x)] = L+M = \lim_{x\to a} f(x) + \lim_{x\to a} g(x).$$

Sequences

Definition of sequence

A sequence is an ordered list of real numbers

$$a_1, a_2, a_3, a_4, \ldots, a_n \ldots$$

The sequence is denoted by (a_n), where a_n is the nth term.

Sometimes the notation $\{a_n\}$ is also used.

A sequence can also be thought of as a function $f : \mathbb{N} \to \mathbb{R}$, where $f(n) = a_n$.

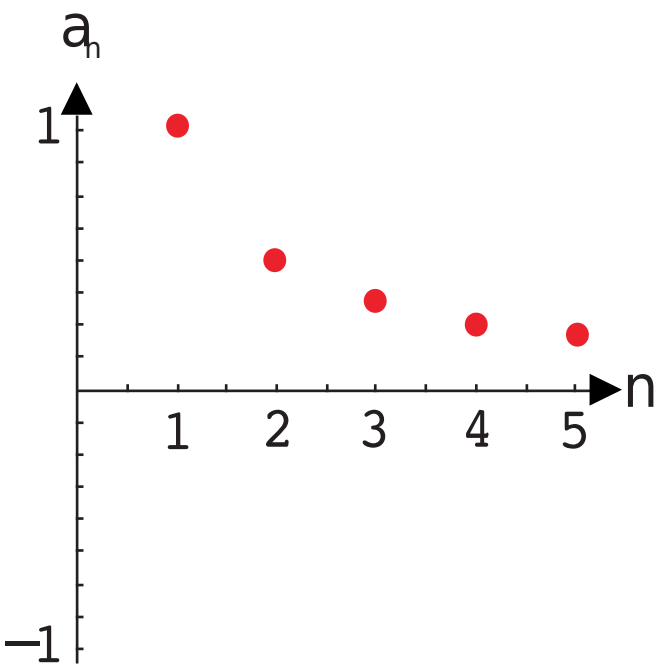
Examples

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots \qquad a_n = \frac{1}{n}$$

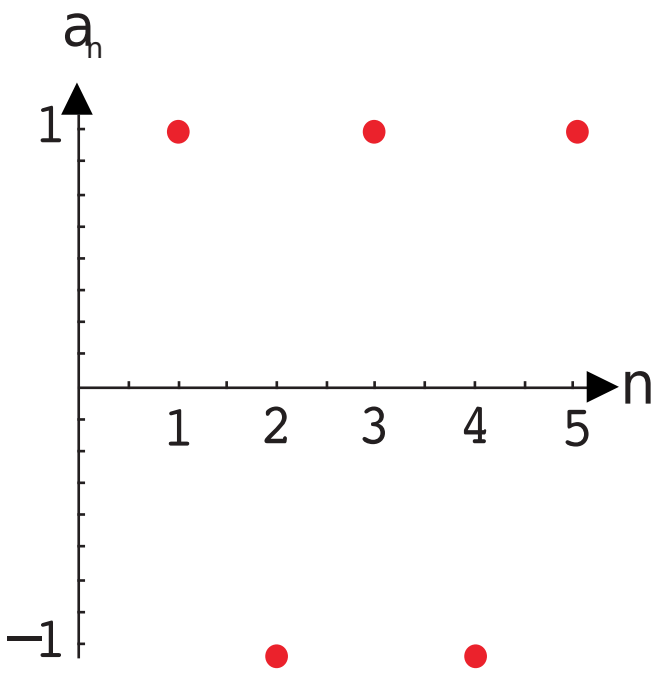
$$1, -1, 1, -1, 1, -1, \ldots \qquad a_n = (-1)^{n-1}$$

The graph of a sequence (a_n) can be plotted on a set of axes with n on the x-axis and a_n on the y-axis.

Example: $a_n = \frac{1}{n}$



Example: $a_n = (-1)^{n-1}$



Limits of Sequences

Definition of limit of sequence

The limit of a sequence (a_n) as $n \to \infty$ is L,

$$\lim_{n\to\infty} a_n = L$$

if a_n gets arbitrarily close to L whenever n is sufficiently large.

If the limit exists, we say that the sequence converges.

If the limit does not exist, we say that the sequence diverges.

Note:

If it exists, L must be a unique finite real number.

$\lim_{n\to\infty} a_n = L$ can also be written as: $a_n \to L$ as $n \to \infty$.

Example 1.21: Find the limit of these sequences, or explain why they diverge: (a) $\left(\frac{1}{n}\right)$ (b) $\left((-1)^{n-1}\right)$ (c) (n)

Solution:

Theorem (Limit Laws):

Let (a_n) and (b_n) be sequences of real numbers and $c \in \mathbb{R}$ a constant.

If $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, then

1. $\lim_{n\to\infty} [a_n + b_n] = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$.
2. $\lim_{n\to\infty} [ca_n] = c \lim_{n\to\infty} a_n$.
3. $\lim_{n\to\infty} [a_n b_n] = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$.
4. $\lim_{n\to\infty} \left[\frac{a_n}{b_n}\right] = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$ provided $\lim_{n\to\infty} b_n \neq 0$.
5. $\lim_{n\to\infty} c = c$.

The Factorial Function

The factorial function $n!$ $(n = 0, 1, 2, ...)$ is defined by

$$n! = n(n-1)! \ , \quad 0! = 1$$

or

$$n! = n \times (n-1) \times (n-2) \times ... \times 3 \times 2 \times 1$$

Therefore

$$\begin{gathered} 1! = 1 \\ 2! = 2 \times 1 = 2 \\ 3! = 3 \times 2 \times 1 = 6 \\ 4! = 4 \times 3 \times 2 \times 1 = 24 \end{gathered}$$

Example

$$(2n+2)! = (2n+2) \times (2n+1) \times (2n) \times (2n-1) \times ... \times 3 \times 2 \times 1$$

or

$$(2n+2)! = (2n+2) \times (2n+1) \times (2n)!$$

Standard Limits

(1) $\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad (p > 0)$

(2) $\lim_{n\to\infty} r^n = 0 \quad (|r| < 1)$

(3) $\lim_{n\to\infty} a^{\frac{1}{n}} = 1 \quad (a > 0)$

(4) $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$

(5) $\lim_{n\to\infty} \frac{a^n}{n!} = 0 \quad (a \in \mathbb{R})$

(6) $\lim_{n\to\infty} \frac{\log n}{n^p} = 0 \quad (p > 0)$

(7) $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = \mathrm{e}^a \quad (a \in \mathbb{R})$

(8) $\lim_{n\to\infty} \frac{n^p}{a^n} = 0 \quad (p \in \mathbb{R}, a > 1)$

(9) $\lim_{n\to\infty} \arctan(cn) = \frac{\pi}{2} \quad (c > 0)$

Note:

Standard limits (1), (3), (4), (6), (7), (8), (9) also hold for limits of real-valued functions as $x \to \infty$ by replacing n with x.
Standard limit (2) also holds for $x \to \infty$ when $0 \le r < 1$.

Example 1.22: Evaluate $\lim_{n\to\infty}\left[\left(\frac{n-2}{n}\right)^n + \frac{4n^2}{3^n}\right]$.

Solution:

Example 1.23: Find the limit of the sequence

$$a_n = \frac{3^n + 2}{4^n + 2^n}, \quad n \geq 1.$$

Solution:

Note:

The order hierarchy can be used to help identify the fastest growing term in an expression as $n \to \infty$:

$$\log n \ll n^p \ll a^n \ll n!$$

where $p > 0$ and $a > 1$.

Sandwich Theorem for sequences

Let (a_n), (b_n) and (c_n) be sequences of real numbers.

If $a_n \leq c_n \leq b_n$ for all $n > N$ for some N, and

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L$$

then

$$\lim_{n\to\infty} c_n = L.$$

Example 1.24: Evaluate $\lim_{n\to\infty} \dfrac{1+\sin^2\left(\frac{n\pi}{3}\right)}{\sqrt{n}}$.

Solution:

Continuity theorem for sequences

Let $f(x)$ be a real-valued function, (a_n) a sequence of real numbers and $b \in \mathbb{R}$.

Theorem:

If $\lim_{n\to\infty} a_n = b$ and f is continuous at $x = b$ then

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right) = f(b).$$

Example 1.25: Evaluate $\lim_{n\to\infty}\left[\log(3n^2+2)-\log n^2\right]$.

Solution:

The only difference between $\lim_{n\to\infty} a_n = L$ and $\lim_{x\to\infty} f(x) = L$ is that n is a natural number whereas x is a real number.

Theorem:

Let $f(x)$ be a real-valued function and (a_n) be a sequence of real numbers such that $a_n = f(n)$.

$$\text{If} \quad \lim_{x\to\infty} f(x) = L \quad \text{then} \quad \lim_{n\to\infty} a_n = L.$$

This means that we can use the techniques for evaluating limits of functions to evaluate limits of sequences.

Note:

$$\lim_{n\to\infty} a_n = L \quad \not\Longrightarrow \quad \lim_{x\to\infty} f(x) = L.$$

eg. $a_n = \sin(2\pi n)$, $f(x) = \sin(2\pi x)$.

Example 1.26: Prove Standard Limit 6:

$$\lim_{n\to\infty} \frac{\log n}{n^p} = 0 \quad (p > 0)$$

Solution:

Note:

We must change from a limit of a sequence (in terms of n) to a limit of a real-valued function (in terms of x) before applying L'Hôpital's rule.

Adding Infinitely Many Numbers

Series

A series is an infinite sum

$$a_1 + a_2 + a_3 + a_4 + \dots$$

or written in summation notation

$$\sum_{n=1}^{\infty} a_n$$

Examples:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$0.99999\dots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \dots = \sum_{n=1}^{\infty} \frac{9}{10^n}$$

$$1 - 1 + 1 - 1 + 1 - 1 + \dots = \sum_{n=1}^{\infty} (-1)^{n-1}$$

Partial sums of a series

By adding the first n terms of the series we obtain a sequence of partial sums:

$$
\begin{aligned}
s_1 &= a_1, \\
s_2 &= a_1 + a_2, \\
s_3 &= a_1 + a_2 + a_3, \\
\vdots \quad & \quad \vdots \quad \quad \vdots \quad \quad \vdots \\
s_n &= a_1 + a_2 + a_3 + \cdots + a_n
\end{aligned}
$$

Example 1.27: Calculate the first 3 partial sums s_1, s_2, s_3 of the series $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$, and guess an expression for s_n.

Definition of convergence of a series

Let $\sum_{n=1}^{\infty} a_n$ be a series and $s_1, s_2, s_3, \ldots$ be its partial sums.

The sequence $s_1, s_2, s_3, \ldots$ of partial sums may or may not have a limit as $n \to \infty$.

If the sequence of partial sums does have a limit, that is, if $\lim_{n\to\infty} s_n = S$ for some real number S, then the series converges, and the sum of the series is the limit of the partial sums:

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n.$$

We also say the series converges to S, or write

$$\sum_{n=1}^{\infty} a_n = S.$$

If the sequence of partial sums does not have a limit then the series diverges.

Example 1.28: Determine if the series

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

converges, and find its sum if so.

Note:

(a_n) and $a_1, a_2, a_3, \ldots$ denote a *sequence*.

$\lim_{n\to\infty} a_n$ denotes the *limit of a sequence*.

$\sum_{n=1}^{\infty} a_n$ and $a_1 + a_2 + a_3 + \ldots$ denote a *series*.

Example

Let $a_n = 2$.

The sequence with $a_n = 2$ is the constant sequence $2, 2, 2, 2, \ldots$

The series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} 2$ represents the sum $2 + 2 + 2 + \cdots$

The sequence (a_n) converges, since $\lim_{n\to\infty} 2 = 2$.

The series $\sum_{n=1}^{\infty} a_n$ diverges, since $s_n = 2n$ and

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} 2n \text{ diverges.}$$

Geometric Series

A geometric series is a series of the form

$$\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots$$

where $a \in \mathbb{R}\backslash\{0\}$ and $r \in \mathbb{R}$.

The series converges if $|r| < 1$ and diverges if $|r| \geq 1$.

If $|r| < 1$, we have

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

Note:

This follows from the fact that $\displaystyle\sum_{k=0}^{n} ar^k = \frac{a(1-r^{n+1})}{1-r}$ for $r \neq 1$.

Example 1.29: What does the series

$$\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots$$

converge to?

Solution:

Harmonic p Series

A harmonic p series is a series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

The series converges if $p > 1$ and diverges if $p \leq 1$.

Example

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges BUT $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Properties of Series

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be series, and $c \in \mathbb{R}\backslash\{0\}$ a constant.

If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge then

1. $\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$ converges.

2. $\sum_{n=1}^{\infty}(ca_n) = c\sum_{n=1}^{\infty} a_n$ converges.

If $\sum_{n=1}^{\infty} a_n$ diverges then $\sum_{n=1}^{\infty}(ca_n)$ diverges.

Divergence Test

If $\lim_{n\to\infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ diverges.

Note:

$\lim_{n\to\infty} a_n \neq 0$ includes the case that the limit $\lim_{n\to\infty} a_n$ does not exist.

If $\lim_{n\to\infty} a_n = 0$ then

1. $\sum_{n=1}^{\infty} a_n$ may converge or diverge.
2. The Divergence Test is not applicable, so we need to use another test to determine if $\sum_{n=1}^{\infty} a_n$ converges or diverges.

Example 1.30: Does the series $\sum_{n=1}^{\infty} \frac{n+1}{n}$ converge?

Solution:

Comparison Test

Let $\sum_{n=1}^{\infty} a_n$ be a series with non-negative terms (i.e. $a_n \geq 0$).

1. If $\sum_{n=1}^{\infty} b_n$ is another series such that $\sum_{n=1}^{\infty} b_n$ diverges and $0 \leq b_n \leq a_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ also diverges.
2. If $\sum_{n=1}^{\infty} c_n$ is another series such that $\sum_{n=1}^{\infty} c_n$ converges and $c_n \geq a_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ also converges.

To apply the comparison test we compare a given series to a harmonic p series or geometric series.

Example 1.31: Does $\displaystyle\sum_{n=1}^{\infty} \frac{3 + \frac{5}{n}}{2n^2 + n + 2}$ converge or diverge?

Solution:

Example 1.32: Does $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+4}{n^3+5}$ converge or diverge?

Solution:

Ratio Test

Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms (i.e. $a_n > 0$ for all n).
Let

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}.$$

1. If $L < 1$, $\sum_{n=1}^{\infty} a_n$ converges.
2. If $L > 1$, $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $L = 1$, the ratio test is inconclusive.

The ratio test is useful if a_n contains an exponential or factorial function of n.

Example 1.33: Does $\sum_{n=1}^{\infty} \frac{10^n}{n!}$ converge or diverge?

Solution:

Example 1.34: Does $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,n!}$ converge or diverge?

Solution:

Section 2: Hyperbolic Functions

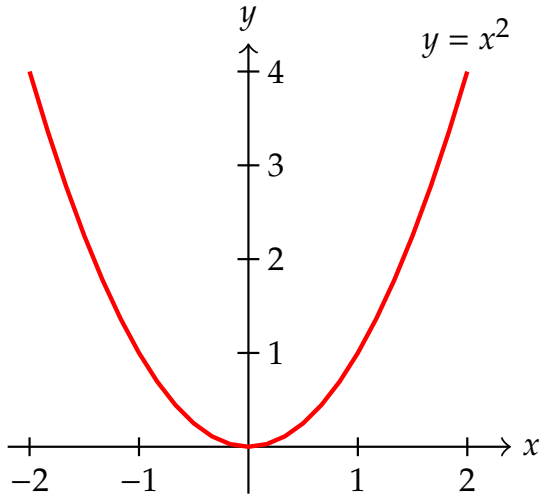
Even Functions

A function f is an even function if

$$f(-x) \;=\; f(x)$$

Example

$f(x) = \cos x$ and $f(x) = x^2$



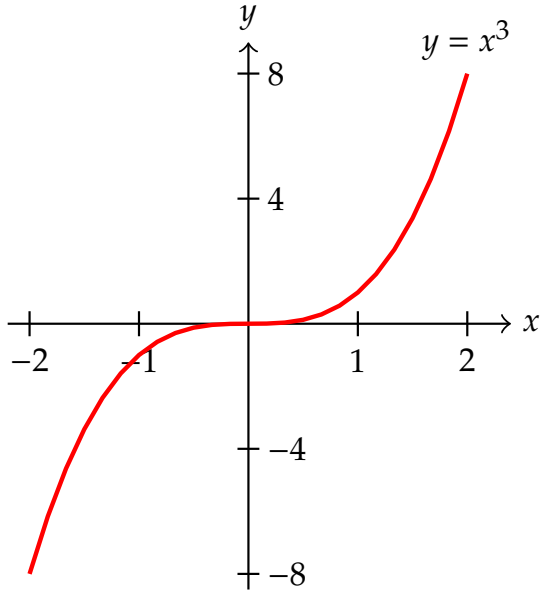
Odd Functions

A function f is an odd function if

$$f(-x) \;=\; -f(x)$$

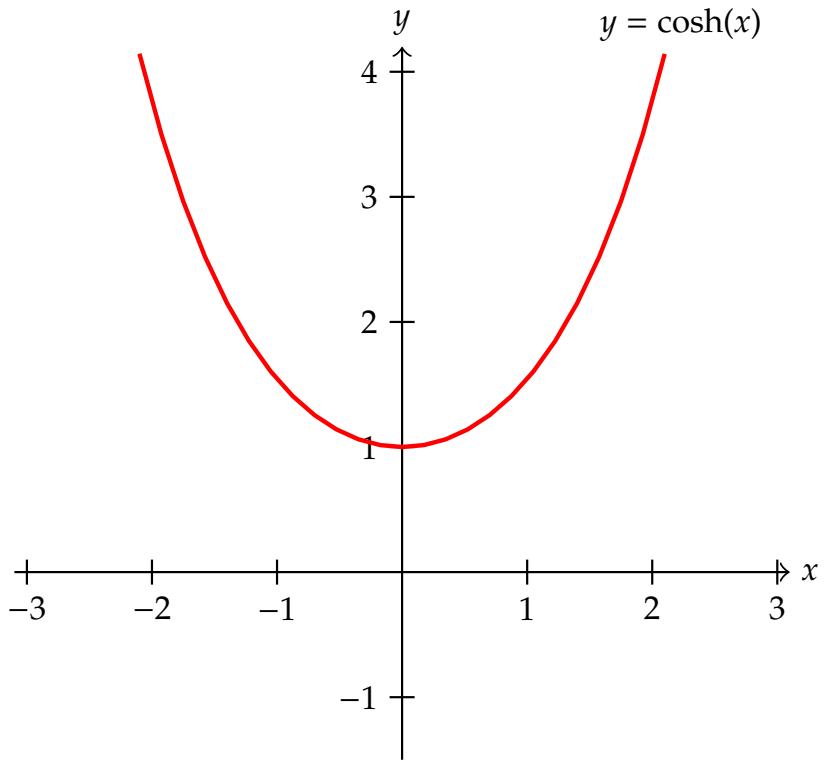
Example

$f(x) = \sin x$ and $f(x) = x^3$



We define the hyperbolic cosine function:

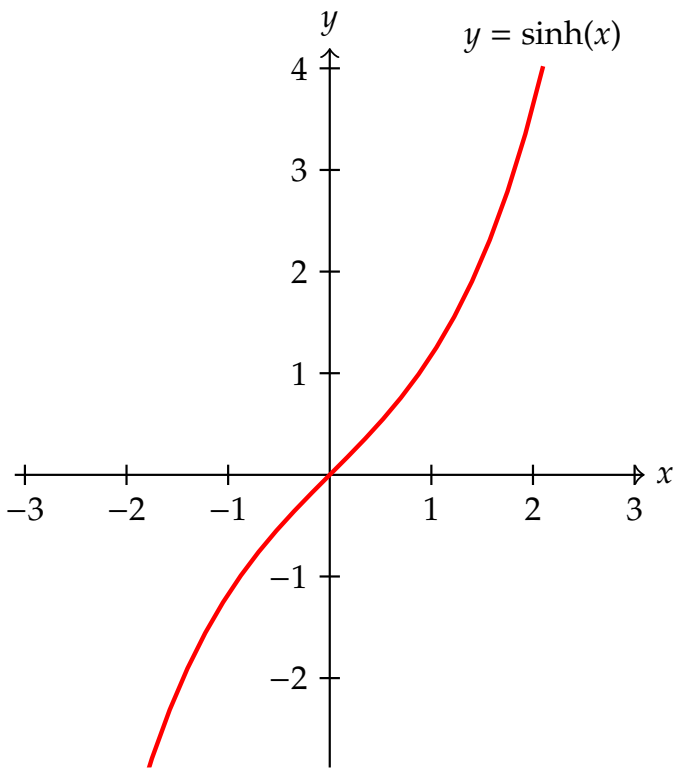
$$\cosh x = \frac{1}{2}\left(e^{x} + e^{-x}\right), \quad x \in \mathbb{R}$$



Properties

We define the hyperbolic sine function:

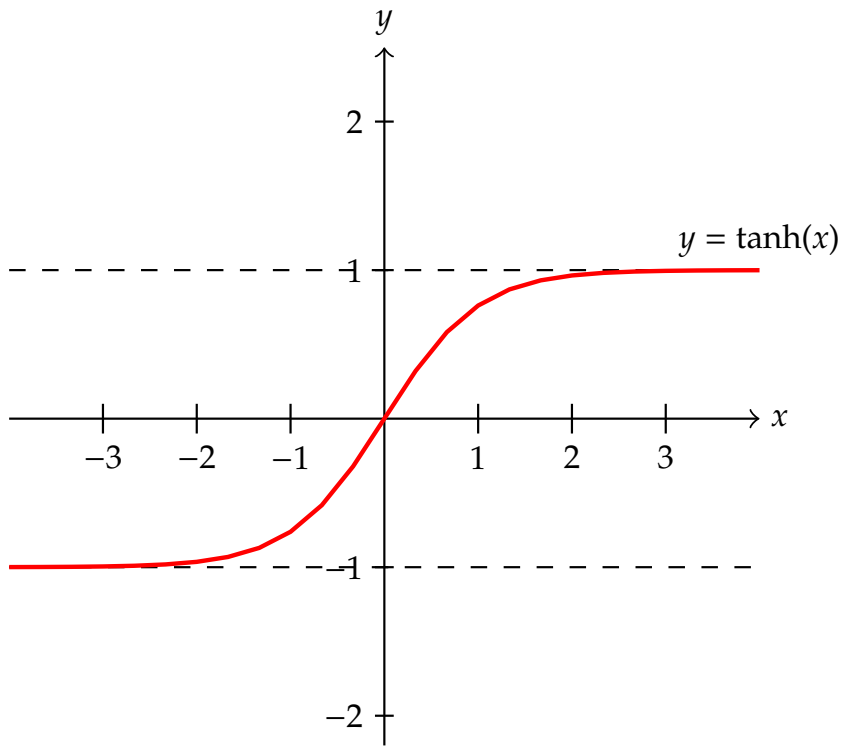
$$\sinh x = \frac{1}{2}\left(e^{x} - e^{-x}\right), \quad x \in \mathbb{R}$$



Properties

We define the hyperbolic tangent function:

$$\begin{aligned}\tanh x &= \frac{\sinh x}{\cosh x} \\ &= \frac{\frac{1}{2}\left(e^{x} - e^{-x}\right)}{\frac{1}{2}\left(e^{x} + e^{-x}\right)} \\ &= \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}, \qquad x \in \mathbb{R}.\end{aligned}$$

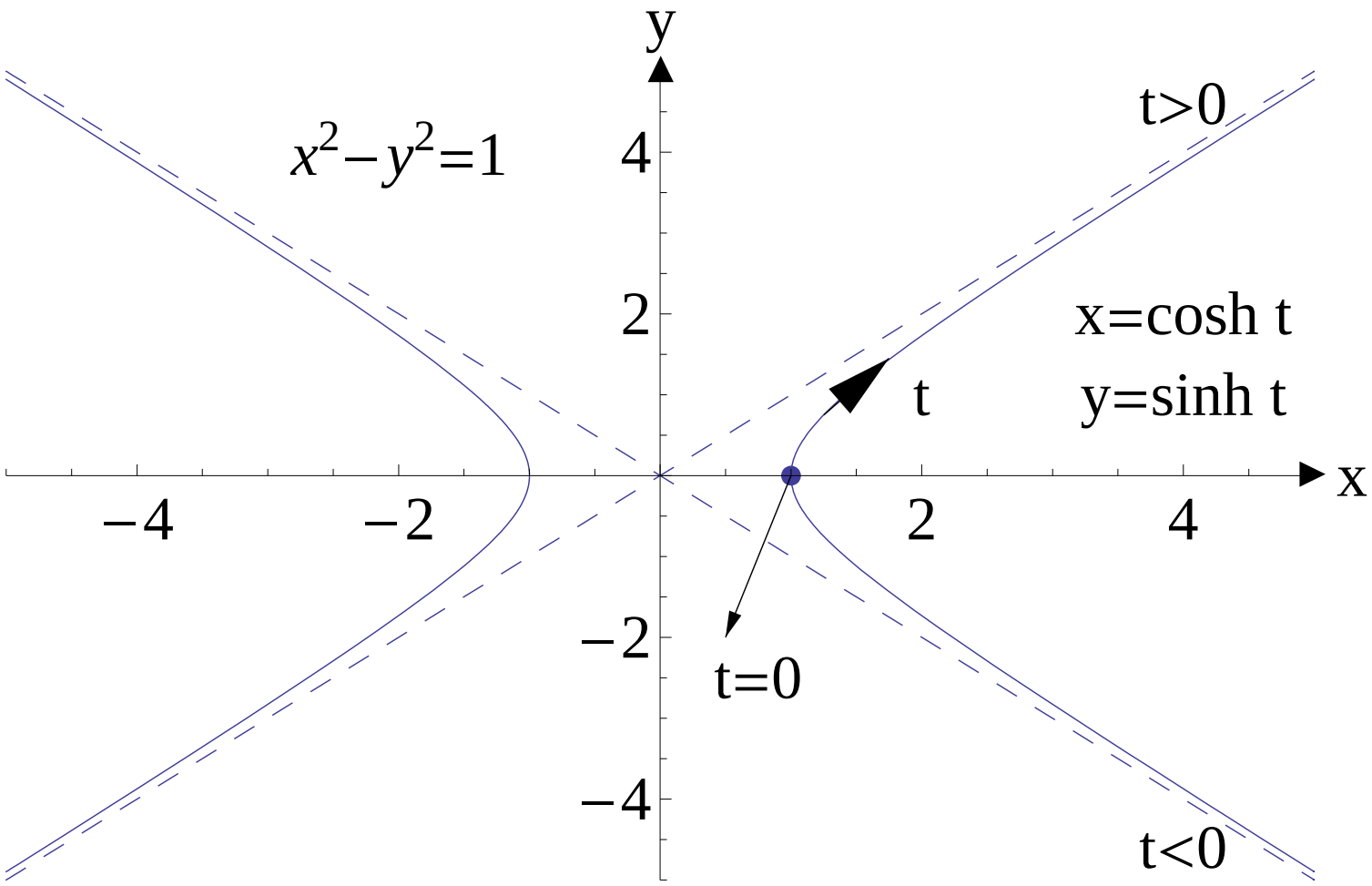


Properties

Why call them hyperbolic functions?

Let $x = \cosh t$ and $y = \sinh t$ then

So $(x, y) = (\cosh t, \sinh t)$ denotes a point on the hyperbola $x^2 - y^2 = 1$. Since $x \geq 1$, the right hand branch of the hyperbola can be parametrised by $x = \cosh t, y = \sinh t, t \in \mathbb{R}$.

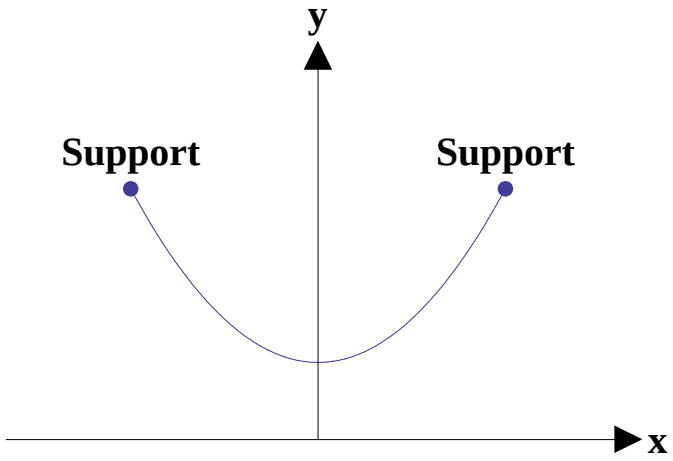


Application: Catenary

A flexible, heavy cable of uniform mass per length ρ and tension T at its lowest point has shape

$$y = \frac{T}{\rho g} \cosh\left(\frac{\rho g x}{T}\right)$$

where g is the acceleration due to gravity.



Example 2.1: If $\cosh x = \frac{13}{12}$ and $x < 0$ find $\sinh x$ and $\tanh x$.

Solution:

Addition Formulae

$$\begin{aligned}
\sinh(x+y) &= \sinh x \cosh y + \cosh x \sinh y \\
\cosh(x+y) &= \cosh x \cosh y + \sinh x \sinh y \\
\sinh(x-y) &= \sinh x \cosh y - \cosh x \sinh y \\
\cosh(x-y) &= \cosh x \cosh y - \sinh x \sinh y
\end{aligned}$$

Example 2.2: Prove the $\sinh(x+y)$ addition formula.

Solution:

Double Angle Formulae

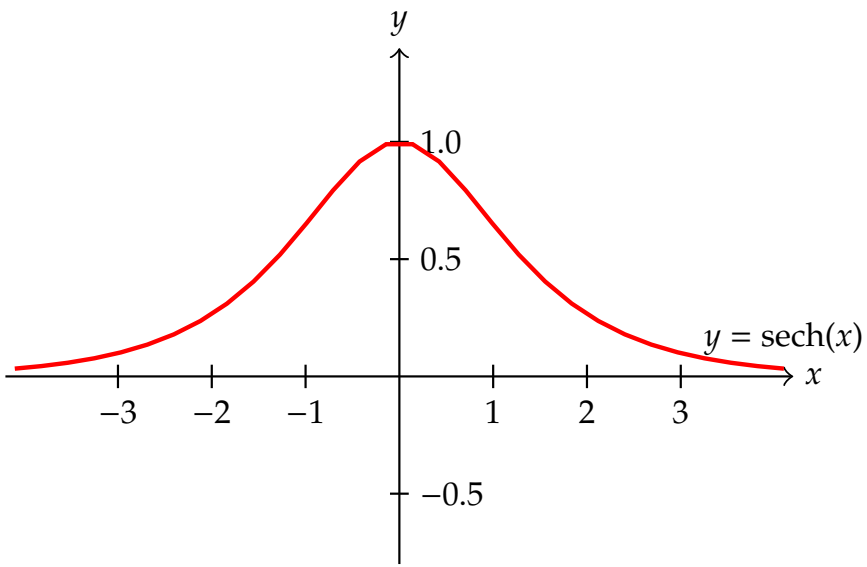
$$\boxed{\begin{aligned} \sinh(2x) &= 2\sinh x\cosh x \\ \cosh(2x) &= \cosh^2 x + \sinh^2 x \\ \cosh(2x) &= 2\cosh^2 x - 1 \\ \cosh(2x) &= 2\sinh^2 x + 1 \end{aligned}}$$

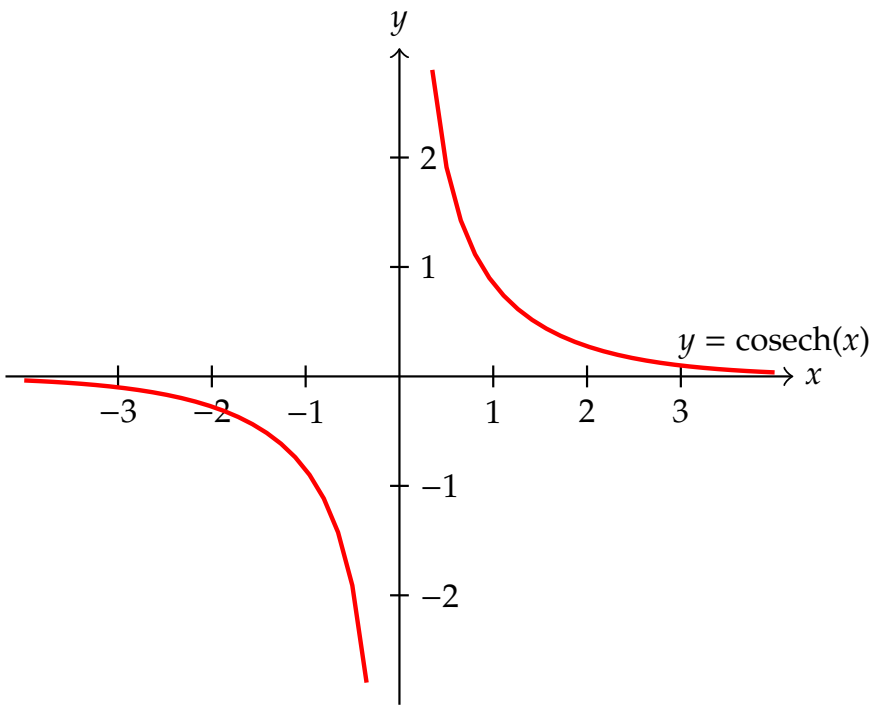
These can be proved using the addition formulae.

Reciprocal Hyperbolic Functions

We define the three reciprocal hyperbolic functions:

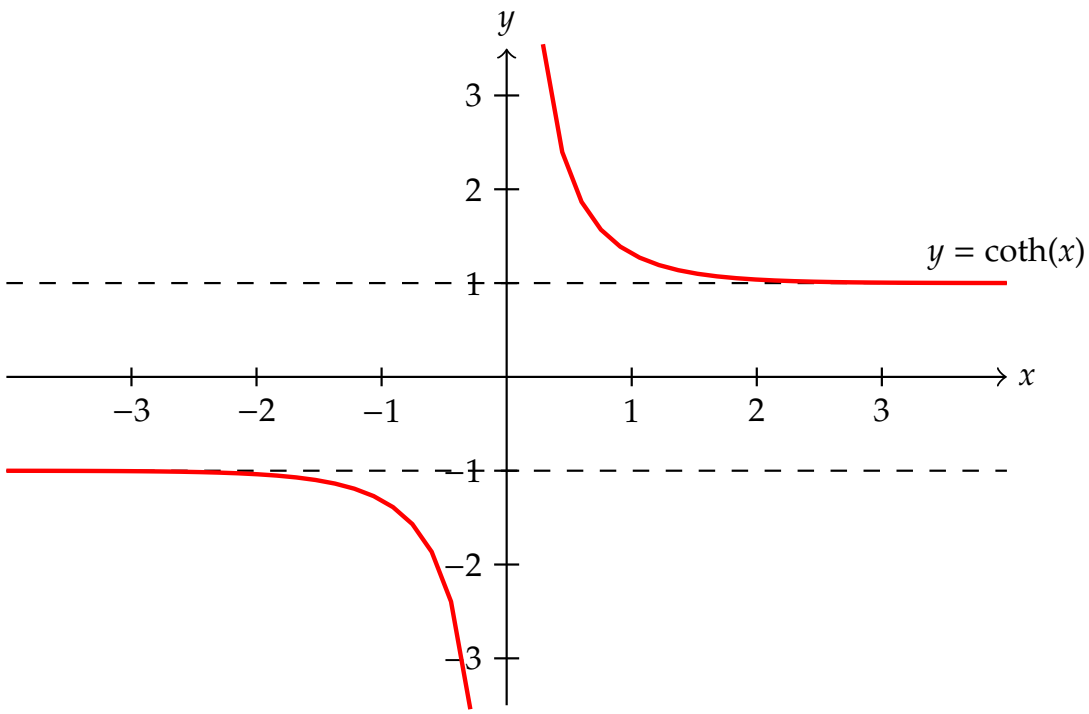
$$\operatorname{sech} x = \frac{1}{\cosh x}, \quad x \in \mathbb{R} \qquad \operatorname{cosech} x = \frac{1}{\sinh x}, x \in \mathbb{R} \setminus \{0\}$$





Reciprocal Hyperbolic Functions

$$\coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}\,,$$
$$x \in \mathbb{R} \setminus \{0\}$$



Basic Identities

$$\boxed{\begin{aligned} \cosh^2 x - \sinh^2 x &= 1 \\ \coth^2 x - 1 &= \operatorname{cosech}^2 x \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x \end{aligned}}$$

Derivatives of Hyperbolic Functions

$$\frac{d}{dx}(\cosh x) = \sinh x, \quad x \in \mathbb{R}$$

$$\frac{d}{dx}(\sinh x) = \cosh x, \quad x \in \mathbb{R}$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x, \quad x \in \mathbb{R}$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x, \quad x \in \mathbb{R}$$

$$\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x, \quad x \in \mathbb{R} \setminus \{0\}$$

$$\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x, \quad x \in \mathbb{R} \setminus \{0\}$$

Example 2.3: Prove that $\dfrac{d\,(\cosh x)}{dx} = \sinh x$.

Solution:

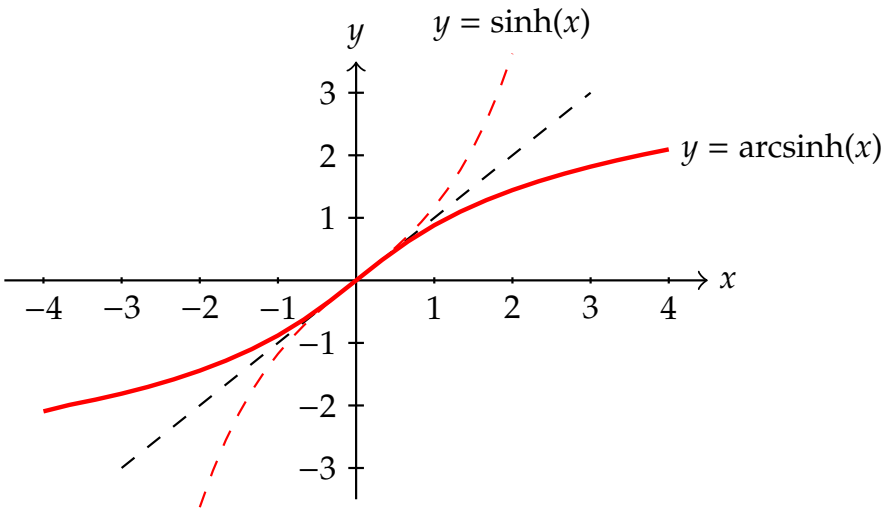
Inverses of Hyperbolic Functions

We define three inverse hyperbolic functions.

1. Inverse hyperbolic sine function: $\operatorname{arcsinh} x$

Since $\sinh x$ is a 1-1 function

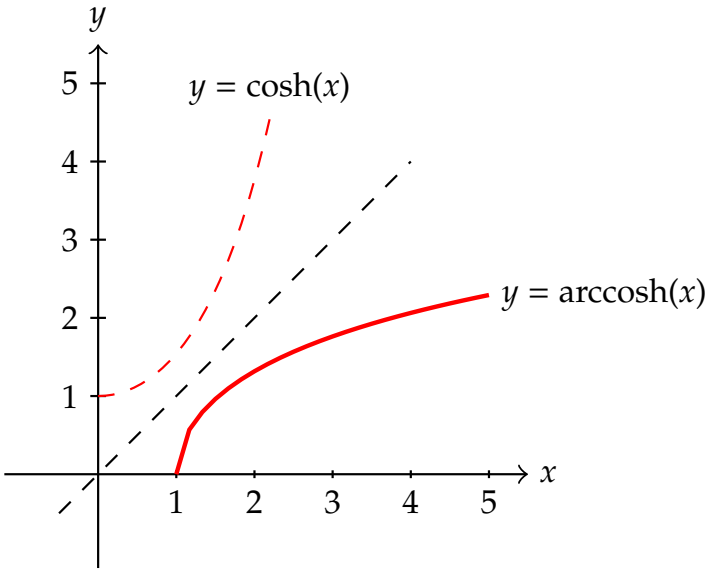
$$\begin{aligned}
\text{domain } \operatorname{arcsinh} x &= \text{range } \sinh x = \mathbb{R}. \\
\text{range } \operatorname{arcsinh} x &= \text{domain } \sinh x = \mathbb{R}. \\
\operatorname{arcsinh}(\sinh x) &= x, \quad x \in \mathbb{R}. \\
\sinh(\operatorname{arcsinh} x) &= x, \quad x \in \mathbb{R}.
\end{aligned}$$



2. Inverse hyperbolic cosine function: $\operatorname{arccosh} x$

Restrict domain of $\cosh x$ to be $[0, \infty)$ to give a 1-1 function. Then

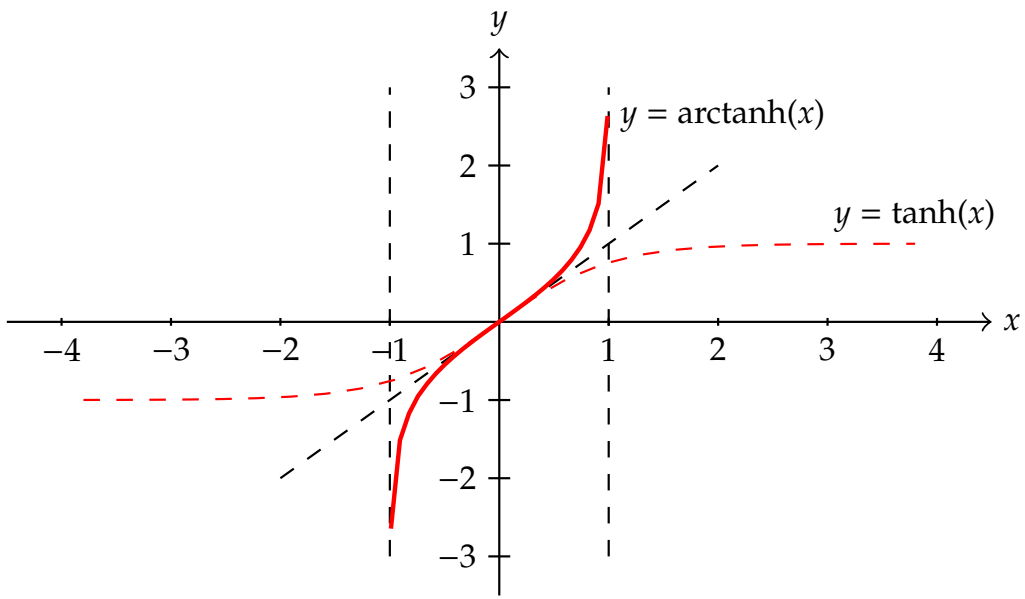
$$\begin{aligned}
\text{domain } \operatorname{arccosh} x &= \text{range } \cosh x = [1, \infty). \\
\text{range } \operatorname{arccosh} x &= \text{restricted domain } \cosh x = [0, \infty). \\
\cosh(\operatorname{arccosh} x) &= x, \quad x \geq 1. \\
\operatorname{arccosh}(\cosh x) &= x, \quad x \geq 0.
\end{aligned}$$



3. Inverse hyperbolic tangent function: $\operatorname{arctanh} x$

Since $\tanh x$ is a 1-1 function

$$\begin{aligned}
\text{domain } \operatorname{arctanh} x &= \text{range } \tanh x = (-1, 1). \\
\text{range } \operatorname{arctanh} x &= \text{domain } \tanh x = \mathbb{R}. \\
\tanh(\operatorname{arctanh} x) &= x, \quad -1 < x < 1. \\
\operatorname{arctanh}(\tanh x) &= x, \quad x \in \mathbb{R}.
\end{aligned}$$



The inverse hyperbolic functions can be expressed in terms of natural logarithms.

$$\boxed{\begin{aligned} \operatorname{arcsinh} x &= \log\left(x + \sqrt{x^2+1}\right), && x \in \mathbb{R} \\ \operatorname{arccosh} x &= \log\left(x + \sqrt{x^2-1}\right), && x \geq 1 \\ \operatorname{arctanh} x &= \frac{1}{2}\log\left(\frac{1+x}{1-x}\right), && -1 < x < 1 \end{aligned}}$$

We can also define inverse reciprocal hyperbolic functions:

- $\operatorname{arcsech} x \qquad (0 < x \leq 1)$
- $\operatorname{arccosech} x \qquad (x \neq 0)$
- $\operatorname{arccoth} x \qquad (x < -1 \text{ or } x > 1)$

Example 2.4: Proof of $\operatorname{arcsinh} x$ relation.

Solution:

Example 2.5: Simplify $\cosh(\operatorname{arcsinh} x)$ for $x \in \mathbb{R}$.

Solution:

Derivatives

$$\frac{d}{dx}(\operatorname{arcsinh} x) = \frac{1}{\sqrt{x^2+1}} \quad (x \in \mathbb{R})$$

$$\frac{d}{dx}(\operatorname{arccosh} x) = \frac{1}{\sqrt{x^2-1}} \quad (x > 1)$$

$$\frac{d}{dx}(\operatorname{arctanh} x) = \frac{1}{1-x^2} \quad (-1 < x < 1)$$

Each formula is derived using implicit differentiation or by differentiating the logarithm definition of each function.

Example 2.6: Prove that $\dfrac{d}{dx}(\operatorname{arcsinh} x) = \dfrac{1}{\sqrt{x^2+1}}$.

Solution:

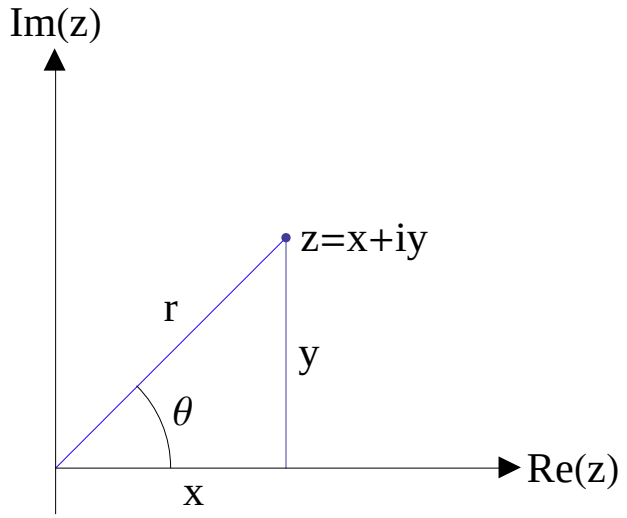
Section 3: Complex Numbers

The Cartesian form of a complex number $z \in \mathbb{C}$ is

$$\boxed{z \;=\; x + iy} \quad \text{where} \quad x, y \in \mathbb{R}$$

and

- $x = \mathrm{Re}(z)$ is the real part of z,
- $y = \mathrm{Im}(z)$ is the imaginary part of z,
- $i^2 = -1$.



The complex number can be written as

$$\boxed{z \;=\; r(\cos\theta + i\sin\theta)}$$

where

- $r = |z| = \sqrt{x^2 + y^2}$
- $\tan\theta = \dfrac{y}{x}$

Note:

The angle θ is not unique – only defined up to multiples of 2π. We choose θ such that $-\pi < \theta \leq \pi$ and call this angle the principal argument of z.

The Complex Exponential

The Complex Exponential

We define the complex exponential using Euler's formula

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

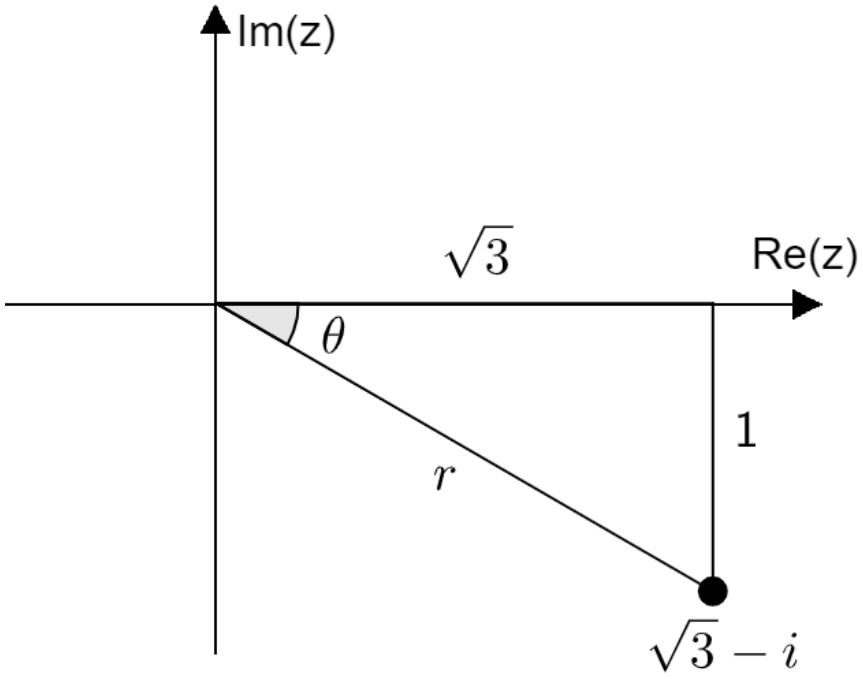
for $\theta \in \mathbb{R}$.

We can then write the polar form of a complex number as

$$\boxed{z = re^{i\theta}}$$

Example 3.1: Simplify $e^{i\pi} + 1$.

Example 3.2: Write $z = \sqrt{3} - i$ in polar form.



Solution:

Properties of the Complex Exponential

1. $\boxed{e^{i0} = 1}$

Proof:

$e^{i0} = \cos 0 + i \sin 0 = 1.$

2. $\boxed{e^{i\theta} e^{i\phi} = e^{i(\theta+\phi)}}$

Proof:

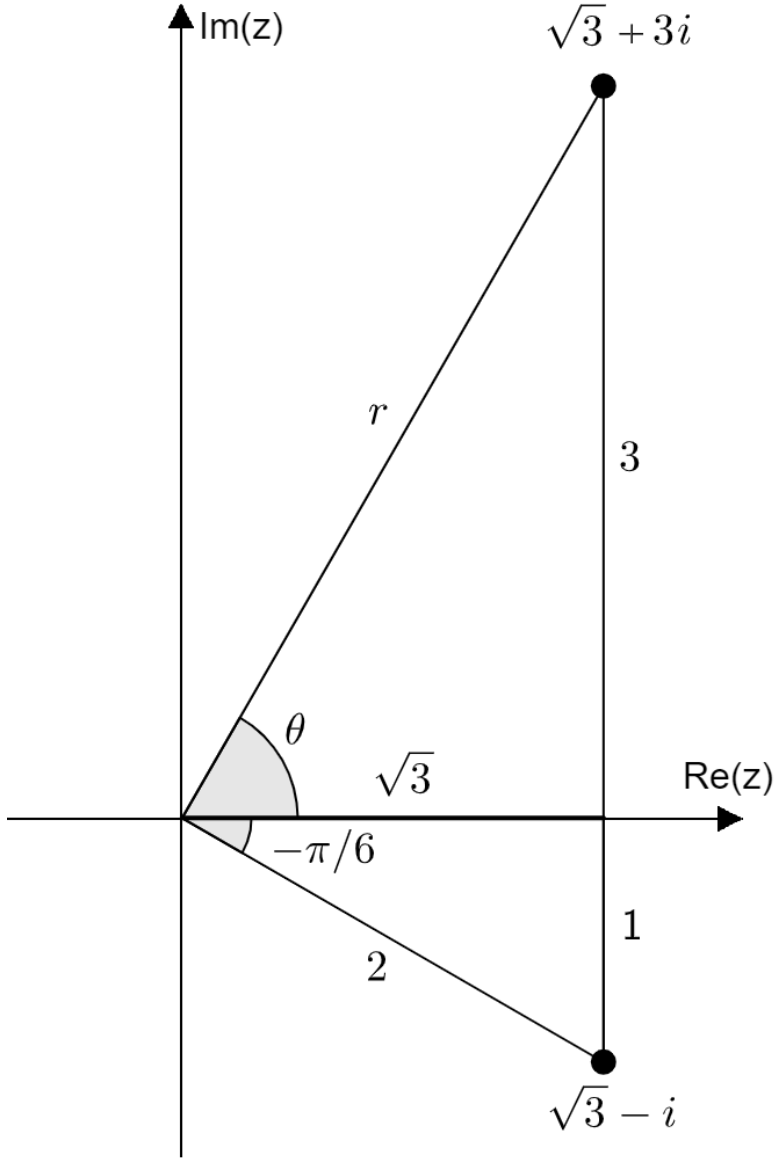
$$\begin{aligned}
e^{i\theta} e^{i\phi} &= (\cos\theta + i\sin\theta)\Big(\cos\phi + i\sin\phi\Big) \\
&= \cos\theta\cos\phi + i\cos\theta\sin\phi + i\sin\theta\cos\phi - \sin\theta\sin\phi \\
&= \Big(\cos\theta\cos\phi - \sin\theta\sin\phi\Big) + i\Big(\cos\theta\sin\phi + \sin\theta\cos\phi\Big) \\
&= \cos\Big(\theta + \phi\Big) + i\sin(\theta + \phi) \\
&= e^{i(\theta+\phi)}.
\end{aligned}$$

Products and Division in Polar Form

If $z = r_1 e^{i\theta}$ and $w = r_2 e^{i\phi}$ then

$$\boxed{\begin{aligned} zw &= r_1 r_2 e^{i(\theta+\phi)} \\ \frac{z}{w} &= \frac{r_1}{r_2} e^{i(\theta-\phi)} \end{aligned}}$$

Example 3.3: Simplify $\left(\sqrt{3}-i\right)\left(\sqrt{3}+3i\right)$ using the complex exponential.



Solution:

De Moivre's Theorem:

If $z = re^{i\theta}$ and n is a positive integer then

$$\boxed{z^n = \left(re^{i\theta}\right)^n = r^n e^{in\theta}}$$

Example 3.4: Evaluate $\left(\sqrt{3}+3i\right)^{30}$.

Solution:

Exponential Form of $\sin\theta$ and $\cos\theta$

$$\text{Now } e^{i\theta} = \cos\theta + i\sin\theta \tag{1}$$

$$\Rightarrow e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$$

$$\Rightarrow e^{-i\theta} = \cos\theta - i\sin\theta \tag{2}$$

Equation (1) + (2) gives

$$e^{i\theta} + e^{-i\theta} = 2\cos\theta$$

$$\Rightarrow \boxed{\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)}$$

Equation (1) – (2) gives

$$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

$$\Rightarrow \boxed{\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)}$$

Note:

These formulae give a connection between the hyperbolic and trigonometric functions.

$$\begin{aligned}\cosh(i\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) = \cos\theta \\ \sinh(i\theta) &= \frac{1}{2}\left(e^{i\theta} - e^{-i\theta}\right) = i\sin\theta\end{aligned}$$

Example 3.5: Express $\cos^3\theta$ in terms of the functions $\cos(n\theta)$ for integers n.

Solution:

Differentiation via the Complex Exponential

If $z = x + yi$ where $x, y \in \mathbb{R}$ then we define

$$e^z = e^{x+iy} = e^x\, e^{iy} = e^x(\cos y + i \sin y).$$

Derivatives of functions from $\mathbb{R}$ to $\mathbb{C}$ are defined similarly as those from $\mathbb{R}$ to $\mathbb{R}$.

Differentiation to functions from $\mathbb{R}$ to $\mathbb{C}$ is also linear and follows the product law.

Show that $\dfrac{d}{dt}\left(e^{kt}\right) = ke^{kt}$ when $k = a + bi \in \mathbb{C}$.

$$\frac{d}{dt}\left[e^{(a+bi)t}\right] = \frac{d}{dt}\left[e^{at}e^{ibt}\right]$$

$$
\begin{aligned}
&= \frac{d}{dt}\left[e^{at}\left(\cos(bt) + i\sin(bt)\right)\right] \\
&= ae^{at}\left[\cos(bt) + i\sin(bt)\right] + e^{at}\left[-b\sin(bt) + bi\cos(bt)\right] \\
&= ae^{at}\left[\cos(bt) + i\sin(bt)\right] + e^{at}\left[bi^2\sin(bt) + bi\cos(bt)\right] \\
&= ae^{at}\left[\cos(bt) + i\sin(bt)\right] + bie^{at}\left[\cos(bt) + i\sin(bt)\right] \\
&= (a + bi)e^{at}\left[\cos(bt) + i\sin(bt)\right] \\
&= (a + bi)e^{at}e^{ibt} \\
&= (a + bi)e^{(a+ib)t}.
\end{aligned}
$$

Example 3.6: Find $\dfrac{d^{40}}{dt^{40}}\left(e^{-t}\cos t\right)$.

Solution:

Note:

Example 3.6 also gives the answer to $\dfrac{d^{40}}{dt^{40}}\left(e^{-t}\sin t\right)$.

Integration via the Complex Exponential

Since $\dfrac{d}{dx}\left(e^{kx}\right) = k\,e^{kx}$ if $k = a + bi$ $(a, b \in \mathbb{R})$, then

$$\begin{aligned} \int k\,e^{kx}\,dx &= e^{kx} + C \\ \Rightarrow \int e^{kx}\,dx &= \frac{1}{k}e^{kx} + D \end{aligned}$$

Example 3.7: Evaluate $\int e^{3x} \sin(2x)\, dx$.

Solution:

Note:

Example 3.7 also gives the answer to $\int e^{3x} \cos(2x)\, dx$.

Section 4: Integral Calculus

Review of integration

Derivative Substitutions

To evaluate

$$\int f[g(x)]g'(x)dx$$

put $u = g(x) \Rightarrow \dfrac{du}{dx} = g'(x)$.

Then

$$\begin{aligned}\int f[g(x)]g'(x)dx &= \int f(u)\frac{du}{dx}\,dx \\ &= \int f(u)\,du\end{aligned}$$

Example 4.1: Evaluate $\int (6x^2 + 10)\sinh(x^3 + 5x - 2)dx$.

Solution:

Example 4.2: Evaluate $\displaystyle\int \frac{\operatorname{sech}^2(3x)}{10 + 2\tanh(3x)}\,dx$.

Solution:

Integration by Parts

The product rule for differentiation is

$$\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$$

Integrate

$$\int \frac{d}{dx}(uv)\,dx = \int \left(\frac{du}{dx}v + u\frac{dv}{dx}\right) dx$$

$$\Rightarrow uv = \int \frac{du}{dx}v\,dx + \int u\frac{dv}{dx}\,dx$$

$$\Rightarrow \quad \boxed{\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx}$$

Example 4.3: Evaluate $\int xe^{5x}\,dx$.

Solution:

Example 4.4: Evaluate $\int_1^2 x^2 \log x \, dx$.

Solution:

Example 4.5: Evaluate $\int \log x \, dx \qquad (x > 0)$.

Solution:

Note:

This technique can also be used to integrate inverse trigonometric functions and inverse hyperbolic functions.

Example 4.6: Evaluate $\int e^{3x} \sin(2x)\, dx$.

Solution:

Trigonometric and Hyperbolic Substitutions

We can use trigonometric and hyperbolic substitutions to integrate expressions containing

$$\sqrt{a^2 - x^2}, \quad \sqrt{a^2 + x^2}, \quad \sqrt{x^2 - a^2},$$

where a is a positive real number.

Method:

Put $x = g(\theta)$. Then

$$\int f(x)\,dx = \int f[g(\theta)]g'(\theta)\,d\theta$$

Integrand	Substitution
$\sqrt{a^2 - x^2}, \quad \dfrac{1}{\sqrt{a^2 - x^2}}, \quad \left(a^2 - x^2\right)^{\frac{3}{2}}$ etc.	$x = a \sin\theta$ or $x = a \cos\theta$
$\sqrt{a^2 + x^2}, \quad \dfrac{1}{\sqrt{a^2 + x^2}}, \quad \left(a^2 + x^2\right)^{-\frac{3}{2}}$ etc.	$x = a \sinh\theta$
$\sqrt{x^2 - a^2}, \quad \dfrac{1}{\sqrt{x^2 - a^2}}, \quad \left(x^2 - a^2\right)^{\frac{5}{2}}$ etc.	$x = a \cosh\theta$
$\dfrac{1}{a^2 + x^2}, \quad \dfrac{1}{(a^2 + x^2)^2}$ etc.	$x = a \tan\theta$

Example 4.7: Evaluate $\int \frac{1}{\sqrt{x^2 - 25}}\, dx$ using a substitution.

Solution:

Note:

Note the identity

$$\sqrt{x^2} = |x| \qquad (x \in \mathbb{R})$$

Example 4.8: Evaluate $\int_0^1 \sqrt{9 - 4x^2}\, dx$.

Solution:

Powers of Hyperbolic Functions

Consider the integral:

$$\int \sinh^m x \cosh^n x \, dx$$

where m, n are integers (≥ 0).

• If m or n is odd, use a "derivative substitution" after rewriting one of the odd power terms using identities if necessary.

• If m and n are even, use double angle formulae.

Example 4.9: Evaluate $\int \cosh^4 \theta \, d\theta$.

Solution:

Example 4.10: Evaluate $\int \sinh^5 x \cosh^6 x \, dx$.

Solution:

Example 4.11: Evaluate $\mathcal{I} = \int \sinh^5 x \cosh^7 x \, dx$.

Solution:

Partial Fractions

Let $f(x)$ and $g(x)$ be polynomials, then

$$\frac{f(x)}{g(x)} \quad \begin{array}{l} \longrightarrow \text{ degree } n \\ \longrightarrow \text{ degree } d \end{array}$$

can be written as the sum of partial fractions.

Case 1: $n < d$

1. Factorise g over the real numbers.
2. Write down partial fraction expansion.
3. Find unknown coefficients

$$A, A_1, A_2, \ldots, A_r, B, B_1, B_2, \ldots, B_r.$$

Denominator Factor	Partial Fraction Expansion
$(x-a)$	$\dfrac{A}{x-a}$
$(x-a)^r$	$\dfrac{A_1}{x-a} + \dfrac{A_2}{(x-a)^2} + \cdots + \dfrac{A_r}{(x-a)^r}$
(x^2+bx+c)	$\dfrac{Ax+B}{x^2+bx+c}$
$(x^2+bx+c)^r$	$\frac{A_1x+B_1}{x^2+bx+c} + \frac{A_2x+B_2}{(x^2+bx+c)^2} + \cdots + \frac{A_rx+B_r}{(x^2+bx+c)^r}$

Example 4.12: Evaluate $\displaystyle\int \frac{4}{x^2(x+2)}\,dx \quad (x \neq 0, -2)$.

Solution:

Example 4.13: Evaluate $\displaystyle\int \frac{4x}{(x^2+4)(x-2)}\,dx \quad (x \neq 2)$.

Solution:

Case 2: $n \geq d$

Use long division, then apply case 1.

Example 4.14: Find

$$\int \frac{5x^4 + 13x^3 + 6x^2 + 4}{x^3 + 2x^2}\, dx \quad (x \neq 0, -2).$$

Solution:

Section 5: First Order Differential Equations

Ordinary Differential Equations

An ordinary differential equation (ODE) is an equation of the form

$$f\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0$$

The order of an ODE is the order of the highest derivative occurring in the ODE.

Example 5.1: What order is $3\frac{d^4y}{dx^4} = \left(\frac{dy}{dx}\right)^2 + 2x^2y$?

A solution of an ODE is a function $y(x)$ that satisfies the ODE for all x in some interval.

Example 5.2: Verify by substitution that $y(x) = x^2 + \dfrac{2}{x}$ is a solution of $\dfrac{dy}{dx} + \dfrac{y}{x} = 3x$ for all $x \in \mathbb{R} \setminus \{0\}$.

Solution:

First Order ODEs

The general form of a first order ODE is $\dfrac{dy}{dx} = f(x, y)$.

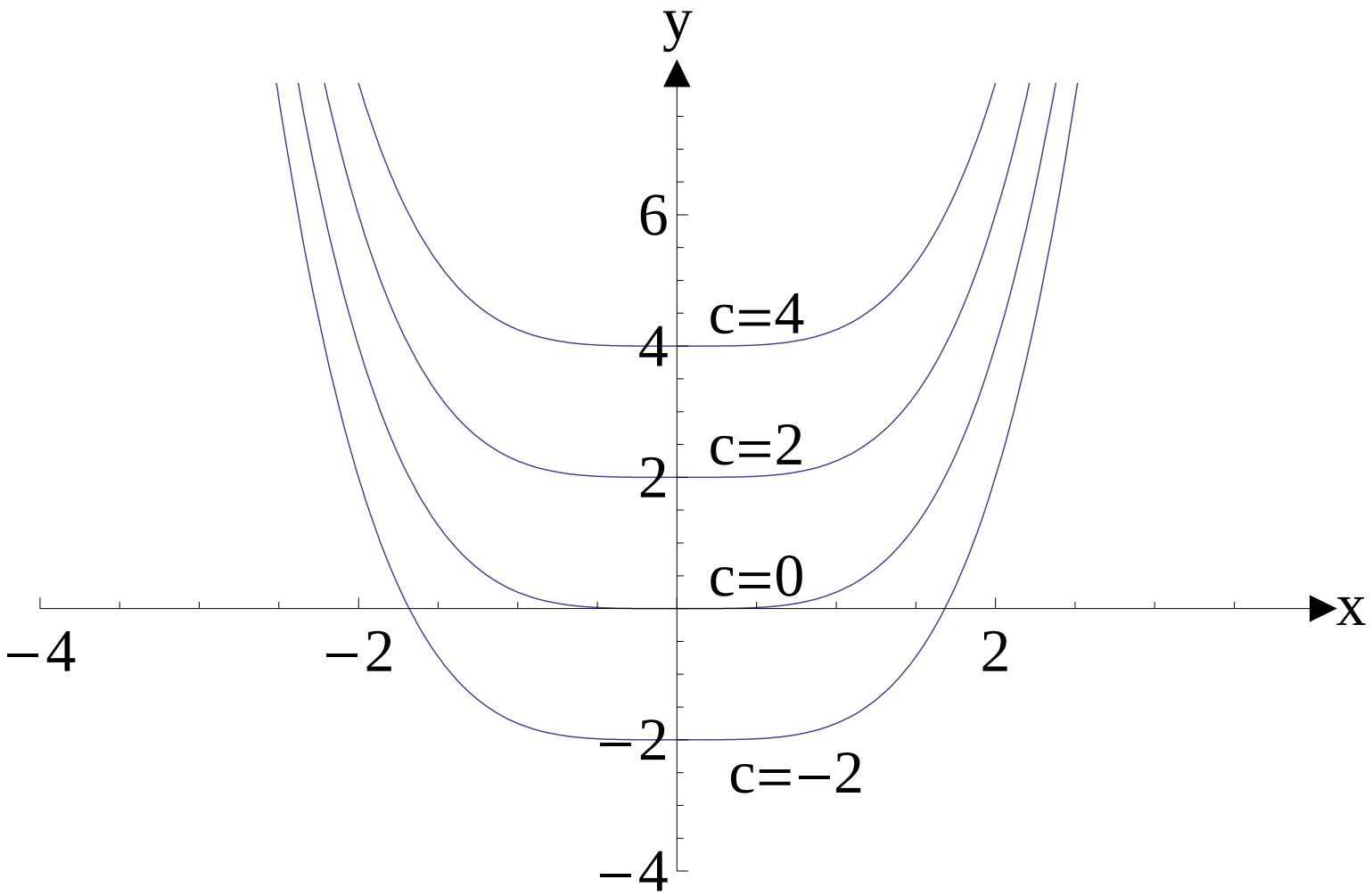
Example 5.3: Solve $\dfrac{dy}{dx} = x^3$.

Solution:

A solution like this which contains an arbitrary constant $c \in \mathbb{R}$ is called a general solution.

The general solution represents a family of solutions, where each value of c corresponds to a different solution of the ODE.

Sketch of the family of solutions of $\dfrac{dy}{dx} = x^3$



Initial value problem for a first order ODE

Solve $\dfrac{dy}{dx} = f(x, y)$ subject to the condition $y(x_0) = y_0$.

Example 5.4: Solve $\dfrac{dy}{dx} = x^3$ given $y(0) = 2$.

Solution:

Separable ODEs

A separable first order ODE has the form:

$$\frac{dy}{dx} = \mathcal{M}(x)\mathcal{N}(y), \quad (\mathcal{M}(x) \neq 0, \;\; \mathcal{N}(y) \neq 0)$$

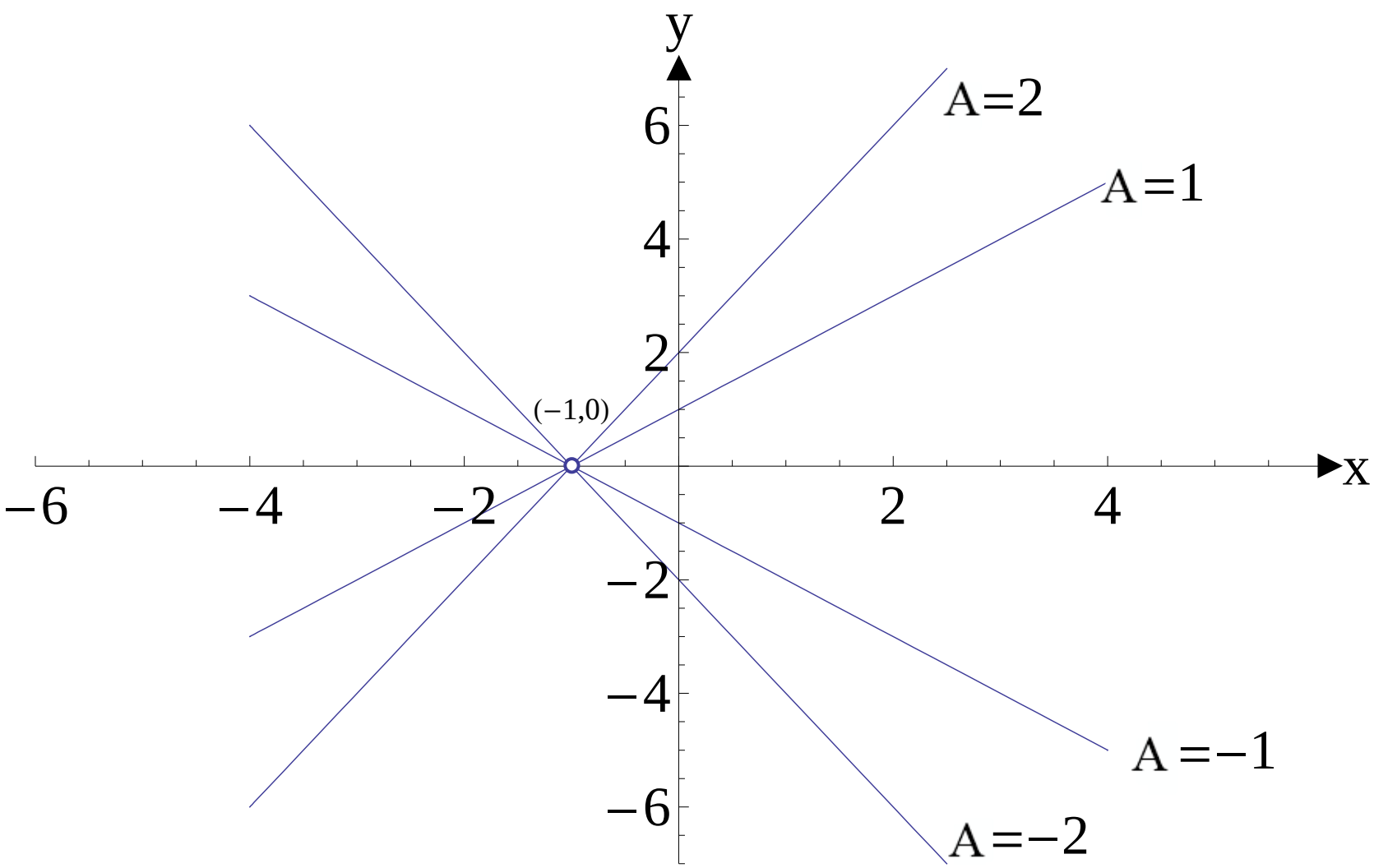
To solve, use *separation of variables*

$$\begin{aligned}
& \frac{dy}{dx} && = \mathcal{M}(x)\mathcal{N}(y) \\
\Rightarrow \quad & \frac{1}{\mathcal{N}(y)}\frac{dy}{dx} && = \mathcal{M}(x) \qquad (\mathcal{N}(y) \neq 0) \\
\Rightarrow \quad & \int \frac{1}{\mathcal{N}(y)}\frac{dy}{dx}\,dx && = \int \mathcal{M}(x)dx \\
\Rightarrow \quad & \int \frac{1}{\mathcal{N}(y)}dy && = \int \mathcal{M}(x)\,dx
\end{aligned}$$

Example 5.5: Solve $\dfrac{dy}{dx} = \dfrac{y}{1+x} \qquad (x \neq -1)$.

Solution:

Family of solutions

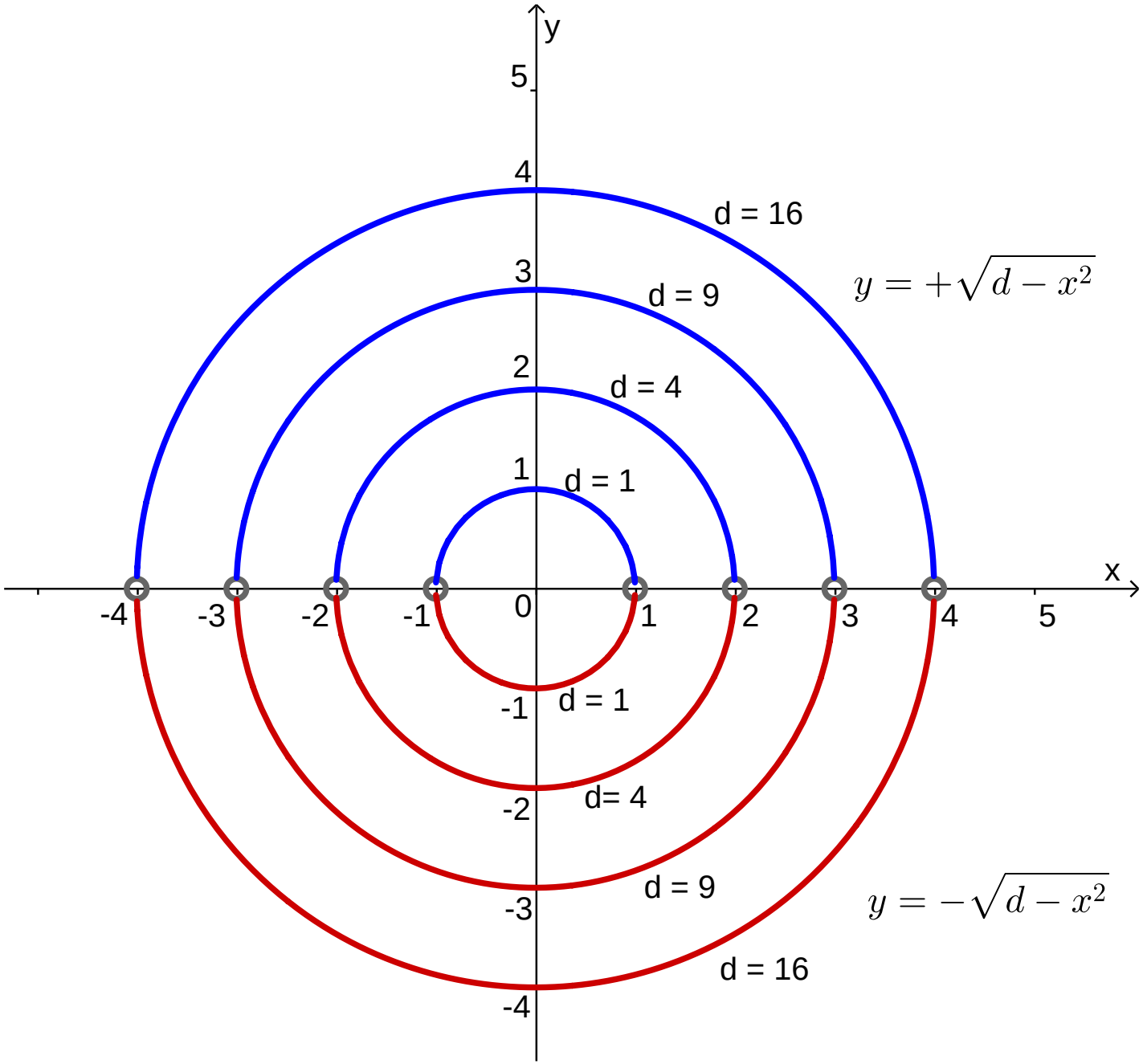


Example 5.6: Solve $\dfrac{dy}{dx} = \dfrac{-x}{y} \qquad (y \neq 0)$

if $y(0) = 3$.

Solution:

Family of solutions



Linear First Order ODEs

Example 5.7: Solve $x\dfrac{dy}{dx} + y = e^x$.

Solution:

A linear first order ODE has the form:

$$\frac{dy}{dx} + \mathcal{P}(x)y = \mathcal{Q}(x)$$

To solve:

Multiply ODE by $\mathcal{I}(x)$

$$\mathcal{I}(x)\frac{dy}{dx} + \mathcal{P}(x)\mathcal{I}(x)y = \mathcal{Q}(x)\mathcal{I}(x)$$

If the left side can be written as the derivative of $y(x)\mathcal{I}(x)$, then

$$\frac{d}{dx}\left[y(x)\mathcal{I}(x)\right] \quad = \quad \mathcal{Q}(x)\mathcal{I}(x)$$

which can be solved by integrating with respect to x.

Aim:

Find an integrating factor $\mathcal{I}$ so the left side will be the derivative of $y\mathcal{I}$. Then

$$\frac{d}{dx}(y\mathcal{I}) \equiv \mathcal{I}\frac{dy}{dx} + \mathcal{P}\mathcal{I}y$$

$$\Rightarrow \quad \frac{dy}{dx}\mathcal{I} + y\frac{d\mathcal{I}}{dx} = \mathcal{I}\frac{dy}{dx} + \mathcal{P}\mathcal{I}y$$

$$\Rightarrow \quad y\frac{d\mathcal{I}}{dx} = \mathcal{P}\mathcal{I}y$$

To solve for all y

$$\Rightarrow \quad \boxed{\frac{d\mathcal{I}}{dx} = \mathcal{P}\mathcal{I}} \qquad \text{(separable)}$$

$$
\begin{aligned}
\Rightarrow \frac{1}{\mathcal{I}}\frac{d\mathcal{I}}{dx} &= \mathcal{P} \\
\Rightarrow \int \frac{1}{\mathcal{I}} d\mathcal{I} &= \int \mathcal{P} dx \\
\Rightarrow \log|\mathcal{I}| &= \int \mathcal{P}\, dx + c \\
\Rightarrow |\mathcal{I}| &= e^{\int \mathcal{P} dx + c} \\
&= e^{\int \mathcal{P} dx} \cdot e^{c} \\
\Rightarrow \mathcal{I} &= \underbrace{\pm e^{c}}_{\text{constant}} \cdot e^{\int \mathcal{P} dx}
\end{aligned}
$$

So one integrating factor is

$$\boxed{\mathcal{I}(x) = e^{\int \mathcal{P} dx}}$$

Note:

Since we only need one integrating factor $\mathcal{I}$, we can neglect the '$+c$' and modulus signs when calculating $\mathcal{I}$.

Example 5.8: Find the general solution of

$$\frac{dy}{dx} + \frac{y}{x} = \sin x \qquad (x \neq 0).$$

Solution:

Example 5.9: Solve $\dfrac{1}{2}\dfrac{dy}{dx} - xy = x$ if $y(0) = -3$.

Solution:

Note:

Solving ODEs by substitution

Sometimes it is possible to make a substitution to reduce a general first order ODE to a separable or linear ODE.

- A homogeneous type ODE has the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Substituting $u = \dfrac{y}{x}$ reduces the ODE to a separable ODE.

- Bernoulli's equation has the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Substituting $u = y^{1-n}$ reduces the ODE to a linear ODE.

Example 5.10: Solve the homogeneous type differential equation

$$\frac{dy}{dx} = \frac{y}{x} + \cos^2\left(\frac{y}{x}\right) \qquad \left(-\frac{\pi}{2} < \frac{y}{x} < \frac{\pi}{2}\right)$$

by substituting $u = \dfrac{y}{x}$.

Solution:

Qualitative analysis

Qualitative analysis is a way of determining the shape and long-term behaviour of solutions of a 1st order ODE without needing to solve it analytically.

Qualitative analysis typically involves:

- finding the equilibrium solutions of an ODE
- drawing a phase plot
- determining the stability of the equilibrium solutions
- using the above information to draw the family of solutions

Equilibrium Solutions

Definition

An equilibrium solution is a constant solution of an ODE.

Note:

For the ODE $\dfrac{dx}{dt} = f(x,t)$, this means

- $x(t) = C$ where C is a constant
- $\dfrac{dx}{dt} = 0$

Terminology

We often simply say equilibrium instead of *equilibrium solution*.
The plural form of equilibrium is equilibria.

Example 5.11: Find all equilibrium solutions of the ODE

$$\frac{dx}{dt} = 3x - 2.$$

Solution:

Phase plots

A phase plot is a plot of $\frac{dx}{dt}$ as a function of x.

A phase plot will give

- the equilibria
- the behaviour of solutions close to the equilibria

Note:

Phase plots are only useful for ODEs of the form

$$\frac{dx}{dt} = f(x)$$

i.e., when the right-hand side has no explicit dependence on t.

ODEs of this form are called autonomous.

Example 5.12: For the ODE

$$\frac{dx}{dt} = 3x - 2,$$

(a) Draw a phase plot

(b) Sketch the family of solutions of the ODE, including any equilibria.

(c) Describe the long-term behaviour of solutions with initial conditions

i. $x(0) = \frac{1}{2}$

ii. $x(0) = 1$

Solution:

(a). The phase plot is

(b).

Family of solutions:

(c) From the phase plot and family of solutions, we can see that

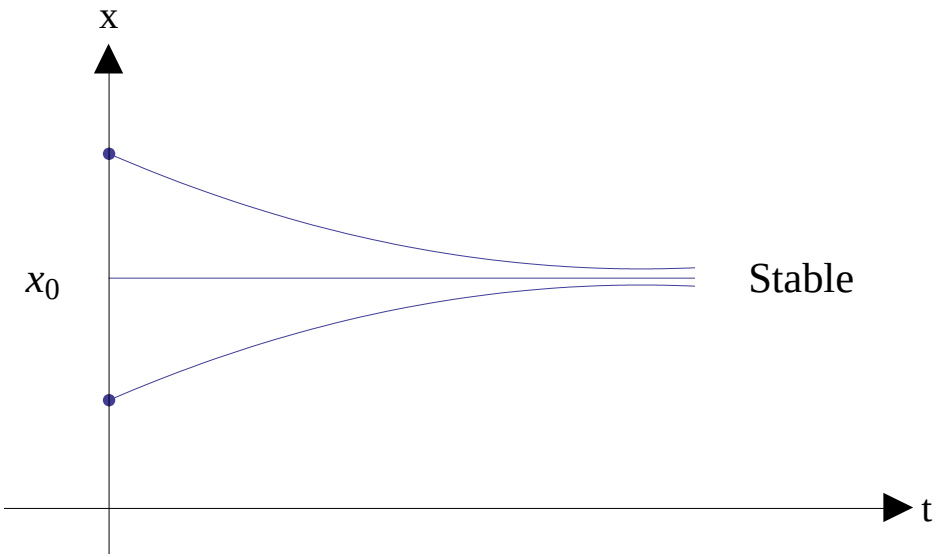
Note:

When drawing a family of solutions, we include at least one example of a solution with each possible qualitative behaviour (unless stated otherwise in the question).

Two solutions have the same qualitative behaviour if their shape and long-term behaviour are the same. Otherwise they are qualitatively different.

Stability of equilibria

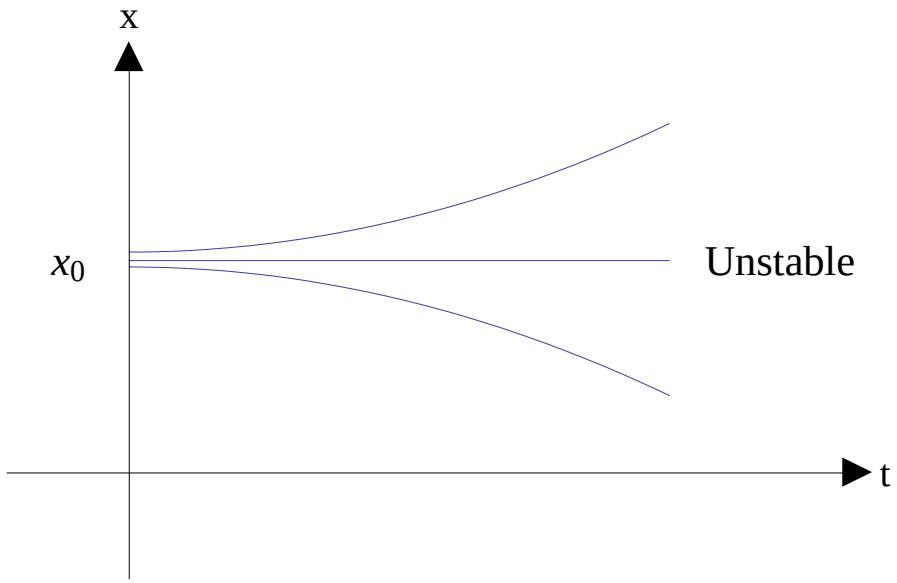
An equilibrium is stable if solutions that start nearby move closer to the equilibrium as t increases.



On a phase plot:

Stability of equilibria

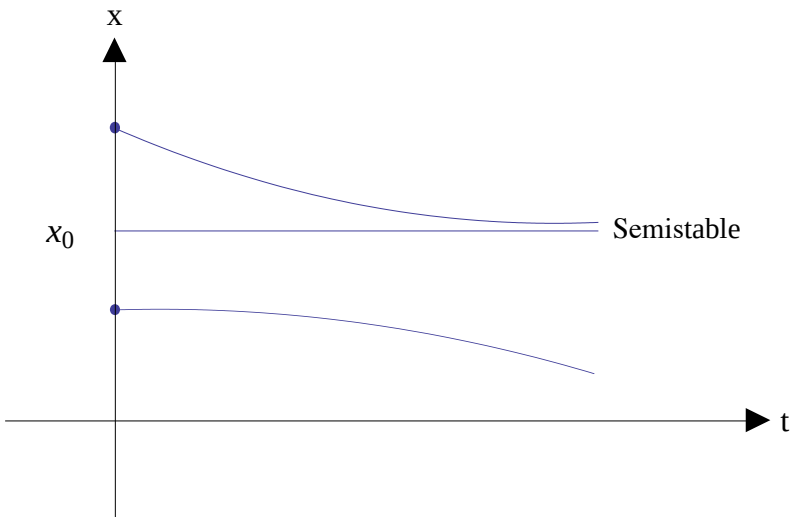
An equilibrium is unstable if solutions that start nearby move further away as t increases.

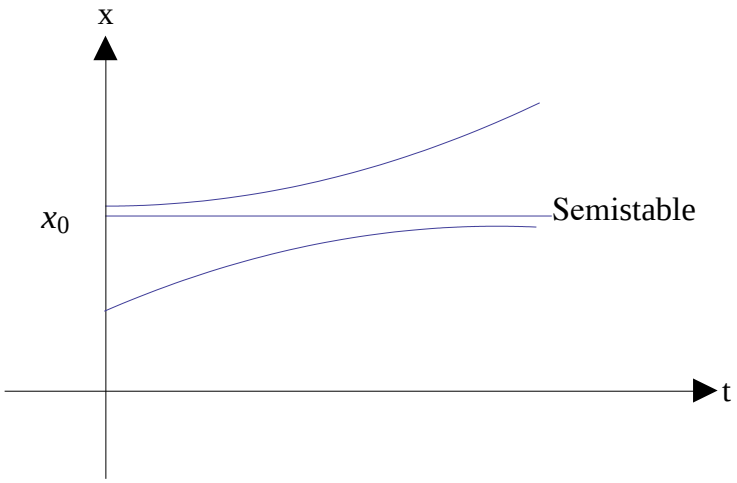


On a phase plot:

Stability of equilibria

An equilibrium is semistable if on one side of the equilibrium, solutions that start nearby move closer as t increases, whereas on the other side, solutions move further away as t increases.





On a phase plot:

Example 5.12 (continued):

(d) Determine the stability of the equilibrium.

Solution:

Population Models

Malthus (Doomsday) model

Rate of growth is proportional to the population p at time t.

$$\begin{aligned} \frac{dp}{dt} &\propto p \\ \Rightarrow \frac{dp}{dt} &= kp \qquad \text{(separable/linear)} \end{aligned}$$

where k is a constant of proportionality representing net births per unit population per unit time.

If the initial population is $p(0) = p_0$, then the solution is

$$p(t) = p_0 e^{kt}$$

You should check that you can derive this on your own!

Note:

The Doomsday model predicts:

- $k > 0$: unbounded exponential growth
- $k < 0$: population dies out
- $k = 0$: population stays constant

Unbounded exponential growth is unrealistic in the long term.

Doomsday model with harvesting.

Remove some of the population at a constant rate.

$$\frac{dp}{dt} = kp - h, \quad h > 0.$$

Example 5.13: A pharmaceutical company grows engineered yeast to produce a drug. The yeast is continuously harvested to collect the drug.

The population p (in millions of yeast cells) at time t days is described by

$$\frac{dp}{dt} = 3p - 2 \qquad (p \geq 0, t \geq 0)$$

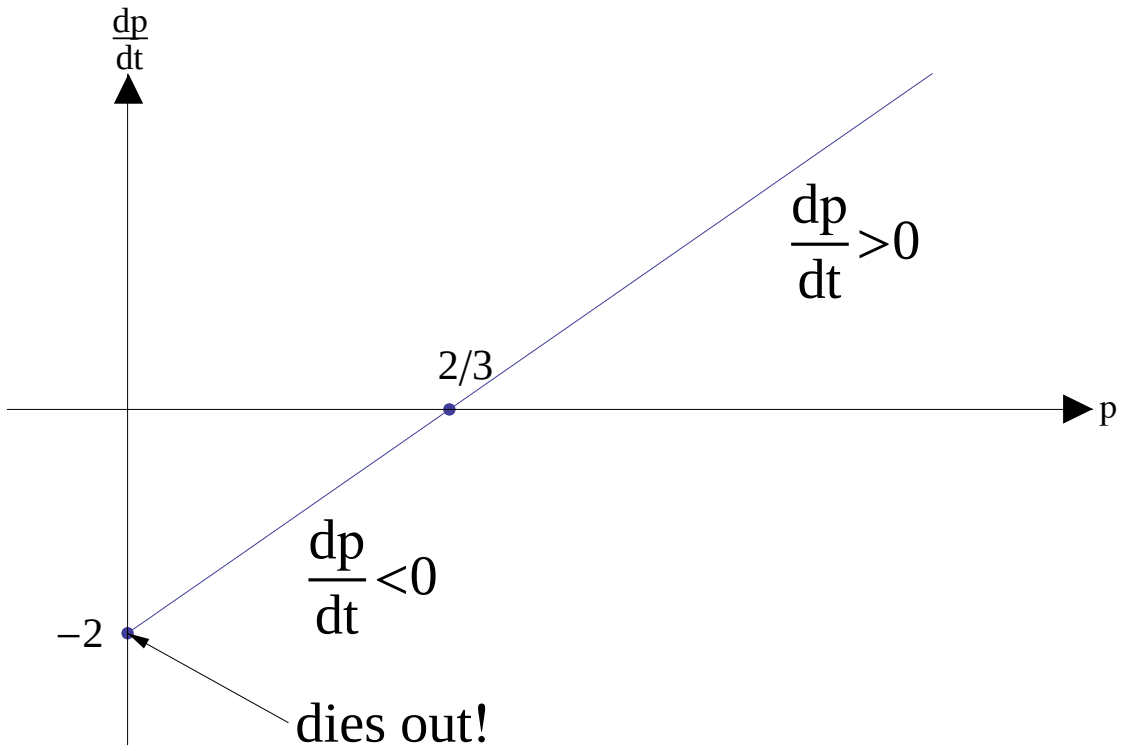
(a) For what initial population sizes $p(0)$ will the yeast population eventually die out?

(b) Find the time taken for the population to die out, if the initial population size is $p(0) = \frac{1}{2}$.

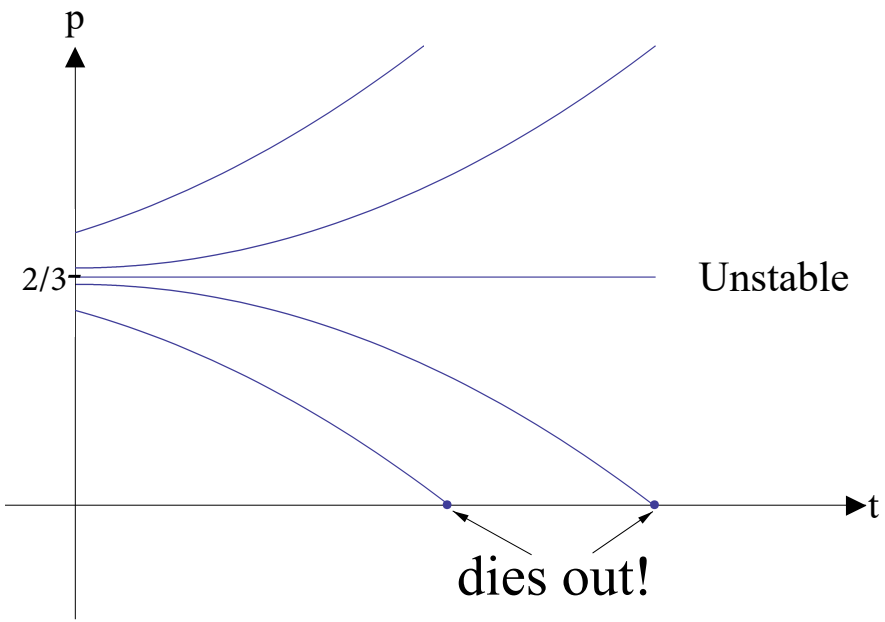
Solution:

From Example 5.12 we know

- the equilibrium is
- the phase plot is



- the family of solutions is



The population will die out if

(b) Find the time taken for the population to die out, if the initial population size is $p(0) = \frac{1}{2}$.

Logistic model.

Include “competition” term in Malthus’ model since overcrowding, disease, lack of food and natural resources will cause more deaths.

$$\frac{dp}{dt} = \underbrace{kp}_{\text{net birth rate}} - \underbrace{\frac{k}{a}p^2}_{\text{competition term}} = \quad kp\left(1 - \frac{p}{a}\right)$$

where $a > 0$ is the carrying capacity.

Example 5.14: A population is modelled by the logistic model

$$\frac{dp}{dt} = p\left(1 - \frac{p}{4}\right) \qquad (k = 1, a = 4)$$

Find the equilibrium solutions, determine their stability, and sketch the family of solutions for the ODE.

Solution:

Find the equilibrium solutions:

Phase plot:

Family of solutions:

Note:

The logistic model accurately describes

- population in a limited space (e.g. bacteria culture).
- population of USA from 1790-1950.

Logistic model with harvesting.

Remove some of the population at constant rate:

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{a}\right) - h, \quad h > 0, \; a > 0$$

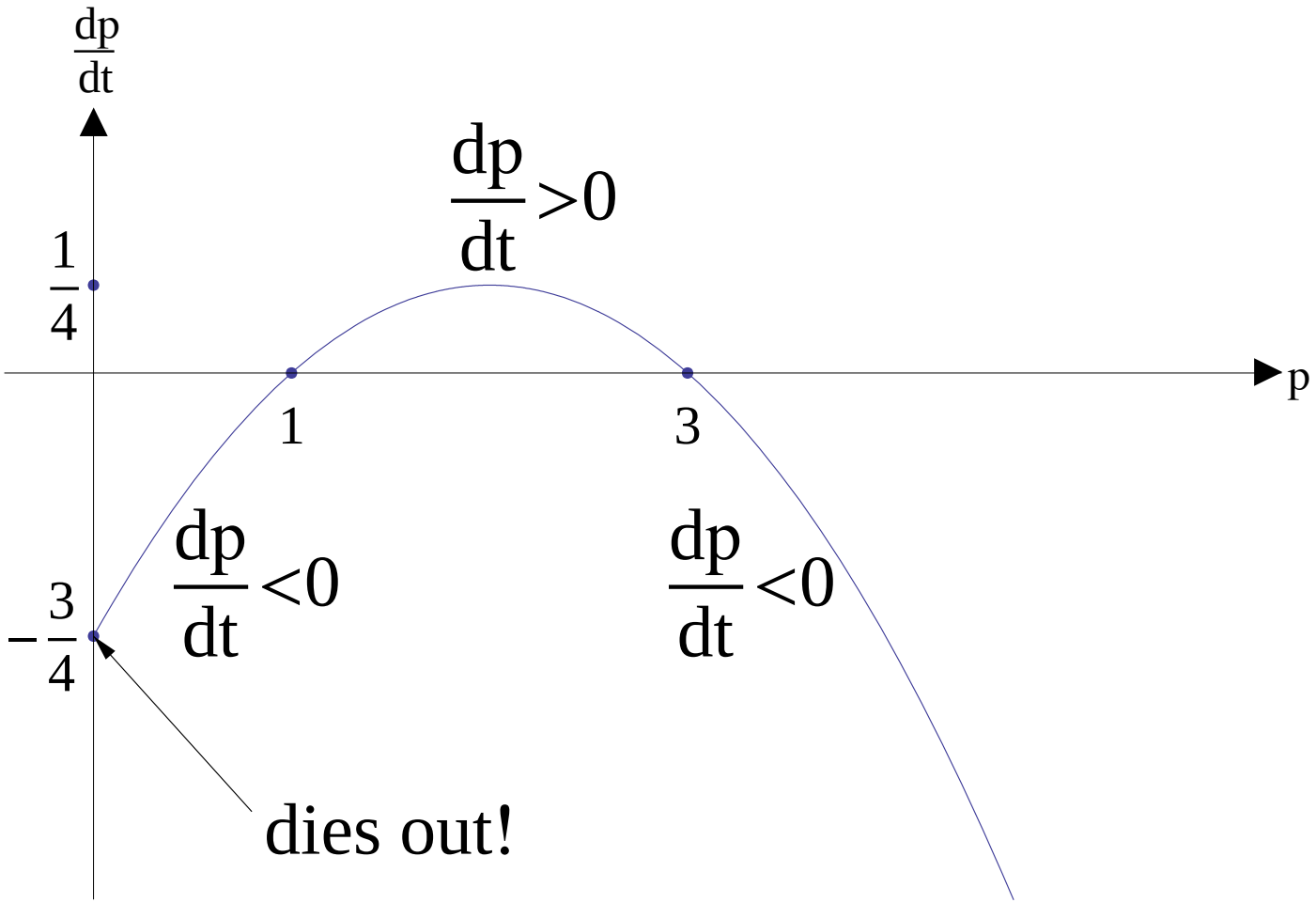
Example 5.15: For a population described by the logistic model with harvesting

$$\frac{dp}{dt} = p\left(1 - \frac{p}{4}\right) - \frac{3}{4} \qquad \left(a = 4,\ k = 1,\ h = \frac{3}{4}\right)$$

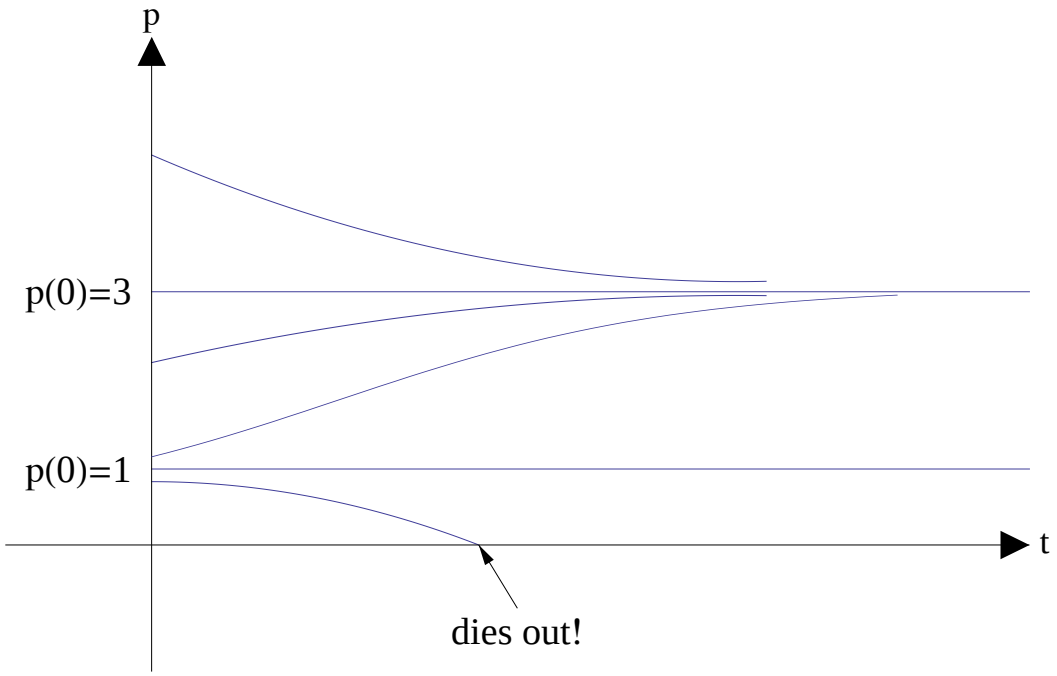
determine the long-term consequences for the population predicted by the model.

Solution:

Phase plot:


$\frac{dp}{dt}$
$\frac{dp}{dt} > 0$
$\frac{1}{4}$
p
1
3
$\frac{dp}{dt} < 0$
$\frac{dp}{dt} < 0$
$-\frac{3}{4}$
dies out!

Stability and family of solutions:



Interpretation:

(1) If $0 < p(0) < 1$ then

(2) If $p(0) = 1$ then

(3) If $1 < p(0) < 3$ then

(4) If $p(0) = 3$ then

(5) If $p(0) > 3$ then

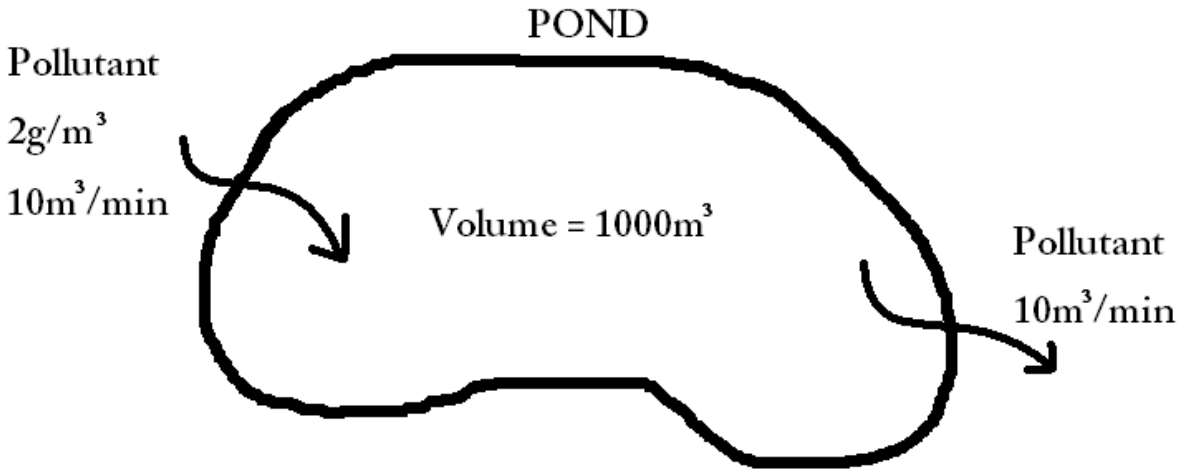
Note:

Case (2) is unrealistic in practice, as even a small deviation from $p(0) = 1$ will result in the solution following case (1) or (3).

Mixing Problems

Example 5.16: Effluent (pollutant concentration $2g/m^3$) flows into a pond (volume $1000m^3$, initially $100g$ pollutant) at a rate of $10m^3/min$. The pollutant mixes quickly and uniformly with pond water and flows out of pond at a rate of $10m^3/min$.

Find the concentration of pollutant in the pond at any time, and interpret the long-term behaviour of the system.



Derive the ODE:

Let x be the amount (grams) of pollutant in pond at time t minutes. Then $C = \dfrac{x}{V}$ is the concentration of pollutant in pond ($grams/m^3$), where V is the volume of the pond (m^3) at time t.

$$\boxed{\frac{dx}{dt} = \text{rate pollutant flows in - rate pollutant flows out}}$$

Solve analytically:

$$\frac{dx}{dt} + \frac{x}{100} = 20 \qquad \text{(Linear/Separable)}$$

General solution is

$$x(t) = 2000 + ce^{\frac{-t}{100}}$$

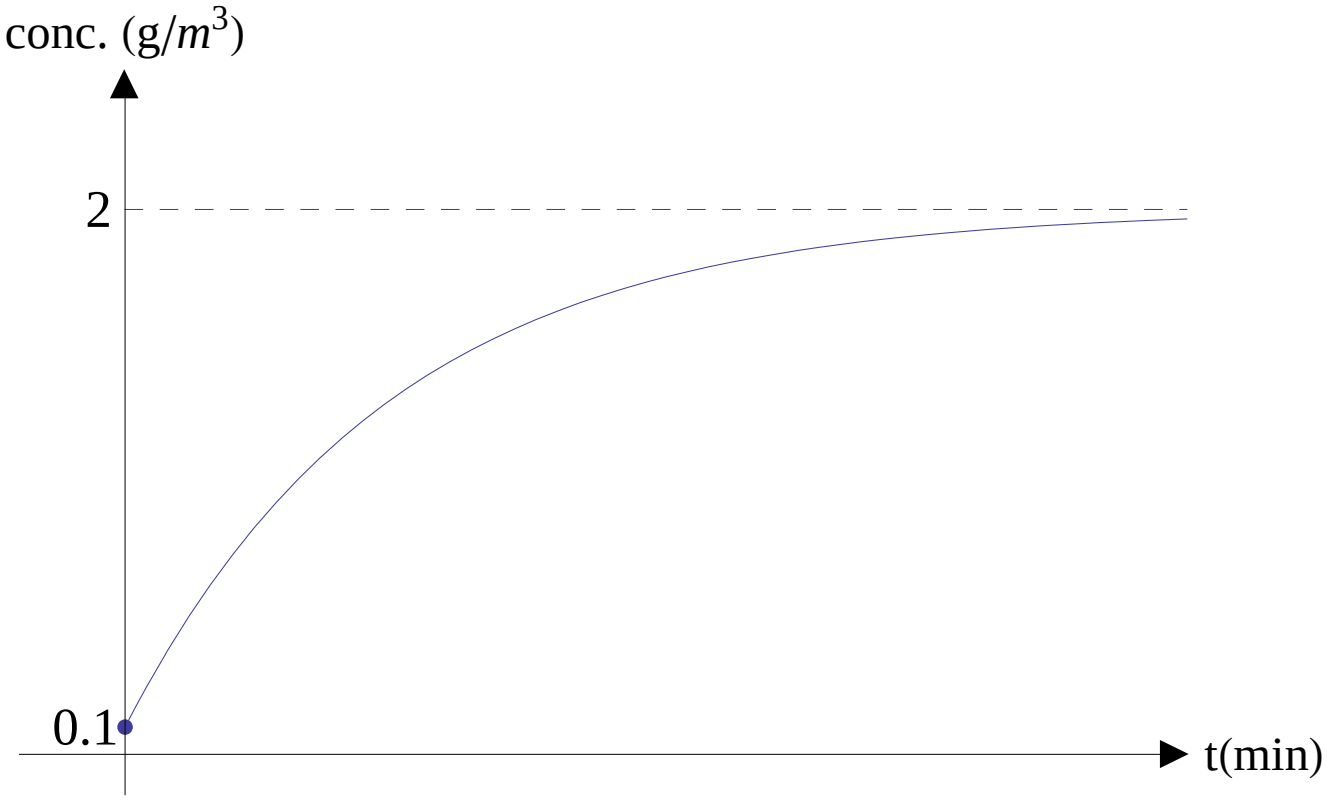
(working omitted).

Initial condition is $x(0) = 100 \quad \Rightarrow \quad c = -1900$

$$\Rightarrow x(t) = 2000 - 1900e^{\frac{-t}{100}}$$

Find concentration:

Interpret:

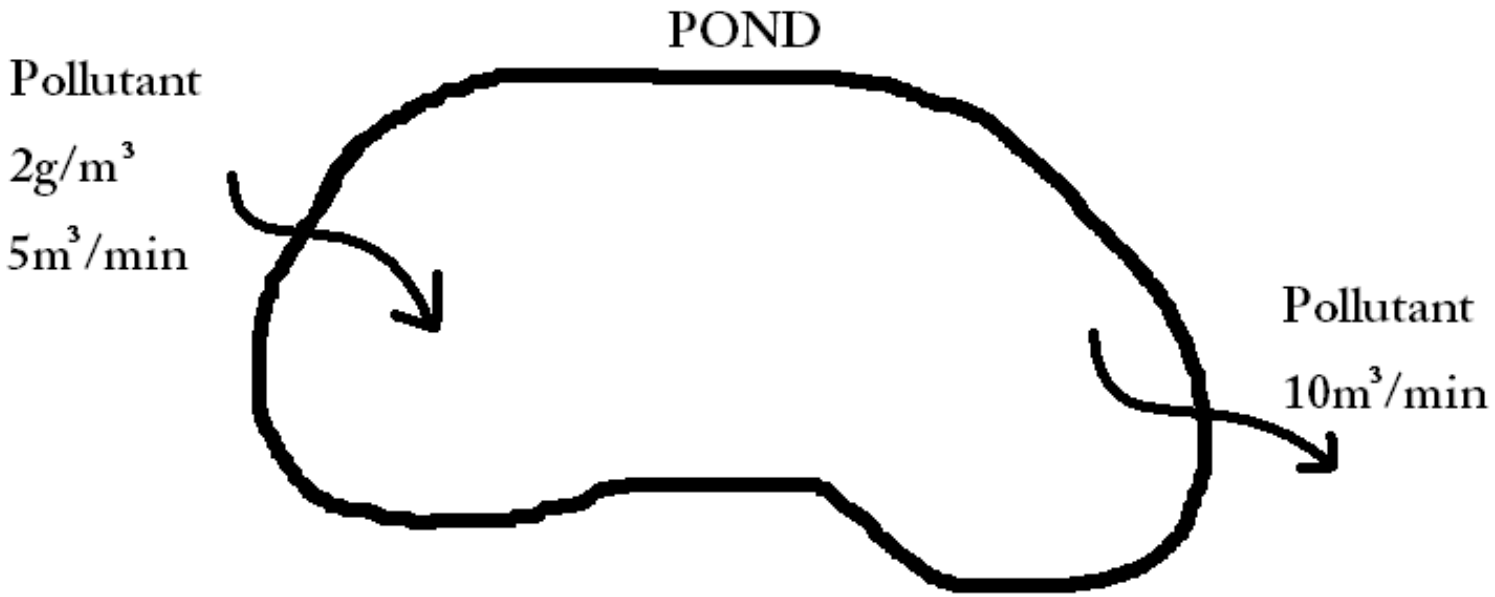


Definitions

1. Transient terms: terms decaying to 0 as $t \to \infty$.
2. Steady state terms: terms NOT decaying to 0 as $t \to \infty$.

The solution for the concentration can be classified as follows.

Example 5.17: Derive an ODE describing the amount x (g) of pollutant in the lake at time t (minutes), if the input flow rate is decreased to $5m^3/min$.



Let V be the volume in pond (m^3) at time t minutes.

Section 6: Second Order Differential Equations

A second order ODE has the form

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}\right) = 0$$

The general form of a linear second order ODE is

$$\frac{d^2y}{dx^2} + \mathcal{P}(x)\frac{dy}{dx} + \mathcal{Q}(x)y = \mathcal{R}(x)$$

- If $\mathcal{R}(x) = 0$, the ODE is homogeneous (H).
- If $\mathcal{R}(x) \neq 0$, the ODE is inhomogeneous (IH).

Note:

A homogeneous linear ODE is different to a homogeneous type first order ODE.

The general solution of a second order ODE typically has two arbitrary constants.

Initial value problem for a second order ODE

Solve

$$\frac{d^2y}{dx^2} + \mathcal{P}(x)\frac{dy}{dx} + \mathcal{Q}(x)y = \mathcal{R}(x)$$

subject to the conditions $y(x_0) = y_0$ and $y'(x_0) = y_1$.

Boundary value problem for a second order ODE

Solve

$$\frac{d^2y}{dx^2} + \mathcal{P}(x)\frac{dy}{dx} + \mathcal{Q}(x)y = \mathcal{R}(x)$$

subject to the conditions $y(a) = y_0$ and $y(b) = y_1$.

Homogeneous 2nd Order Linear ODEs

Theorem:

The general solution of

$$y'' + \mathcal{P}(x)y' + \mathcal{Q}(x)y = 0$$

is the function y given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where

- y_1, y_2 are two linearly independent solutions of the homogeneous ODE,
- $c_1, c_2 \in \mathbb{R}$ are arbitrary constants.

Definition:

Two functions y_1 and y_2 are linearly independent if

$$c_1 y_1(x) + c_2 y_2(x) = 0 \quad \Rightarrow \quad c_1 = c_2 = 0$$

or equivalently, if neither function is a non-zero constant multiple of the other function.

Example 6.1:

(a) Are $y_1(x) = x^2$, $y_2(x) = 2x^2$ linearly independent?

(b) Are $y_1(x) = e^{2x}$, $y_2(x) = xe^{2x}$ linearly independent?

Aside - A connection with Linear Algebra*

Why are these ODEs called linear?

Because they can be written in terms of a linear transformation. The ODE

$$\frac{d^2y}{dx^2} + \mathcal{P}(x)\frac{dy}{dx} + \mathcal{Q}(x)y = \mathcal{R}(x)$$

can be written as

$$T(y) = \mathcal{R}(x)$$

where

$$T(y) = \frac{d^2y}{dx^2} + \mathcal{P}(x)\frac{dy}{dx} + \mathcal{Q}(x)y$$

is a linear transformation on a vector space of functions.

These concepts are covered in MAST10007 Linear Algebra.

** This slide is not examinable in MAST10006.*

Homogeneous 2nd Order Linear ODEs with Constant Coefficients

General form: $$ay'' + by' + cy = 0$$

where a, b, c are constants.

To solve for $y(x)$:

Try $y(x) = e^{\lambda x}$. Then $y'(x) = \lambda e^{\lambda x}$ and $y''(x) = \lambda^2 e^{\lambda x}$.

Substitute into the ODE:

$$\begin{aligned} ay'' + by' + cy = a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} &= 0 \\ \left(a\lambda^2 + b\lambda + c\right) \underbrace{e^{\lambda x}}_{\neq 0} &= 0 \end{aligned}$$

$$\Rightarrow \boxed{a\lambda^2 + b\lambda + c = 0} \qquad \text{Characteristic Equation}$$

$$\Rightarrow \lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Case 1: $b^2 - 4ac > 0$

- 2 distinct real values λ_1, λ_2
- 2 linearly independent solutions

$$e^{\lambda_1 x}, \quad e^{\lambda_2 x}$$

- General Solution:

$$\boxed{y(x) = Ae^{\lambda_1 x} + Be^{\lambda_2 x}}$$

Example 6.2: Solve $y'' + 7y' + 12y = 0$ for $y(x)$.

Solution:

Case 2: $b^2 - 4ac = 0$

- 1 real value $\lambda = \dfrac{-b}{2a}$
- 1 solution is $e^{\lambda x}$
- 2nd linearly independent solution is $xe^{\lambda x}$ (found using variation of parameters — not in syllabus).
- General Solution:

$$\boxed{y(x) = Ae^{\lambda x} + Bxe^{\lambda x}}$$

We now verify that $xe^{\lambda x}$ is a solution:

$$\text{If } y(x) = xe^{\lambda x}, \text{ then}$$

$$\begin{aligned} y'(x) &= (\lambda x + 1)\, e^{\lambda x}, \\ y''(x) &= \left(\lambda^2 x + 2\lambda\right) e^{\lambda x}. \end{aligned}$$

So $ay'' + by' + cy$

$$\begin{aligned} &= a\left(\lambda^2 x + 2\lambda\right) e^{\lambda x} + b\,(\lambda x + 1)\, e^{\lambda x} + cxe^{\lambda x} \\ &= xe^{\lambda x} \underbrace{\left(a\lambda^2 + b\lambda + c\right)}_{=0} + \underbrace{(2\lambda a + b)}_{=0} e^{\lambda x} \\ &= 0 \end{aligned}$$

So $y(x) = xe^{\lambda x}$ is a solution.

Example 6.3: Solve $y'' + 2y' + y = 0$ for $y(x)$.

Solution:

Case 3: $b^2 - 4ac < 0$

- 2 complex conjugate values

$$\lambda_1 = \alpha + i\beta, \quad \lambda_2 = \alpha - i\beta$$

- 2 linearly independent complex solutions

$$e^{(\alpha+i\beta)x}, \quad e^{(\alpha-i\beta)x}$$

- General solution over the complex numbers:

$$y(x) = C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x} \qquad \text{where } C_1, C_2 \in \mathbb{C}$$

To find the general solution over the real numbers, consider

$$y_c = e^{(\alpha+i\beta)x} = e^{\alpha x}\Big(\cos(\beta x) + i\sin(\beta x)\Big).$$

Because y_c is a solution of $ay'' + by' + cy = 0$ over the complex numbers, we have

$$ay_c'' + by_c' + cy_c = 0 + 0i$$

Take the real part of this equation:

$$\mathrm{Re}\,(ay_c'' + by_c' + cy_c) = 0$$

$$a\mathrm{Re}\Big(y_c''\Big) + b\mathrm{Re}\Big(y_c'\Big) + c\mathrm{Re}\Big(y_c\Big) = 0$$

$$a\Big(\mathrm{Re}(y_c)\Big)'' + b\Big(\mathrm{Re}(y_c)\Big)' + c\Big(\mathrm{Re}(y_c)\Big) = 0$$

So $Re(y_c) = e^{\alpha x}\cos(\beta x)$ is a real solution of the ODE.

Similarly, $\mathrm{Im}(y_c) = e^{\alpha x}\sin(\beta x)$ is a real solution of the ODE.

$y_1 = e^{\alpha x}\cos(\beta x)$ and $y_2 = e^{\alpha x}\sin(\beta x)$ are two linearly independent real solutions of the ODE.

Therefore the general solution over the real numbers is

$$y = Ae^{\alpha x}\cos(\beta x) + Be^{\alpha x}\sin(\beta x) \quad \text{where } A, B \in \mathbb{R}.$$

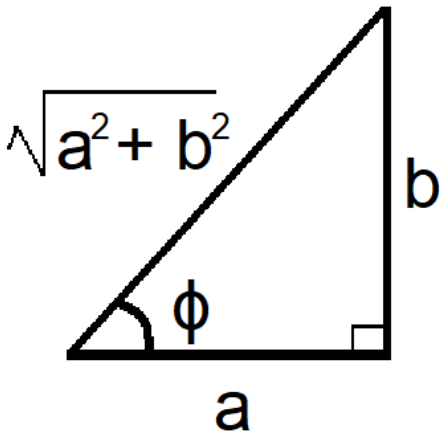
Example 6.4: Solve $y'' - 4y' + 13y = 0$ subject to $y(0) = 1$ and $y'(0) = 6$.

Solution:

This solution has the form $y(x) = e^{2x}\,(a\cos\theta + b\sin\theta)$.

In general, for $a, b > 0$,

$$a\cos\theta + b\sin\theta = \sqrt{a^2+b^2}\left(\frac{a}{\sqrt{a^2+b^2}}\cos\theta + \frac{b}{\sqrt{a^2+b^2}}\sin\theta\right)$$



$$\phi = \arctan\left(\frac{b}{a}\right)$$

$$= \sqrt{a^2+b^2}\left(\cos\phi\cos\theta + \sin\phi\sin\theta\right)$$

$$= \sqrt{a^2+b^2}\cos(\theta-\phi).$$

Hence we can rewrite the solution from Example 6.4 as

This form is sometimes preferable for graphing or further manipulation.

Inhomogeneous 2nd Order Linear ODEs

Theorem:

The general solution of

$$y'' + \mathcal{P}(x)y' + \mathcal{Q}(x)y = \mathcal{R}(x)$$

is the function y given by

$$y(x) = y_{\mathcal{H}}(x) + y_{\mathcal{P}}(x)$$

where

- $y_{\mathcal{H}}(x) = c_1 y_1(x) + c_2 y_2(x)$ is the general solution of the homogeneous ODE (called the homogeneous solution, GS(H)),
- $y_{\mathcal{P}}(x)$ is a solution of the inhomogeneous ODE (called a particular solution, PS(IH)),

Inhomogeneous 2nd Order Linear ODEs with Constant Coefficients

General form:

$$ay'' + by' + cy = \mathcal{R}(x)$$

where a, b, c are constants.

Example 6.5: Solve $y'' + 2y' - 8y = \mathcal{R}(x)$ where

(a) $\mathcal{R}(x) = 1 - 8x^2$

(b) $\mathcal{R}(x) = e^{3x}$

(c) $\mathcal{R}(x) = 85 \cos x$

(d) $\mathcal{R}(x) = 3 - 24x^2 + 7e^{3x}$.

Solution:

Step 1: Find the general solution of $y'' + 2y' - 8y = 0$.

Step 2: Find a particular solution of $y'' + 2y' - 8y = \mathcal{R}(x)$.

(a) $\mathcal{R}(x) = 1 - 8x^2$: $\quad y'' + 2y' - 8y = 1 - 8x^2$

(b) $\mathcal{R}(x) = e^{3x}$: $\quad y'' + 2y' - 8y = e^{3x}$

$\left(e^{3x} \text{ is NOT part of } GS(H)\right)$

(c) $\mathcal{R}(x) = 85\cos x$: $\quad y'' + 2y' - 8y = 85\cos x$

Superposition of Particular Solutions

Theorem:

A particular solution of

$$ay'' + by' + cy = \alpha\,\mathcal{R}_1(x) + \beta\,\mathcal{R}_2(x)$$

is

$$y_{\mathcal{P}}(x) = \alpha y_1(x) + \beta y_2(x)$$

where

- $y_1(x)$ is a particular solution of $ay'' + by' + cy = \mathcal{R}_1(x)$,
- $y_2(x)$ is a particular solution of $ay'' + by' + cy = \mathcal{R}_2(x)$,
- a, b, c, α, β are constants.

Example 6.5 (d): $\mathcal{R}(x) = 3 - 24x^2 + 7e^{3x}$.

Solution:

Example 6.6: Solve $y'' - y = e^x$.

Solution:

$GS(H) : y_{\mathcal{H}}(x) = Ae^x + Be^{-x}$

Example 6.7: Solve $y'' + 2y' + y = e^{-x}$.

Solution:

$GS(H): \; y_{\mathcal{H}}(x) = Ae^{-x} + Bxe^{-x}$

Example 6.8: Solve $y'' + 49y = 28\sin(7t)$.

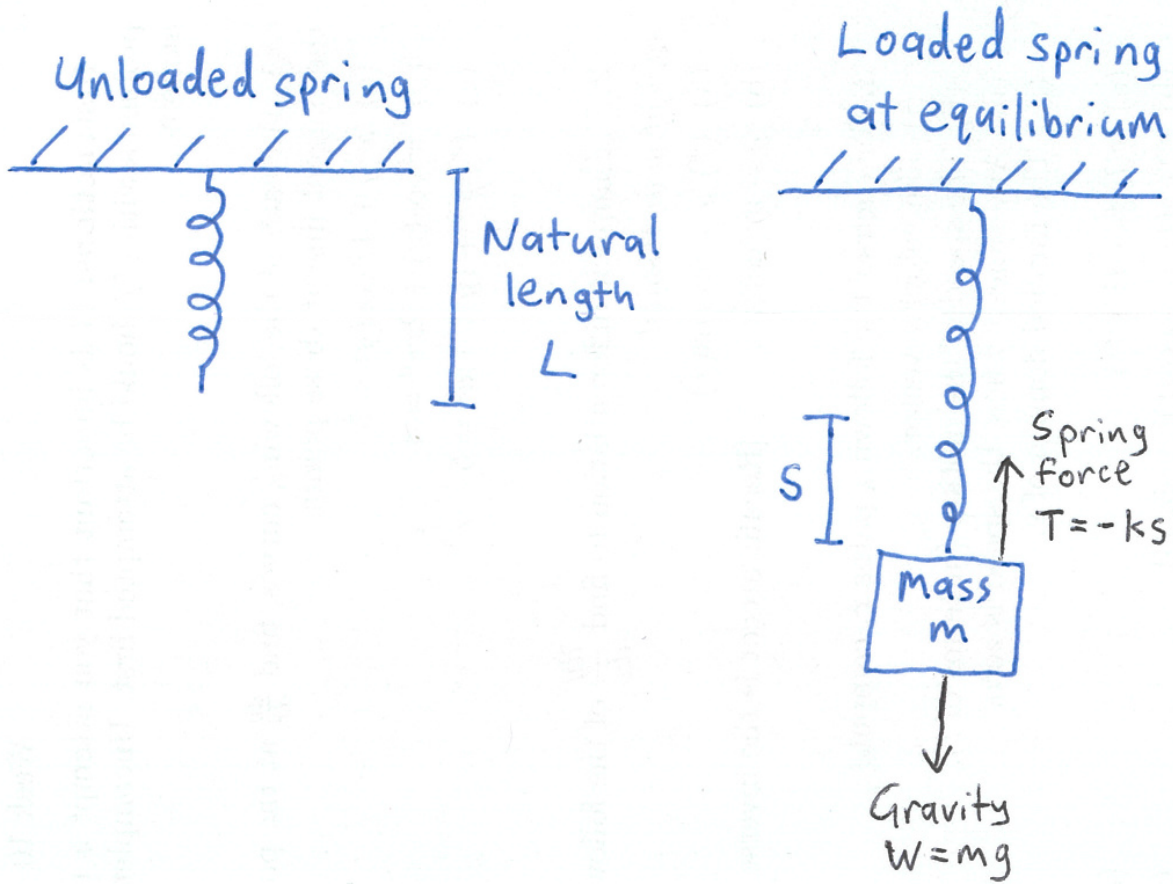
Solution:

$GS(H) : y_{\mathcal{H}}(t) = A\cos(7t) + B\sin(7t)$

Springs - Free Vibrations

Consider an object (of mass m kg) attached to a spring hanging from a fixed support.

Suppose that the natural length of the spring when unloaded is L m, and that when the object is attached, it causes the spring to stretch by s m downwards at equilibrium.



The forces are:

- gravitational force, given by the gravitational law:

$$W = mg \qquad (g = 9.8 \text{ m s}^{-2} \text{ on Earth})$$

- restoring force in spring, given by Hooke's Law:

$$T = \underbrace{-k}_{\text{spring constant}} \cdot \text{extension} \qquad (k > 0)$$

Note:

The negative sign in Hooke's law is because the direction of the force T is opposite to the direction of extension:

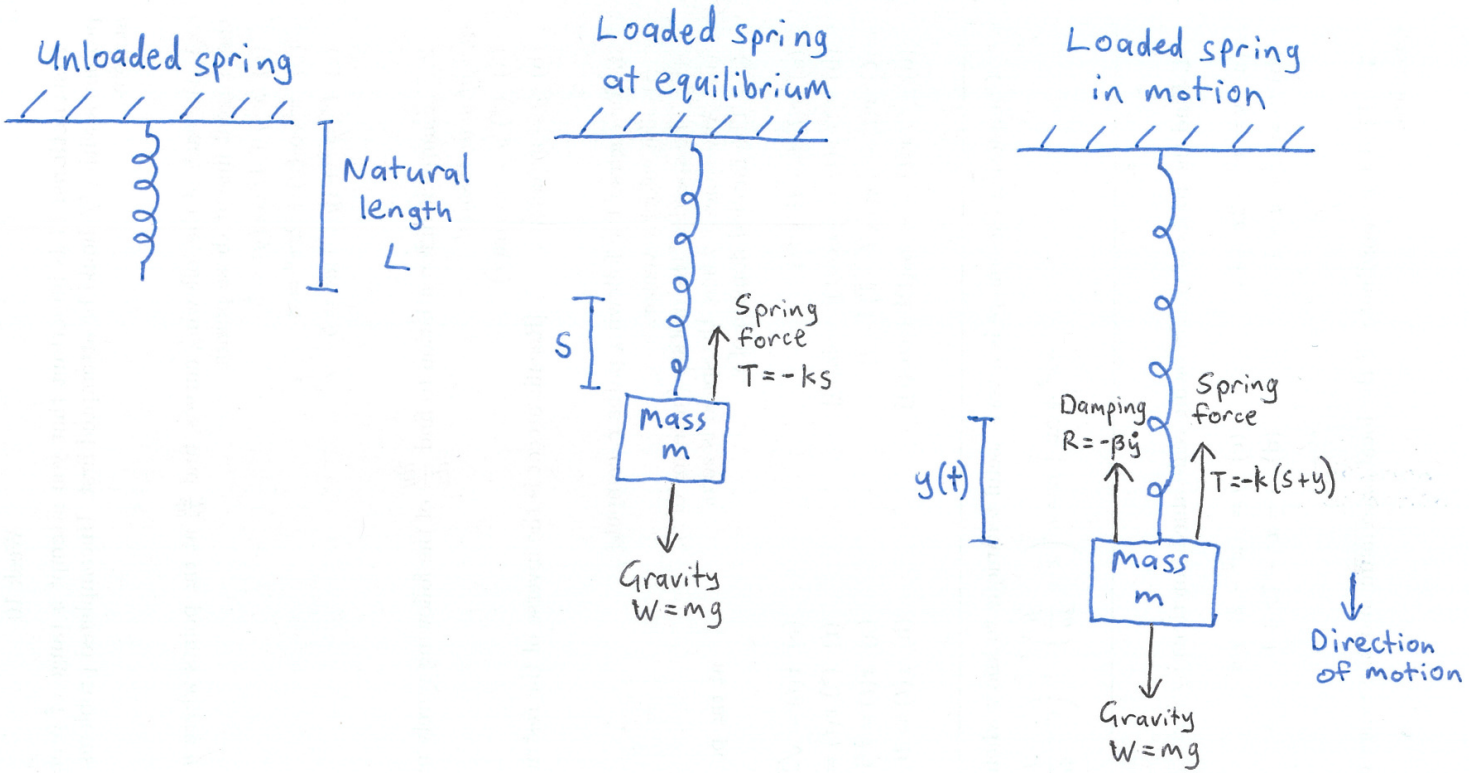
- if the spring is stretched downwards, then it pulls up
- if the spring is compressed upwards, then it pushes down.

The units for k are N m^{-1} (Newton per metre)

At equilibrium, the net force is zero, so

Suppose the object is set in motion.

Let $y(t)$ be the displacement of the object below the equilibrium position ($y = 0$) at time t seconds.



Note: Since displacement y is measured below the equilibrium position, y is positive when the object is below the equilibrium position, and y is negative when the object is above the equilibrium position.

When the object is moving, there is an extra force acting:

- damping force is proportional to velocity

$$R = -\beta \dot{y} \qquad (\beta \geq 0)$$

$\uparrow$

damping constant

Note:

The negative sign is because the direction of the damping force is opposite the direction of motion.

The units for β are N / (m s^{-1}) = N s m^{-1}

Derive the equation of motion

Newton's 2nd Law states that

$$\text{net force} = \text{mass} \cdot \text{acceleration}$$

$$F_{net} = \quad m \quad \cdot \quad \ddot{y}$$

Find the net force:

$$\begin{aligned} F_{net} &= W + T + R \\ &= mg - k(s + y) - \beta\dot{y} \end{aligned}$$

Apply Newton's Law:

$$
\begin{aligned}
F_{net} &= m\ddot{y} \\
mg - k(s+y) - \beta\dot{y} &= m\ddot{y} \\
mg - ks - ky - \beta\dot{y} &= m\ddot{y} \\
-ky - \beta\dot{y} &= m\ddot{y} \qquad \text{(as } mg = ks\text{)}
\end{aligned}
$$

$$\Rightarrow \boxed{m\ddot{y} + \beta\dot{y} + ky = 0}$$

Equation of motion

To solve, try $y(t) = e^{\lambda t}$

$$\Rightarrow \quad m\lambda^2 + \beta\lambda + k = 0$$

$$\Rightarrow \quad \lambda = \frac{-\beta \pm \sqrt{\beta^2 - 4mk}}{2m}$$

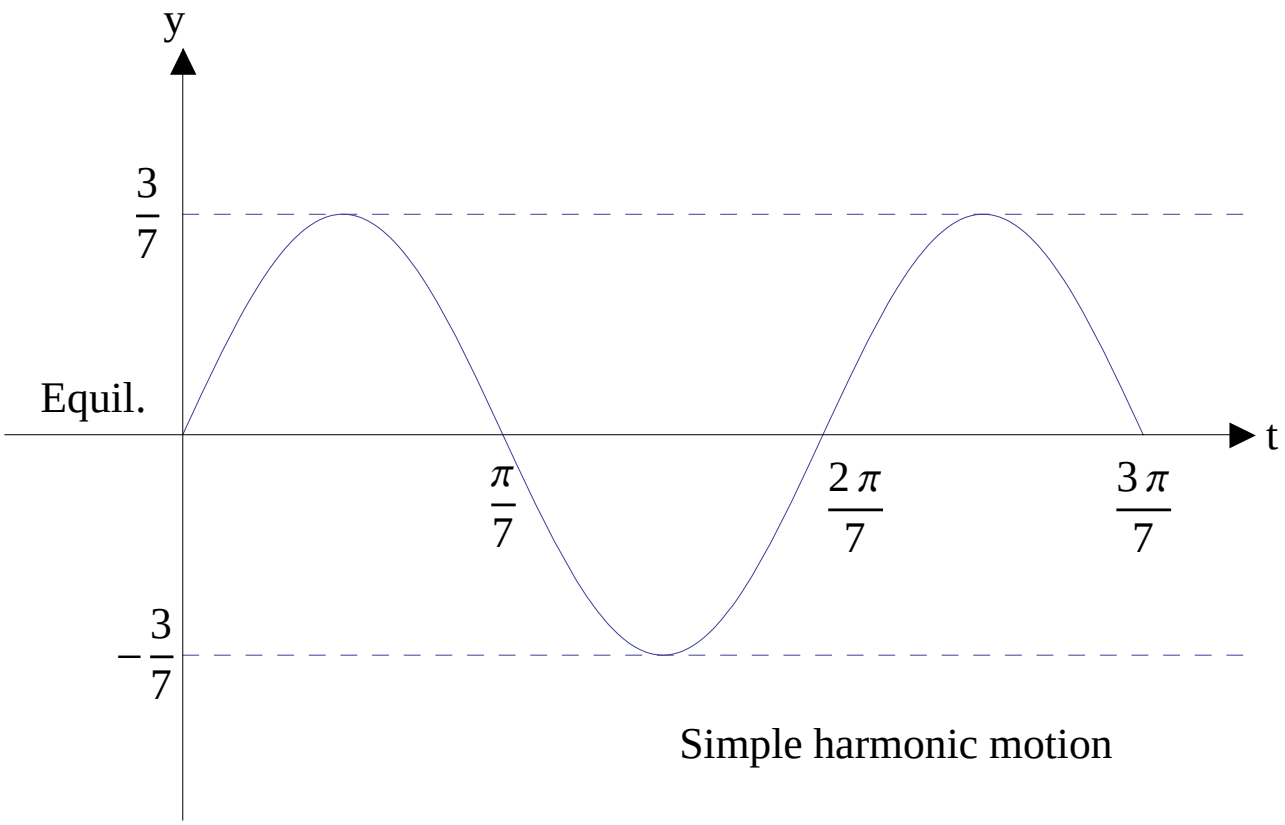
- If $\beta = 0$: $\quad \lambda = \pm ib$ simple harmonic motion
- If $0 < \beta < 2\sqrt{mk}$: $\quad \lambda = a \pm ib$ underdamped, weak damping
- If $\beta = 2\sqrt{mk}$: $\quad \lambda = a, a$ critical damping
- If $\beta > 2\sqrt{mk}$: $\quad \lambda = a, b$ overdamped, strong damping

Example 6.9: A $\frac{40}{49}$ kg mass stretches a spring hanging from a fixed support by 0.2m. The mass is released from the equilibrium position with a downward velocity of 3m/s. Find the position of the mass y below equilibrium at any time t, if the damping constant β is:

(a) 0 (b) $\frac{160}{49}$ (c) $\frac{80}{7}$ (d) $\frac{2000}{49}$

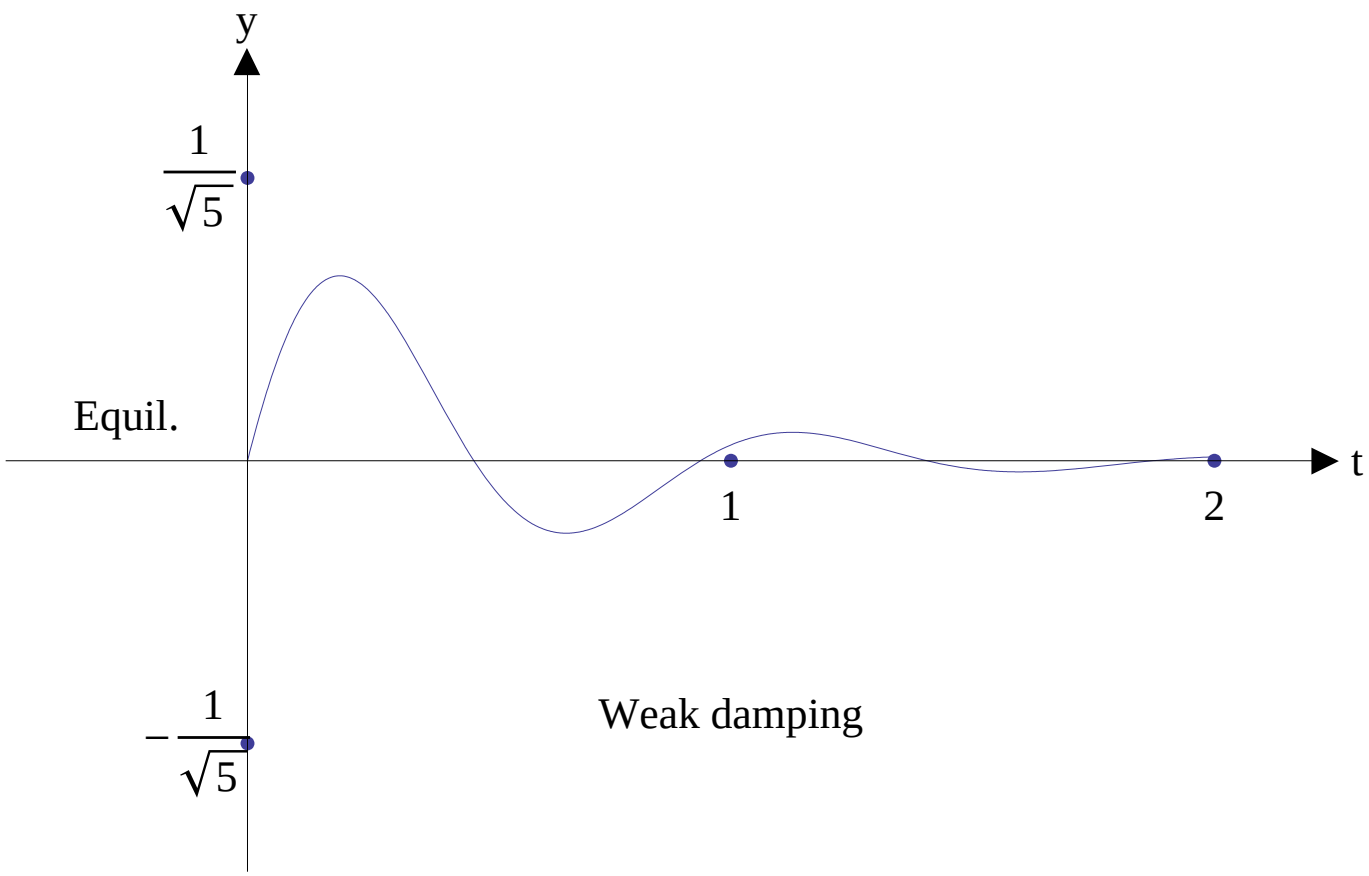
Solution:

(a) $\beta = 0 : \quad \ddot{y} + 49y = 0$

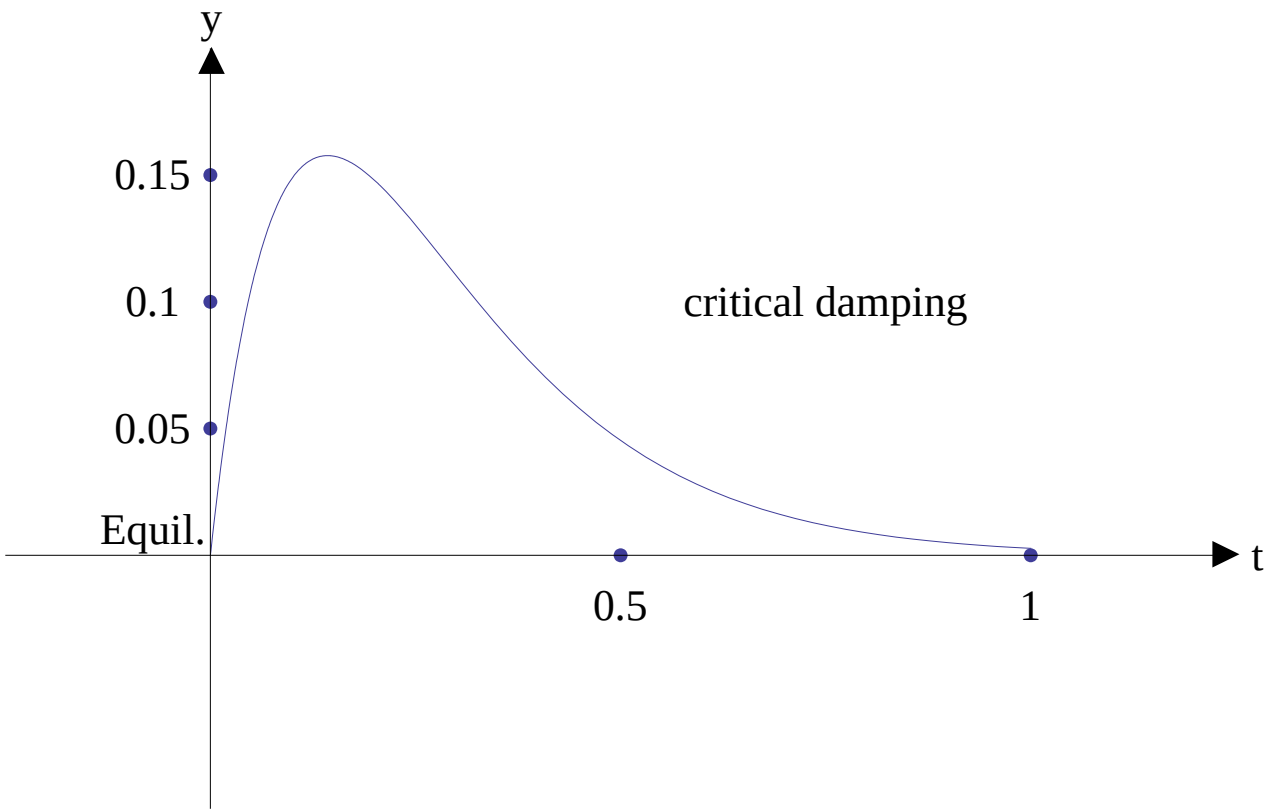


Simple harmonic motion

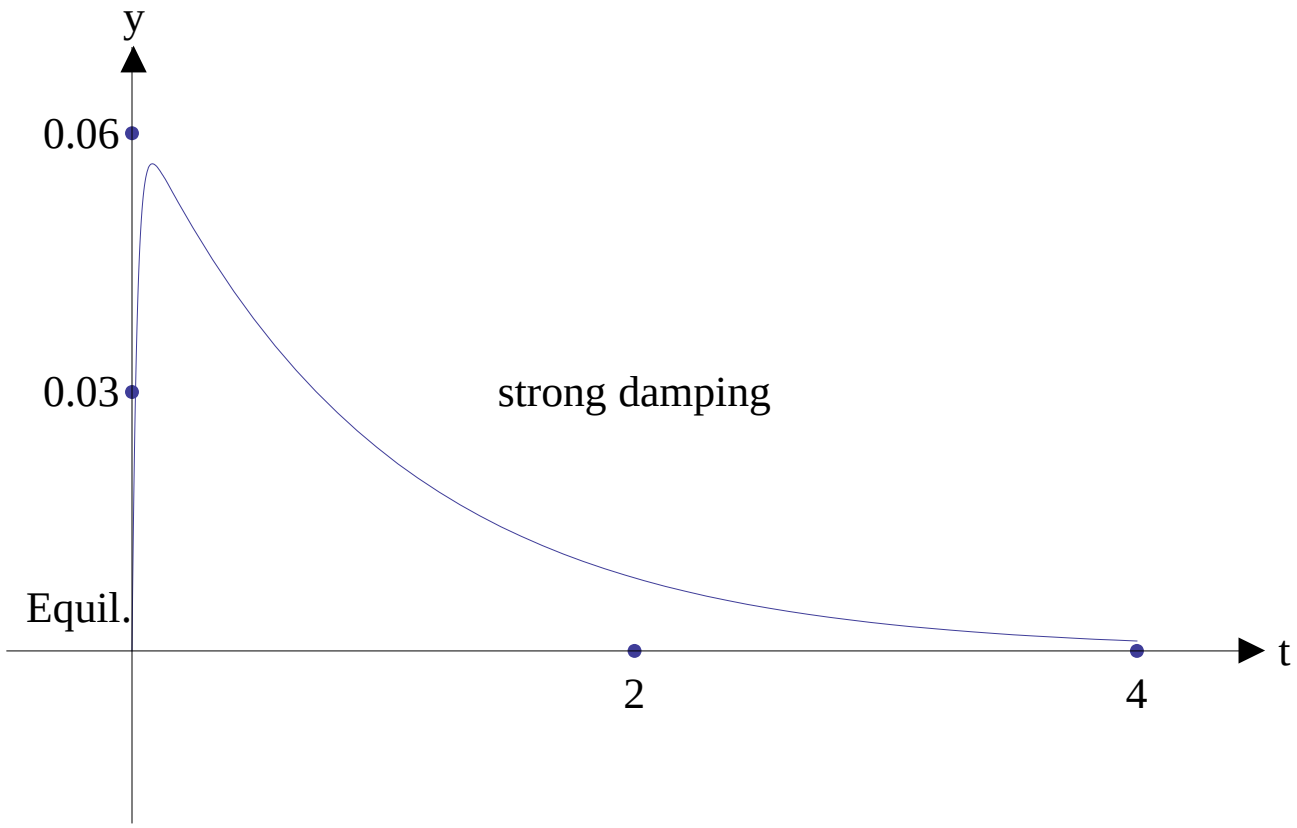
(b) $\beta = \frac{160}{49}$: $\quad \ddot{y} + 4\dot{y} + 49y = 0$


y
$\frac{1}{\sqrt{5}}$
Equil.
t
1
2
$-\frac{1}{\sqrt{5}}$
Weak damping

(c) $\beta = \frac{80}{7}$: $\quad \ddot{y} + 14\dot{y} + 49y = 0$

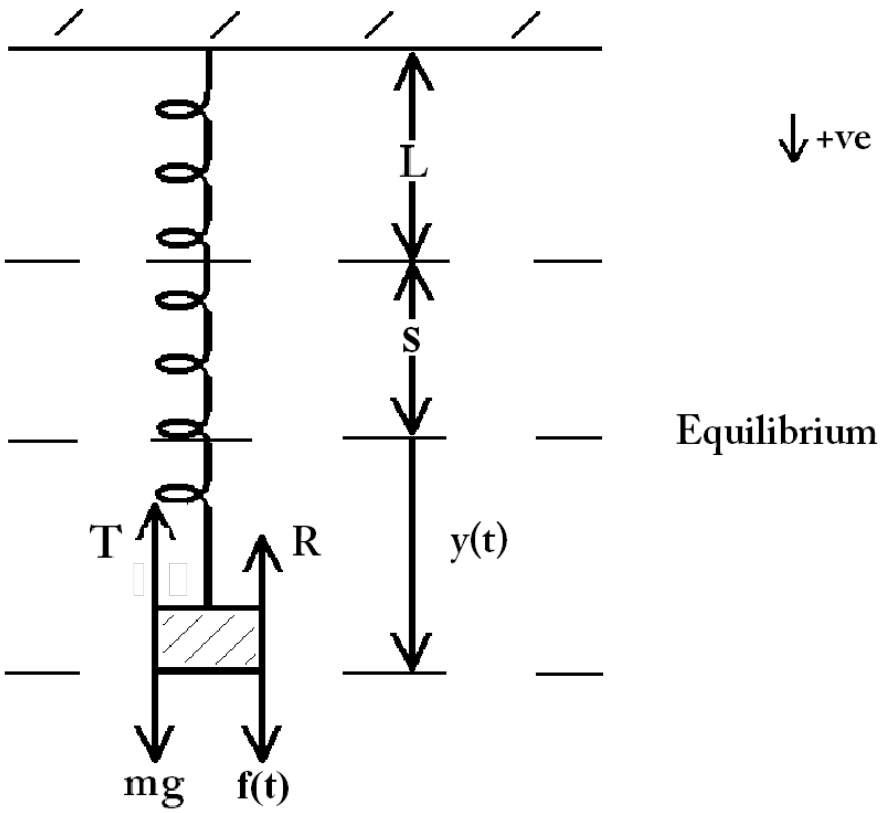

y
0.15
0.1
0.05
Equil.
critical damping
t
0.5
1

(d) $\beta = \frac{2000}{49}$: $\quad \ddot{y} + 50\dot{y} + 49y = 0$


y
0.06
0.03
Equil.
strong damping
2
4
t

Springs - Forced Vibrations

Suppose an external force $f(t)$ is applied to the spring-mass system at time t.



Derive the equation of motion

The net force is

$$F_{net} = W + T + R + f$$

$$= mg - k(s + y) - \beta\dot{y} + f$$

Apply Newton's Law:

$$F_{net} = m \cdot \ddot{y}$$

$$mg - k(s + y) - \beta\dot{y} + f = m\ddot{y}$$

$$mg - ks - ky - \beta\dot{y} + f = m\ddot{y}$$

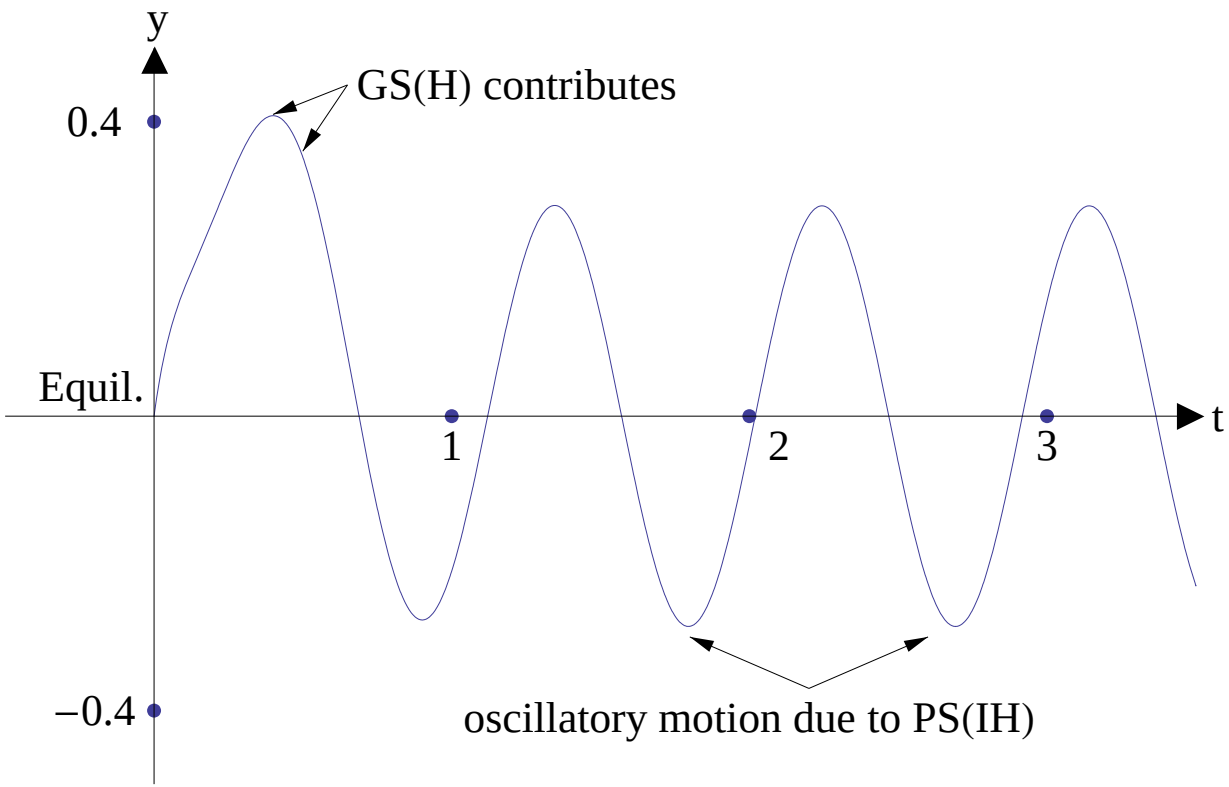
$$\boxed{m\ddot{y} + \beta\dot{y} + ky = f(t)} \qquad \text{(as } mg = ks\text{)}$$

Example 6.10: Apply an external downwards force $f(t) = \frac{160}{7}\sin(7t)$ in Example 6.9 (c) and (a).

Solution:

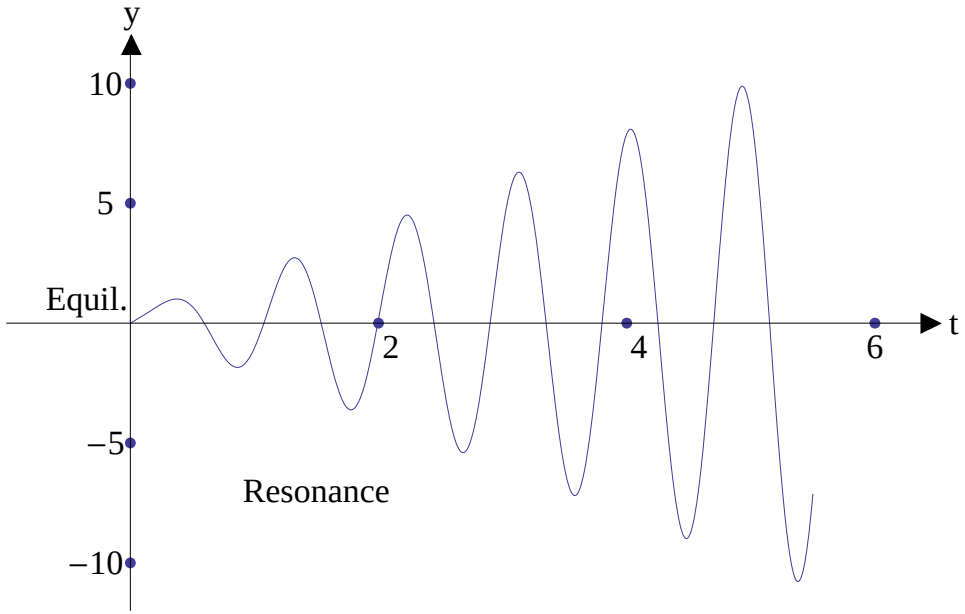
$(a) \quad \beta = \frac{80}{7}: \quad \ddot{y} + 14\dot{y} + 49y = 28\sin(7t)$

$GS(IH): \quad y(t) = (A + Bt)\,e^{-7t} - \frac{2}{7}\cos(7t)$ (working omitted)


y
GS(H) contributes
0.4
Equil.
1
2
3
t
−0.4
oscillatory motion due to PS(IH)

$(b) \quad \beta = 0: \quad \ddot{y} + 49y = 28\sin(7t)$

$GS(IH): \quad y(t) = A\cos(7t) + B\sin(7t) - 2t\cos(7t)$
(working omitted)



Definition

Resonance: Resonance occurs when the external force f has the same form as one of the terms in the $GS(H)$.

If $\beta = 0$, then the $PS(IH)$ will grow without bound as $t \to \infty$.

Section 7: Functions of Two Variables

Example

The temperature T at a point on the Earth's surface at a given time depends on the latitude x and the longitude y. We think of T being a function of the variables x, y and write $T = f(x, y)$.

In general

A function of two variables is a mapping f that assigns a unique real number $z = f(x, y)$ to each pair of real numbers (x, y) in some subset D of the xy plane $\mathbb{R}^2$. We also write

$$f : D \to \mathbb{R}$$

where D is called the domain of f.

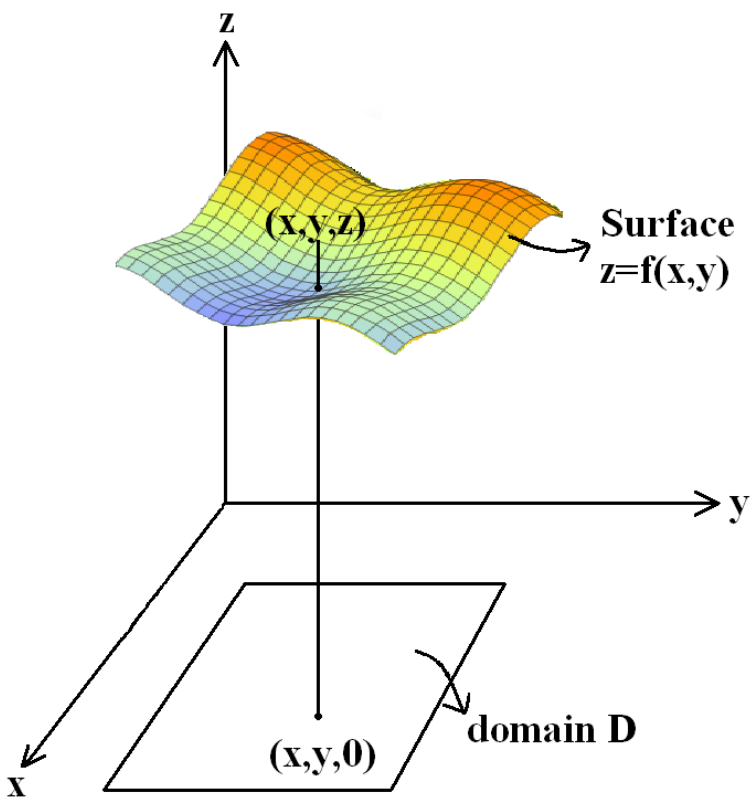
Example

If $f(x, y) = x^2 + y^3$ then $f(2, 1) = 4 + 1 = 5$.

We can represent the function f by its graph in $\mathbb{R}^3$. The graph of f is:

$$\left\{(x,y,z) \in \mathbb{R}^3 : (x,y) \in D \text{ and } z = f(x,y)\right\}.$$

This is a surface lying directly above the domain D. The x and y axes lie in the horizontal plane and the z axis is vertical.



Equations of a Plane

The Cartesian equation of a plane has the form

$$ax + by + cz = d$$

where a, b, c, d are real constants.

$\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is a normal vector to the plane.

In fact, the plane passing through a point (x_0, y_0, z_0) with a normal vector (a, b, c) consists of the points (x, y, z) such that (a, b, c) is perpendicular to $(x - x_0, y - y_0, z - z_0)$ and thus has equation

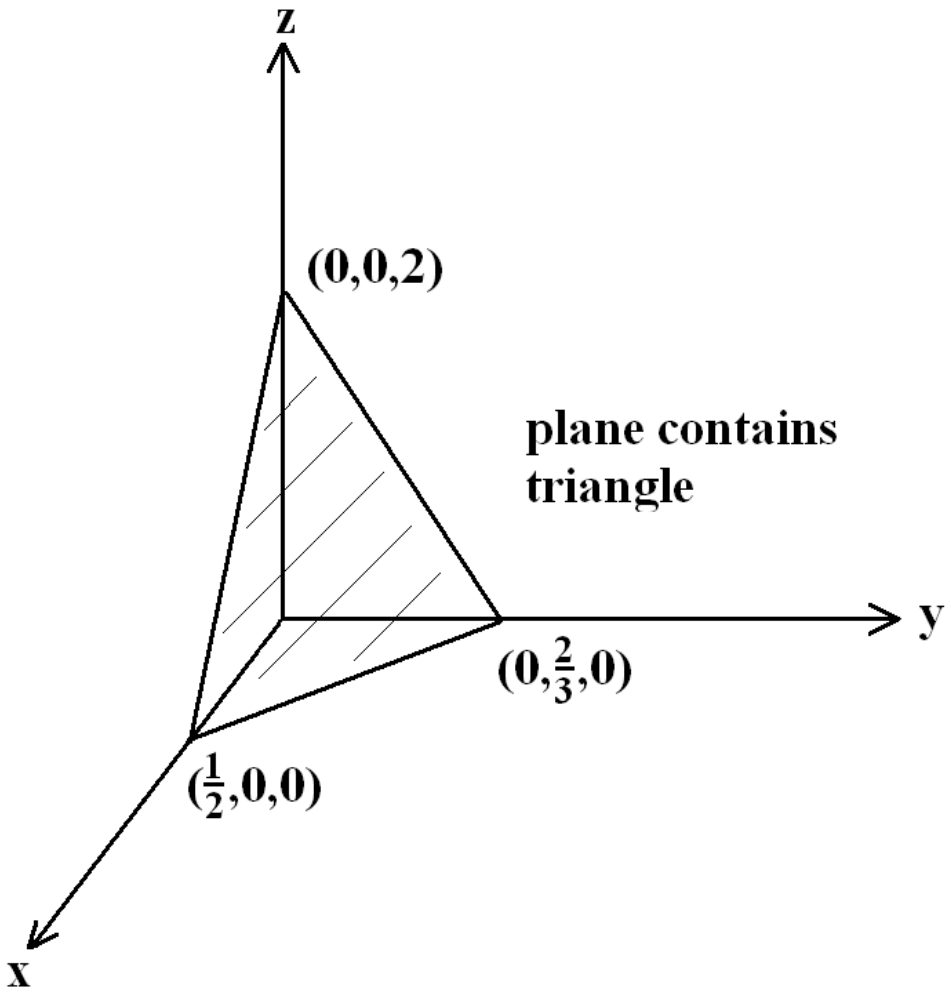
$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

that is,

$$ax + by + cz = ax_0 + by_0 + cz_0.$$

Example 7.1: Sketch the plane $4x + 3y + z = 2$.

Solution:


z
(0,0,2)
plane contains
triangle
y
(0,$\frac{2}{3}$,0)
($\frac{1}{2}$,0,0)
x

Level Curves

A curve on the surface $z = f(x, y)$ for which z is a constant is a contour.

The same curve drawn in the xy plane is a level curve.

So a level curve of f has the form

$$\{(x, y) : f(x, y) = c\}$$

where $c \in \mathbb{R}$ is a constant.

Sketching Functions of Two Variables

The key steps in drawing a graph of a function of two variables $z = f(x, y)$ are:

1. Draw some level curves.
2. Draw the x-z and y-z cross sections.
3. Draw the x, y, z axes.
   For right handed axes: the positive x axis is towards you, the positive y axis points to the right, and the positive z axis points upward.
4. Draw the contours for the level curves you found earlier.
5. Add the cross sections you drew earlier.
6. Label any x, y, z axis intercepts and key points.

Example 7.2: Let $f(x, y) = \sqrt{1 - x^2 - y^2}$.
(a) Find the maximal domain and range of f.
(b) Find the level curves and cross sections in the x-z and y-z planes of the surface $z = f(x, y)$.
(c) Sketch the surface and identify it.

Solution:

Find the maximal domain:

Find the range:

Note:

The domain is a subset of $\mathbb{R}^2$. The range is a subset of $\mathbb{R}$.

Find the level curves:

Find the cross sections:

Sketch the surface:

Identify the surface:

Example 7.3: Sketch the graph of $z = \sqrt{4x^2 + y^2}$.

Solution:

The surface is an elliptic cone.

Limits

Let $f(x, y)$ be a function of two variables.

The limit of $f(x, y)$ as (x, y) approaches (x_0, y_0) is L, written

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = L$$

if $f(x, y)$ gets arbitrarily close to L whenever (x, y) is close enough to (x_0, y_0) but $(x, y) \neq (x_0, y_0)$.

Note:

1. If it exists, L must be a unique finite real number.
2. The limit can exist even if f is undefined at (x_0, y_0).
3. The usual limit laws apply.

Continuity

Let $f(x, y)$ be a function of two variables.

f is continuous at $(x, y) = (x_0, y_0)$ if

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0)$$

Note:

The continuity theorems for functions of one variable can be generalised to functions of two variables.

Example 7.4: Let $f(x, y) = x^2 + y^2$. At which points (x, y) is f continuous?

Solution:

Example 7.5: Evaluate $\lim_{(x,y)\to(2,1)} \log(1 + 2x^2 + 3y^2)$.

Solution:

First Order Partial Derivatives

Let $f(x, y)$ be a function of two variables. The first order partial derivatives of f with respect to the variables x and y are defined by the limits:

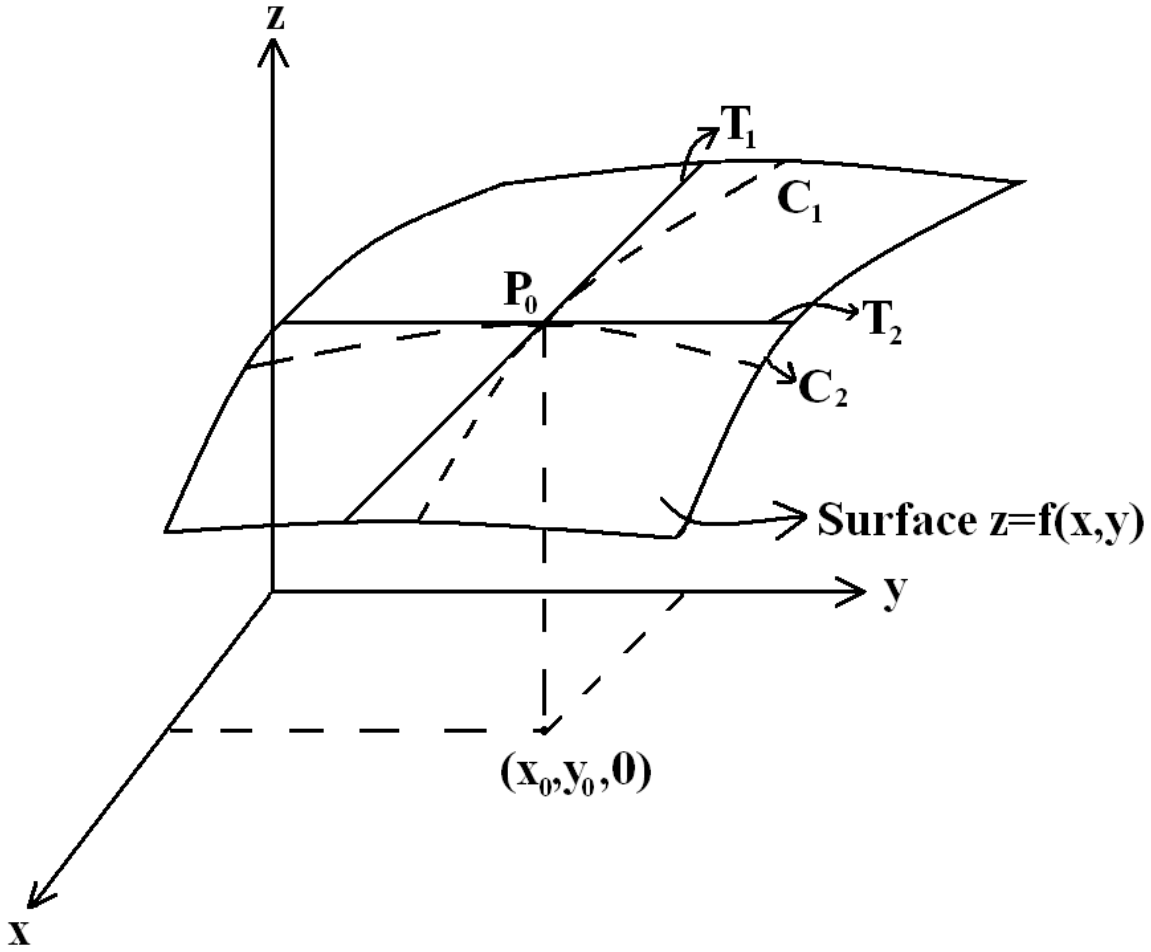
$$f_x = \frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

$$f_y = \frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

Note:

- $\dfrac{\partial f}{\partial x}$ measures the rate of change of f with respect to x when y is held constant.
- $\dfrac{\partial f}{\partial y}$ measures the rate of change of f with respect to y when x is held constant.

Geometric Interpretation of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$



Let C_1 be the curve where the vertical plane $y = y_0$ intersects the surface. Then $\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}$ gives the slope of the tangent to C_1 at (x_0, y_0, z_0).

Let C_2 be the curve where the vertical plane $x = x_0$ intersects the surface. The $\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)}$ gives the slope of the tangent to C_2 at (x_0, y_0, z_0).

- T_1 and T_2 are the tangent lines to C_1 and C_2.

Example 7.6: Let $f(x, y) = xy^2$. Find $\dfrac{\partial f}{\partial y}$ from first principles.

Solution:

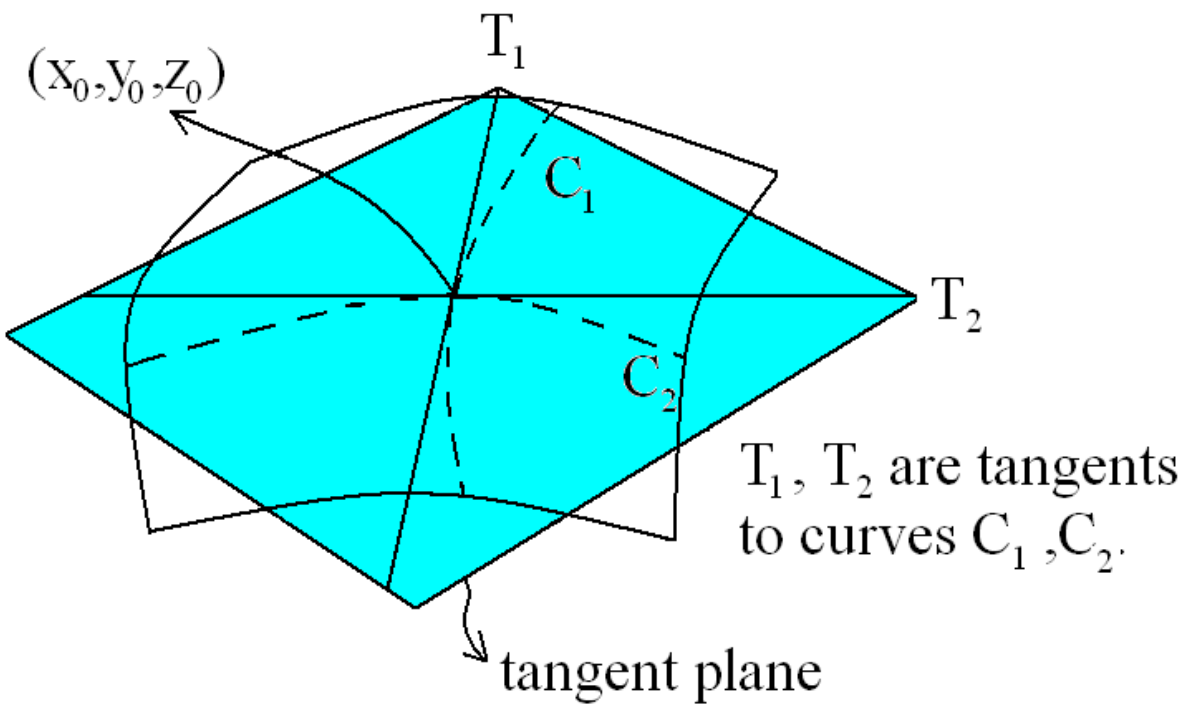
Example 7.7: Let $f(x, y) = 3x^3y^2 + x^2 + y \log x$.
Find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

Solution:

Tangent Planes and Differentiability

Let $f(x, y)$ be a function of two variables. We say that f is differentiable at (x_0, y_0) if the tangent lines to all curves on the surface $z = f(x, y)$ passing through (x_0, y_0, z_0) form a plane, called the tangent plane.

This holds if f_x and f_y exist and are continuous near (x_0, y_0).



The tangent line T_1 has equation ($y = y_0$ fixed):

$$z - z_0 = \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)}(x - x_0)$$

The tangent line T_2 has equation ($x = x_0$ fixed):

$$z - z_0 = \frac{\partial f}{\partial y}\Big|_{(x_0,y_0)}(y - y_0)$$

Since a plane passing through (x_0, y_0, z_0) has the form

$$z - z_0 = \alpha(x - x_0) + \beta(y - y_0)$$

the tangent plane has equation

$$z - z_0 = \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)}(x - x_0) + \frac{\partial f}{\partial y}\Big|_{(x_0,y_0)}(y - y_0)$$

or equivalently,

$$\boxed{z = z_0 + \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)}(x - x_0) + \frac{\partial f}{\partial y}\Big|_{(x_0,y_0)}(y - y_0).}$$

Linear Approximations

If f is differentiable at (x_0, y_0), we can approximate $z = f(x, y)$ by its tangent plane at (x_0, y_0, z_0), when (x, y) is close to (x_0, y_0).

That is:

$$\boxed{f(x,y) \approx \underbrace{z_0 + \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}(x - x_0) + \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)}(y - y_0)}_{\text{equation of tangent plane}}}$$

when (x, y) is close to (x_0, y_0).

This is called the linear approximation to f near (x_0, y_0).

Example 7.8: Let

$$f(x, y) = xe^{xy}.$$

(a) Find the equation of the tangent plane to the surface $z = f(x, y)$ at the point where $(x, y) = (2, 0)$.

(b) Hence, approximate $f(2.1, -0.1)$.

Solution:

Note:

The actual value is $(2.1)e^{-0.21} \approx 1.70223$

Approximate Change

Rearranging the linear approximation equation, we get

$$f(x,y) - f(x_0,y_0) \approx \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)}(x - x_0) + \frac{\partial f}{\partial y}\Big|_{(x_0,y_0)}(y - y_0).$$

Let $\Delta x = x - x_0, \quad \Delta y = y - y_0, \quad \Delta f = z - z_0 = f(x,y) - f(x_0,y_0)$.

The approximate change in f near (x_0,y_0), for small changes Δx and Δy in x and y, is:

$$\boxed{\Delta f \approx \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)}\Delta x + \frac{\partial f}{\partial y}\Big|_{(x_0,y_0)}\Delta y}$$

Example 7.9: For a differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$, it is known that

$$f_x(3,8) = 3 \quad \text{and} \quad f_y(3,8) = -2.$$

Estimate the change in f as (x, y) changes from $(3, 8)$ to $(3.2, 7.9)$.

‘

Second Order Partial Derivatives

Let $f(,y)$ be a function of two variables. The second order partial derivatives of f with respect to x and y are defined by:

- $f_{xx} = \left(f_x\right)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial x^2}$
- $f_{yy} = \left(f_y\right)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial y^2}$
- $f_{xy} = \left(f_x\right)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial y \partial x}$
- $f_{yx} = \left(f_y\right)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial x \partial y}$

Theorem:

If the second order partial derivatives of f exist and are continuous then $f_{xy} = f_{yx}$.

Example 7.10: Find the second order partial derivatives of $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x \sin(x + 2y)$.

Solution:

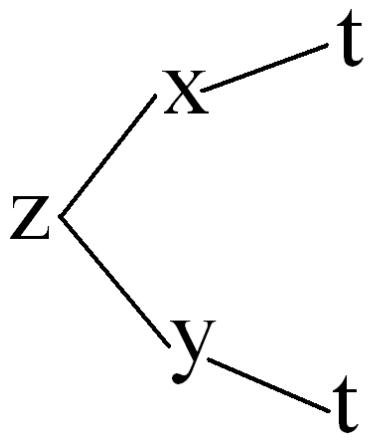
Note:

$f_{xy} = f_{yx}$ as expected since trigonometric functions and polynomials are continuous for all $(x, y) \in \mathbb{R}^2$.

Chain Rule

1. If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$ are differentiable functions, then $z = f(g(t), h(t))$ is a function of t, and

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}}$$

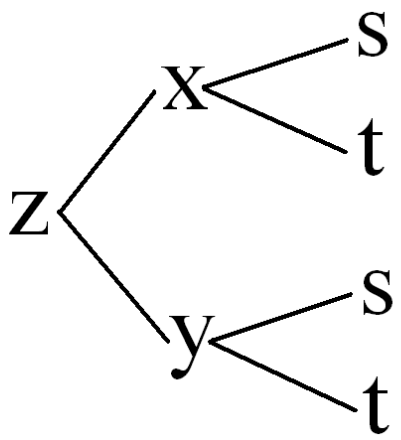


Example 7.11: If $z = x^2 - y^2$, $x = \sin t$, $y = \cos t$. Find $\dfrac{dz}{dt}$ at $t = \dfrac{\pi}{6}$.

Solution:

2. If $z = f(x, y)$ and $x = g(s, t)$, $y = h(s, t)$ are differentiable functions, then z is a function of s and t with

$$\boxed{\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \end{aligned}}$$



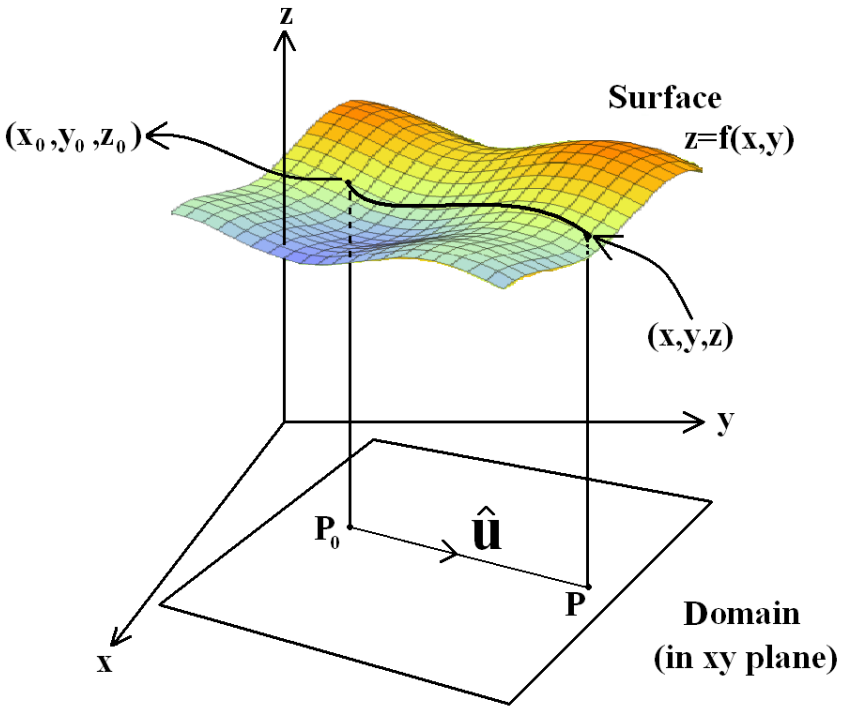
Example 7.12: If $z = e^x \sinh y$, $x = st^2$, $y = s^2 t$.
Find $\dfrac{\partial z}{\partial s}$.

Solution:

Directional Derivatives

Let $\hat{\mathbf{u}} = (u_1, u_2)$ be a unit vector in the xy-plane (so $u_1^2 + u_2^2 = 1$). The rate of change of f at $P_0 = (x_0, y_0)$ in the direction $\hat{\mathbf{u}}$ is the directional derivative $D_{\hat{\mathbf{u}}} f \big|_{P_0}$.

Geometrically this represents the slope of the surface $z = f(x, y)$ above the point P_0 in the direction $\hat{\mathbf{u}}$.



The straight line starting at $P_0 = (x_0, y_0)$ with velocity $\hat{\mathbf{u}} = (u_1, u_2)$ has parametric equations:

$$x = x_0 + tu_1, \quad y = y_0 + tu_2.$$

Hence,

$$\begin{aligned}
D_{\hat{\mathbf{u}}} f \Big|_{P_0} &= \text{rate of change of } f \text{ along the straight line at } t = 0 \\
&= \text{value of } \frac{d}{dt} f(x_0 + tu_1, y_0 + tu_2) \text{ at } t = 0 \\
&= f_x(x_0, y_0) x'(0) + f_y(x_0, y_0) y'(0) \qquad \text{by the chain rule} \\
&= f_x(x_0, y_0)\, u_1 + f_y(x_0, y_0)\, u_2.
\end{aligned}$$

We can also write this as a dot product

$$D_{\hat{\mathbf{u}}} f \Big|_{P_0} = \left(\frac{\partial f}{\partial x}\bigg|_{P_0}, \frac{\partial f}{\partial y}\bigg|_{P_0} \right) \cdot (u_1, u_2).$$

Gradient Vectors

If $f : \mathbb{R}^2 \to \mathbb{R}$ is a differentiable function, we can define the gradient of f to be the vector

$$\boxed{\text{grad}\, f = \nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)}$$

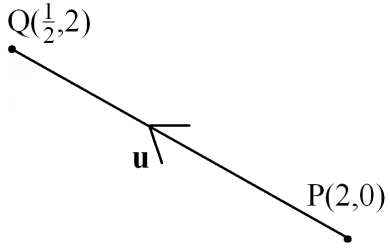
Then the directional derivative of f at the point P_0 in the direction $\hat{\mathbf{u}}$ is the dot product

$$\boxed{D_{\hat{\mathbf{u}}} f \Big|_{P_0} = \nabla f \Big|_{P_0} \cdot \hat{\mathbf{u}}}$$

Example 7.13: Find the directional derivative of $f(x, y) = xe^y$ at $(2, 0)$ in the direction from $(2, 0)$ towards $\left(\frac{1}{2}, 2\right)$.

Solution:

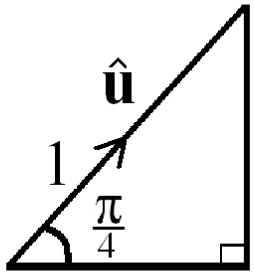
- direction $\hat{\mathbf{u}}$



Example 7.14: Find the directional derivative of $f(x,y) = \arcsin\left(\frac{x}{y}\right)$ at $(1,2)$ in the direction $\frac{\pi}{4}$ anticlockwise from the positive x axis.

Solution:

- direction $\hat{\mathbf{u}}$

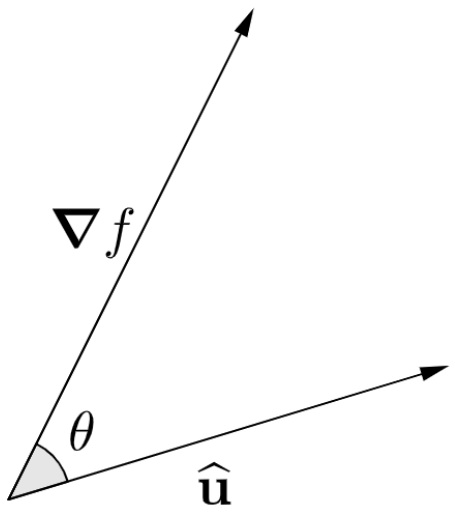


Properties of ∇f

The directional derivative of f is

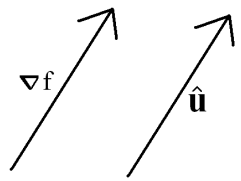
$$\begin{aligned} D_{\hat{\mathbf{u}}} f &= \nabla f \cdot \hat{\mathbf{u}} \\ &= \|\nabla f\| \, \|\hat{\mathbf{u}}\| \cos\theta \\ &= \|\nabla f\| \cos\theta \end{aligned}$$

where θ is the angle between ∇f and $\hat{\mathbf{u}}$.



So for fixed ∇f:

- $D_{\hat{\mathbf{u}}} f$ is maximum when $\cos\theta = 1$ so $\theta = 0$.

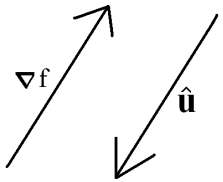


$\Rightarrow \nabla f$ points in the direction in which f increases the fastest

In this direction, $D_{\hat{\mathbf{u}}} f = \|\nabla f\|$

$\Rightarrow \|\nabla f\|$ is the fastest rate of increase of f.

- $D_{\hat{\mathbf{u}}}f$ is minimum when $\cos\theta = -1$ so $\theta = \pi$



$\Rightarrow -\nabla f$ points in the direction in which f decreases the fastest

In this direction, $D_{\hat{\mathbf{u}}}f = -\|\nabla f\|$

- $D_{\hat{\mathbf{u}}}f = 0$ when $\cos\theta = 0$ so $\theta = \dfrac{\pi}{2}$ and $\nabla f \perp \hat{\mathbf{u}}$.

But $D_{\hat{\mathbf{u}}}f = 0$, whenever $\hat{\mathbf{u}}$ is tangent to a level curve of f (where f = constant).

$$\Rightarrow \nabla f \perp \text{ level curves of } f$$

This gives a geometrical interpretation of ∇f:

- the *direction* of ∇f is the direction of steepest ascent of f.
- the *length* of ∇f, $\|\nabla f\|$, is the slope of the surface in the direction of steepest ascent.
- the direction of $-\nabla f$ is the direction of steepest descent of f.
- ∇f is perpendicular to the level curves of f.

Note:

The direction of steepest ascent is sometimes also called:

- the direction of fastest increase
- the direction of steepest increase

and similarly for the direction of steepest descent.

Example 7.15: In what direction does $f(x, y) = xe^y$

(a) increase (b) decrease

most rapidly at $(2, 0)$? Express direction as a unit vector.

Solution:

From Example 7.13

$$\nabla f(2, 0) = \mathbf{i} + 2\mathbf{j}$$

Stationary Points

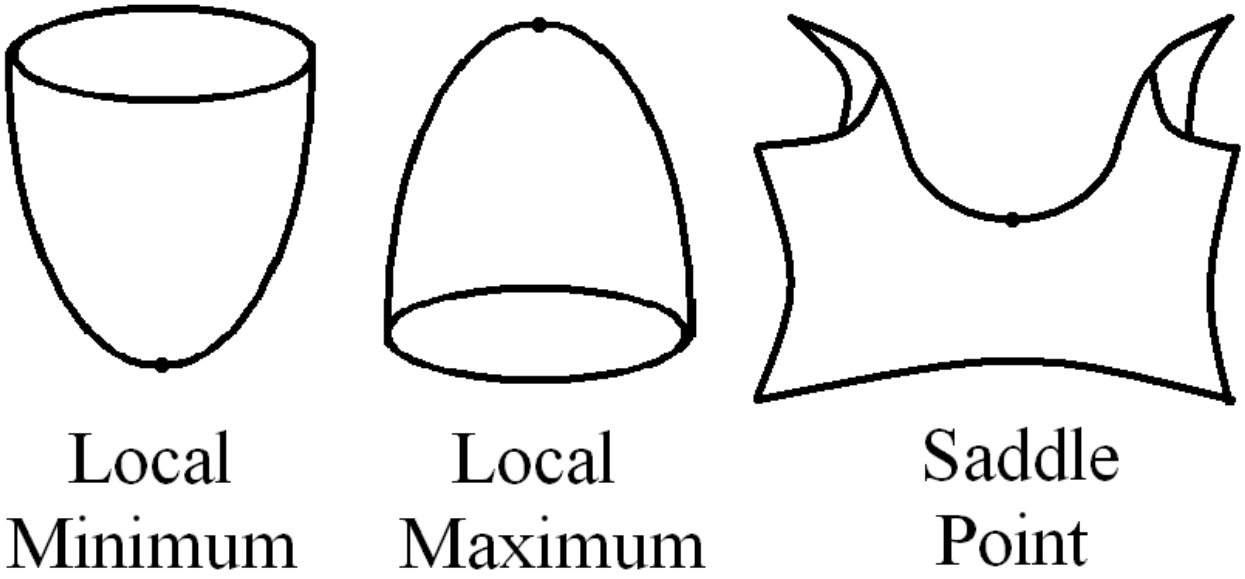
A stationary point of f is a point (x_0, y_0) at which

$$\nabla f = \mathbf{0}$$

So $\dfrac{\partial f}{\partial x} = 0$ and $\dfrac{\partial f}{\partial y} = 0$ simultaneously at (x_0, y_0).

Geometrically, this means that the tangent plane to the graph $z = f(x, y)$ at (x_0, y_0) is horizontal, i.e. parallel to the xy-plane.

Three important types of stationary points are



A function f has a

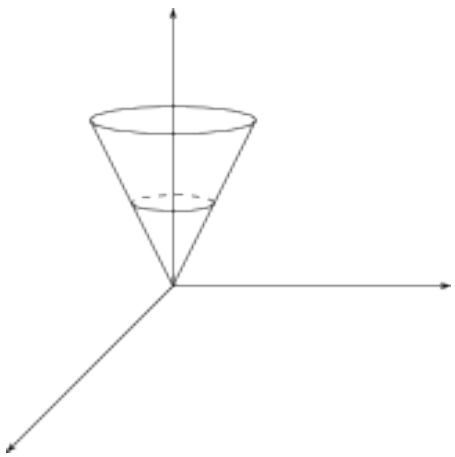
1. local maximum at (x_0, y_0) if $f(x, y) \leq f(x_0, y_0)$ for all (x, y) in some disk centred at (x_0, y_0),

2. local minimum at (x_0, y_0) if $f(x, y) \geq f(x_0, y_0)$ for all (x, y) in some disk centred at (x_0, y_0),

3. saddle point at (x_0, y_0) if (x_0, y_0) is a stationary point, and there are points near (x_0, y_0) with $f(x, y) > f(x_0, y_0)$ and other points near (x_0, y_0) with $f(x, y) < f(x_0, y_0)$.

Any local maximum or minimum of f will occur at a critical point (x_0, y_0) such that

1. $\nabla f(x_0, y_0) = \mathbf{0}$ <u>or</u>
2. $\dfrac{\partial f}{\partial x}$ and/or $\dfrac{\partial f}{\partial y}$ do not exist at (x_0, y_0).



$z = \sqrt{x^2 + y^2}$. Minimum at $(0, 0)$ BUT ∇f does not exist at $(0, 0)$.

Second Derivative Test

If $\nabla f(x_0, y_0) = \mathbf{0}$ and the second partial derivatives of f are continuous on an open disk centred at (x_0, y_0), consider the Hessian function

$$H(x, y) = f_{xx} f_{yy} - (f_{xy})^2$$

evaluated at (x_0, y_0).

Then (x_0, y_0) is a

1. local minimum if $H(x_0, y_0) > 0$ and $f_{xx}(x_0, y_0) > 0$.
2. local maximum if $H(x_0, y_0) > 0$ and $f_{xx}(x_0, y_0) < 0$.
3. saddle point if $H(x_0, y_0) < 0$.

Note: Test is inconclusive if $H(x_0, y_0) = 0$.

Example 7.16: Find and classify the stationary points of $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 + 1$.

Solution:

Example 7.17: Find and classify the stationary points of $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = y \sin x$.

Solution:

Partial Integration

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a continuous function over a domain D in $\mathbb{R}^2$.

The partial indefinite integrals of f with respect to the first and second variables (say x and y) are denoted by:

$\int f(x,y)\,dx$ and $\int f(x,y)\,dy$.

- $\int f(x,y)\,dx$ is evaluated by holding y fixed and integrating with respect to x.
- $\int f(x,y)\,dy$ is evaluated by holding x fixed and integrating with respect to y.

Example 7.18: Evaluate $\int (3x^2y + 12y^2x^3)\,dx$.

Solution:

Note:

Example 7.19: Evaluate $\int_0^1 (3x^2y + 12y^2x^3)\,dy$.

Solution:

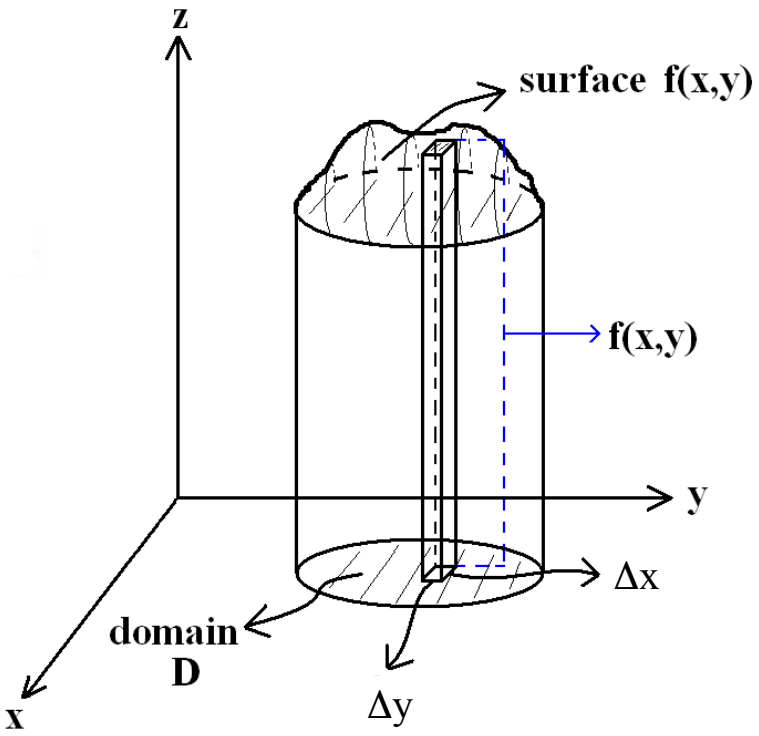
Double Integrals

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a continuous function over a domain D in $\mathbb{R}^2$.

We can evaluate the double integral:

$$\boxed{\iint_D f(x,y)\,dA = \iint_D f(x,y)\,dx\,dy}$$

$\iint_D f(x,y)\,dA$ is the volume under the surface $z = f(x,y)$ that lies above the domain D in the xy plane, if $f(x,y) \geq 0$ in D.



$$\text{Volume of thin rod} \quad = \quad \underbrace{(\text{Area base})}_{\substack{\| \\ \Delta x \Delta y}} \cdot \underbrace{(\text{height})}_{\substack{\| \\ f(x,y)}}$$

The double integral is defined as the limit of sums of the volumes of the rods:

$$\begin{aligned}\iint_D f(x,y)\,dA &= \iint_D f(x,y)\,dx\,dy \\ &= \lim_{\Delta x\to 0}\lim_{\Delta y\to 0}\sum_{i=1}^{n}[f(x,y)\Delta x\Delta y]_i\end{aligned}$$

Note:

If $f(x,y)=1$ then

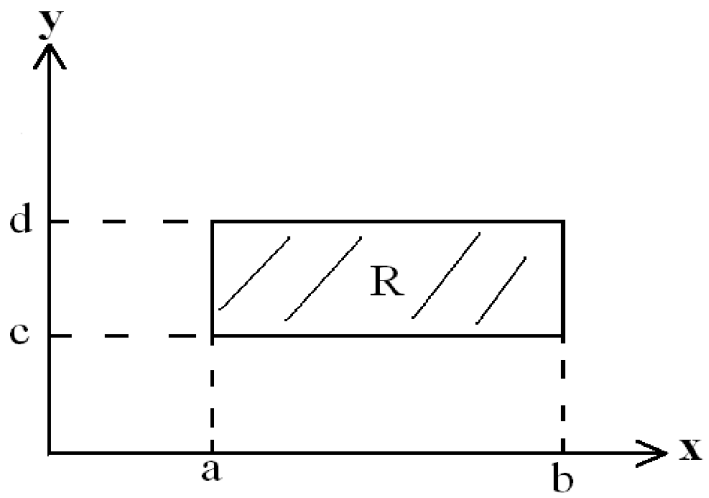
$$\iint_D dA = \iint_D dx\,dy$$

gives the area of the domain D.

Double Integrals Over Rectangular Domains

Definitions

1. $R = [a, b] \times [c, d]$ is a rectangular domain defined by $a \le x \le b, \quad c \le y \le d$.



2. $\int_c^d \int_a^b f(x, y)\, dx\, dy = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy$ means integrate with respect to x first and then integrate with respect to y.

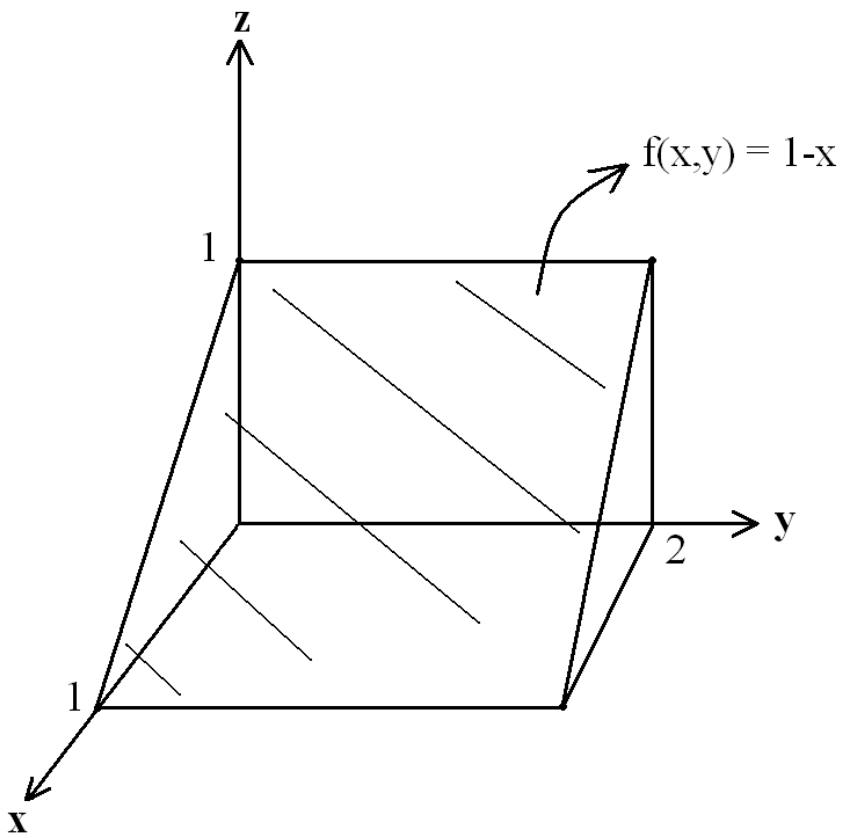
Fubini's Theorem

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a continuous function over the domain $R = [a, b] \times [c, d]$. Then

$$\begin{aligned} \iint_R f(x, y) dA &= \int_c^d \int_a^b f(x, y)\, dx\, dy \\ &= \int_a^b \int_c^d f(x, y)\, dy\, dx \end{aligned}$$

So order of integration is not important.

Example 7.20: Using double integrals, find the volume of the wedge shown below.



Solution:

This can also be calculated as

$$\int_0^1 \int_0^2 \left(1 - x\right) dy\, dx$$

This gives the same answer, as expected by Fubini's Theorem. (Working omitted.)